MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, BIPOLAR, LOW-POWER SCHOTTKY TTL, FLIP-FLOPS, CASCADABLE, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, low-power Schottky TTL, flip-flops, bistable logic microcircuits. Two product assurance classes and a choice of case outline and lead finish are provided for each type and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510.

1.2.1 Device types. The device types shall be as follows:

Device type	Circuit
01	Dual J-K flip-flop with clear
02	Dual D type flip-flop with clear and preset
03	Dual J-K flip-flop with clear and preset
04	Dual J-K flip-flop with preset
05	Dual J-K flip-flop with preset and common clear and common clock
06	Hex D type flip-flop with common clear and common clock
07	Quad D type flip-flop with common clear and common clock
08	Dual J-$\overline{K}$ flip-flop with clear
09	Dual J-$\overline{K}$ flip-flop with clear and preset
10	Dual J-K flip-flop with clear and preset

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outlines. The case outlines shall be designated as follows:

Letter	Case outline (see MIL-M-38510, appendix C)
A	F-1 (14-lead, .280" x .260" x .085"), flat package
B	F-3 (14-lead, .280" x .200" x .070"), flat package
C	D-1 (14-lead, .785" x .310" x .200"), dual-in-line package
D	F-2 (14-lead, .390" x .260" x .085"), flat package
E	D-2 (16-lead, .840" x .310" x .200"), dual-in-line package
F	F-5 (16-lead, .440" x .285" x .085"), flat package
2	C-2 (20-terminal, .358" x .358" .100"), square chip carrier package
X	C-2A (20-terminal, .358" x .358" .075"), square chip carrier package

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.3 Absolute maximum ratings.

Supply voltage range	-0.5 V dc to 7.0 V dc
Input voltage range	-1.5 V dc at -18 mA to 5.5 V dc
Storage temperature range	-65°C to +150°C
Maximum power dissipation per flip-flop, (P_D) 1/	25 mW
Lead temperature (soldering, 10 seconds)	300°C
Thermal resistance, junction to case (θ_{JC}):	
Cases A, B, C, D, E, 2, and X	(See MIL-M-38510, appendix C)
Junction temperature (T_J) 2/	+175°C

1.4 Recommended operating conditions.

Supply voltage range (V_{CC})	4.5 V dc mininum to 5.5 V dc maximum
Minimum high level input voltage (V_{IH})	2.0 V dc
Maximum low level input voltage (V_{IL})	0.7 V dc
Case operating temperature range (T_C)	-55°C to +125°C
Input set up time:	
Device types:	
01, 03, 04, 05, 08, 09, and 10	25 ns minimum
02, 06, and 07	20 ns minimum
Input hold time:	
Device types:	
01, 03, 04, 05, 08, and 10	0 ns minimum
02, 06, 07, and 09	5 ns minimum

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standard. The following specification and standard form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of the specification and standard required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

1/ Must withstand the added P_D due to short-circuit test (e.g., I_{OS}).
2/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening conditions in accordance with method 5004 of MIL-STD-883.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions 1/ $-55°C \le T_C \le +125°C$		Device type	Limits		Unit
					Min	Max	
High-level output voltage	V_{OH}	V_{CC} = 4.5 V, I_{OH} = -400 µA		All	2.5		V
Low-level output voltage	V_{OL}	V_{CC} = 4.5 V, I_{OL} = 4 mA		All		0.4	V
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, I_{IN} = -18 mA, T_C = +25°C		All		-1.5	V
Low-level input current	I_{IL1}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	2/	01,03,04, 05,08,10	-.030	-.360	mA
			2/	06,07	-.075	-.400	
	I_{IL2}		3/	02,09	-.030	-.400	mA
			3/	06	-.075	-.420	
			3/	07	-.075	-.420	
	I_{IL3}		4/	01,08	-.06	-.72	mA
			4/	03,04,10	-.06	-.76	
	I_{IL4}		5/	01,02,03, 04,05,08, 09,10	-.06	-.80	mA
	I_{IL5}		6/	02	-.06	-1.20	mA
	I_{IL6}		4/	05	-.12	-1.52	mA
	I_{IL7}		6/	05	-.12	-1.60	mA
			6/	09	-.06	-1.60	
High-level input current	I_{IH1}	V_{CC} = 5.5 V, V_{IN} = 2.7 V	7/	All		20	µA
	I_{IH2}	V_{CC} = 5.5 V, V_{IN} = 5.5 V	7/	All		100	µA
	I_{IH3}	V_{CC} = 5.5 V, V_{IN} = 2.7 V	8/	02,09		40	µA
	I_{IH4}	V_{CC} = 5.5 V, V_{IN} = 5.5 V	8/	02,09		200	µA
	I_{IH5}	V_{CC} = 5.5 V, V_{IN} = 2.7 V	9/	01,02,03, 04,05,08, 10		60	µA
	I_{IH6}	V_{CC} = 5.5 V, V_{IN} = 5.5 V	9/	01,02,03, 04,05,08, 10		300	µA
	I_{IH7}	V_{CC} = 5.5 V, V_{IN} = 2.7 V	10/	01,03,04, 08,09,10		80	µA
	I_{IH8}	V_{CC} = 5.5 V, V_{IN} = 5.5 V	10/	01,03,04, 08,09,10		400	µA
	I_{IH9}	V_{CC} = 5.5 V, V_{IN} = 2.7 V	6/	05		120	µA
	I_{IH10}	V_{CC} = 5.5 V, V_{IN} = 5.5 V	6/	05		600	µA
	I_{IH11}	V_{CC} = 5.5 V, V_{IN} = 2.7 V	4/	05		160	µA
	I_{IH12}	V_{CC} = 5.5 V, V_{IN} = 5.5 V	4/	05		800	µA

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions 1/ $-55°C \le T_C \le +125°C$	Device type	Limits		Unit
				Min	Max	
Short-circuit output current	I_{OS}	V_{CC} = 5.5 V, V_{IN} = 0 V 11/ 11/	01,02,03, 05,06,07, 08,09	-15	-100	mA
			04,10	-15	-130	
Supply current	I_{CC}	V_{CC} = 5.5 V, V_{IN} = 5.5 V	01,02,03, 04,05,08, 09,10		8	mA
			06		26	
			07		18	
Maximum clock frequency	f_{MAX}		01,03,04, 05,06,07, 08,10	25		MHz
			02,09	20		
Propagation delay to high logic level (clear or preset to output)	t_{PLH1}		01,03,04, 05,08,10	5	32	ns
			02,09	5	39	
			07	5	51	
Propagation delay to low logic level (clear or preset to output)	t_{PHL1}	V_{CC} = 5.0 V, C_L = 50 pF ±10%, R_L = 2 kΩ ±5%	01,03,04, 05,08,10	5	40	ns
			02,09	5	59	
			06	5	52	
			07	5	55	
Propagation delay to high logic level (clock to output)	t_{PLH2}		01,03,04, 05,08,10	5	32	ns
			02,09	5	39	
			06	5	47	
			07	5	46	
Propagation delay to low logic level (clock to output)	t_{PHL2}		01,03,04, 05,08,10	5	42	ns
			02,09	5	59	
			06	5	52	
			07	5	55	

1/ See table III for complete terminal conditions.

2/ Input condition - J or K (device types 01, 03, 04, 05, 08, and 10); and D (device types 06 and 07).

3/ Input condition - D (device type 02); clock or clear (device types 06 and 07); and J or $\overline{K}$ (device type 09).

4/ Input condition - Clock.

5/ Input condition - Clear or preset (device types 03 and 10); clear (device types 01 and 08); preset or clock (device types 02 and 09); and preset (device types 04 and 05).

6/ Input condition - Clear.

7/ Input condition - J or K (device types 01, 03, 04, 05, 08, and 10); D (device type 02); J or $\overline{K}$ (device type 09); and D, clear, clock (device types 06 and 07).

8/ Input condition - Preset or clock.

9/ Input condition - Clear or preset (device types 03 and 10); clear (device types 01, 02, and 08); and preset (device types 04 and 05).

10/ Input condition - Clock (device type 01, 03, 04, 08, and 10); and clear (device type 09).

11/ Not more than one output should be shorted at a time.

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3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Terminal connections and logic diagrams. The terminal connections and logic diagrams shall be as specified on figure 1.

3.2.2 Truth tables. The truth tables shall be as specified on figure 2.

3.2.3 Schematic circuits. The schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained by the agent activity and will be available upon request.

3.2.4 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.5).

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 10 (see MIL-M-38510, appendix E).

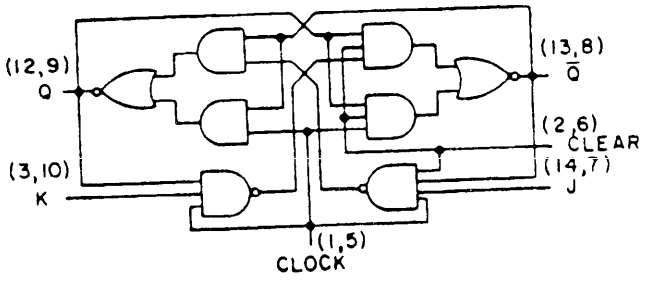
TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters (method 5004)	1	1
Final electrical test parameters (method 5004)	1*,2,3,7,9, 10,11	1*,2,3,9
Group A test requirements (method 5005)	1,2,3,7,8, 9,10,11	1,2,3,7,8, 9, 10, 11
Group B test requirements (method 5005) subgroup 5	1,2,3,9,10, 11	N/A
Group C end-point electrical parameters (method 5005)	N/A	1,2,3
Group D end-point electrical parameters (method 5005)	1,2,3	1,2,3

*PDA applies to subgroup 1 (see 4.2c).

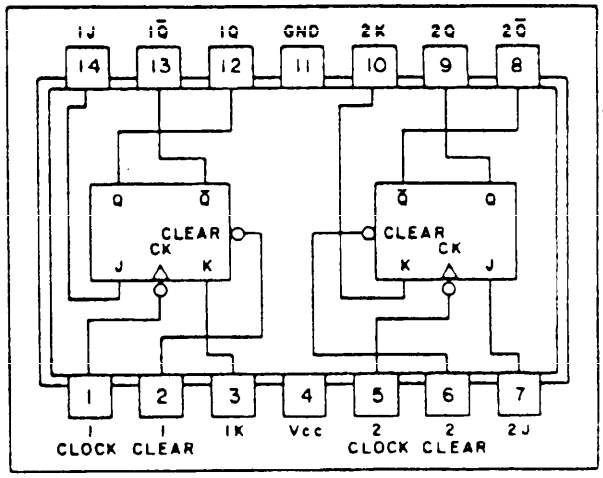
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DEVICE TYPE 01

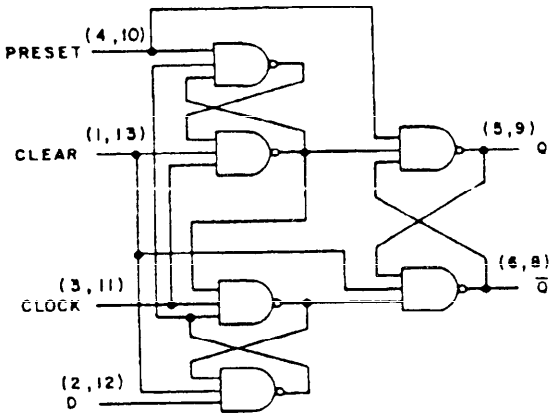


DEVICE TYPE 01

CASES A, B, C, AND D

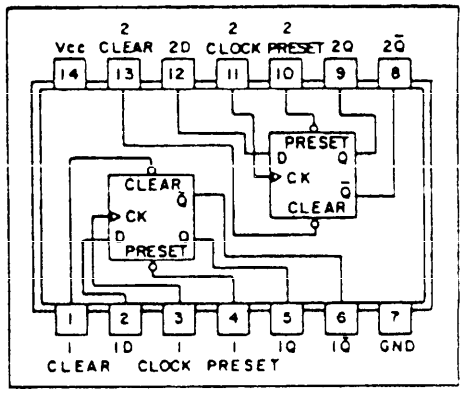


DEVICE TYPE 02

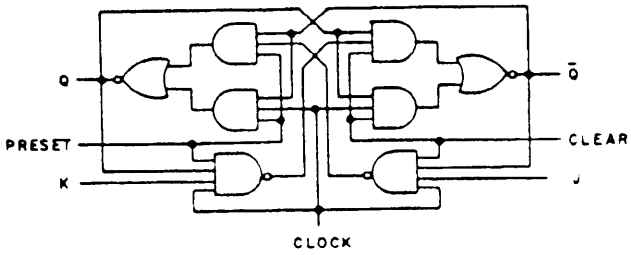


DEVICE TYPE 02

CASES A, B, C, AND D



DEVICE TYPE 03



DEVICE TYPE 03

CASES E AND F

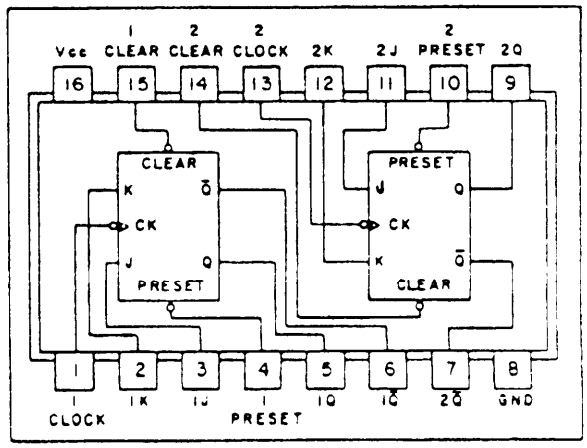
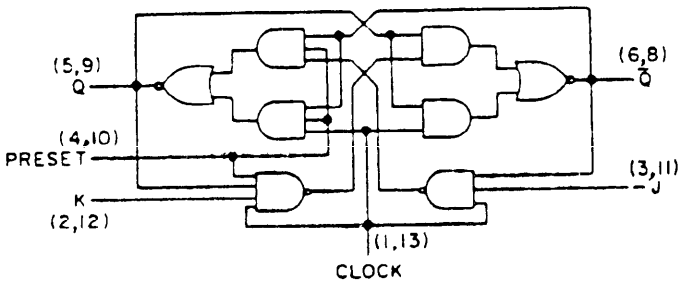


FIGURE 1. Logic diagrams and terminal connections.

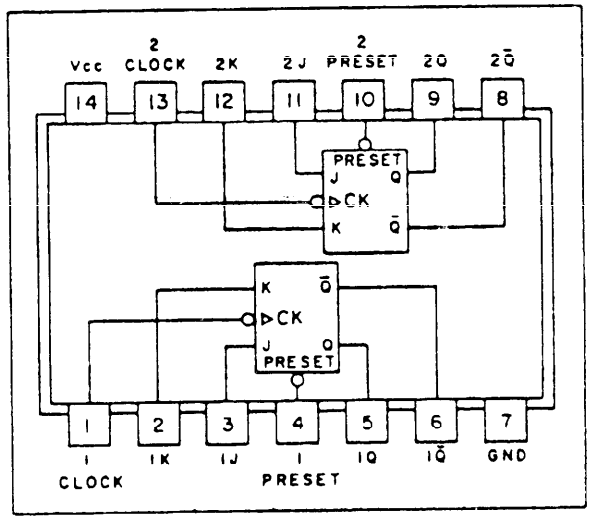
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DEVICE TYPE 04

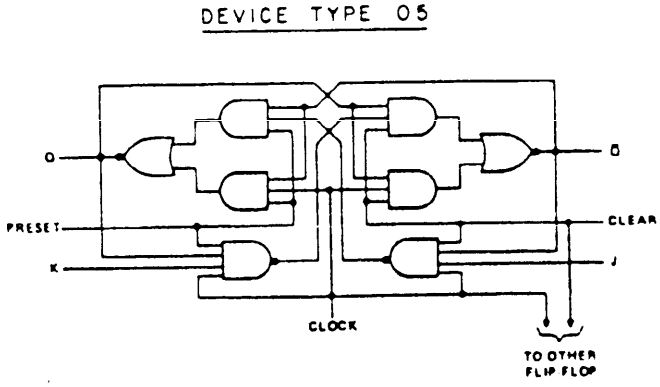


DEVICE TYPE 04

CASES A, B, C, AND D



DEVICE TYPE 05



DEVICE TYPE 05

CASES A, B, C, AND D

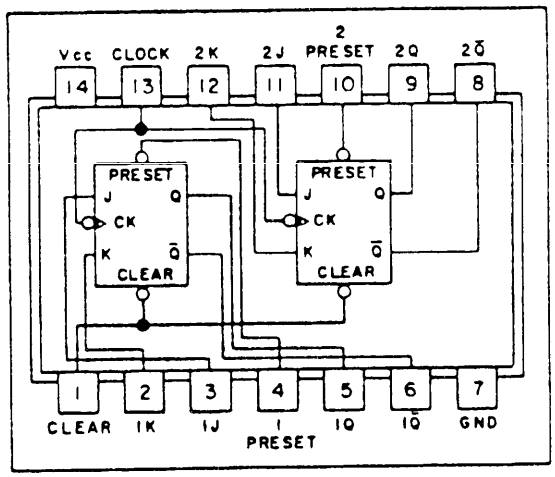
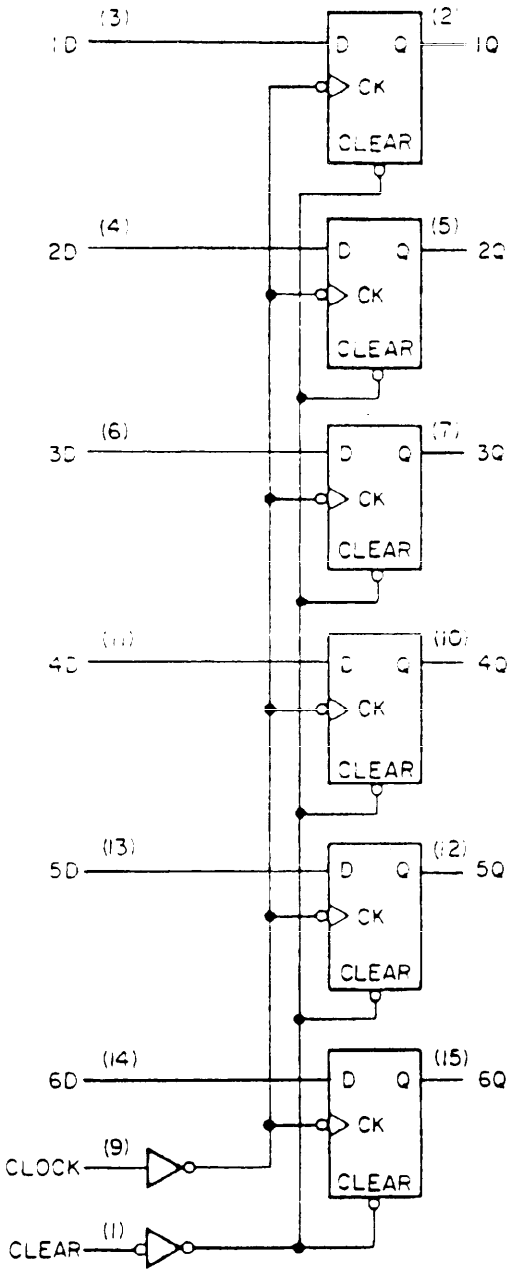


FIGURE 1. Logic diagrams and terminal connections - Continued.

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DEVICE TYPE 06



DEVICE TYPE 06

CASES E AND F

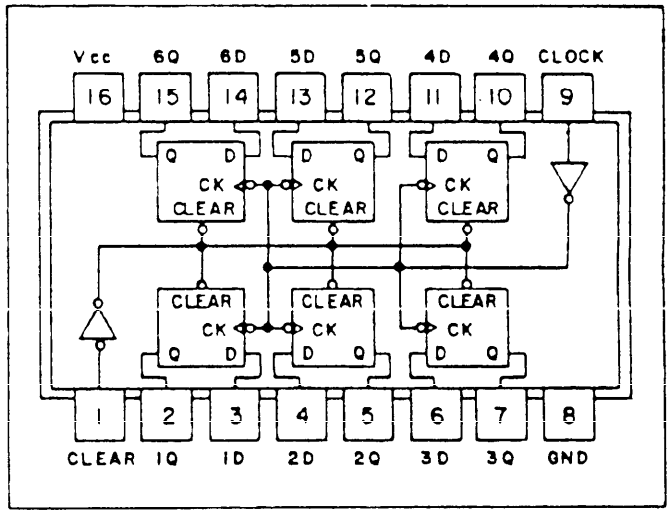
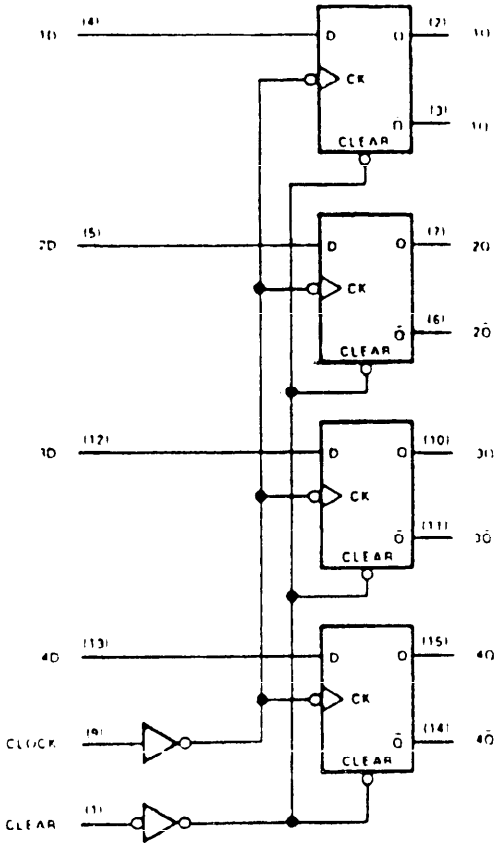


FIGURE 1. Logic diagrams and terminal connections - Continued.

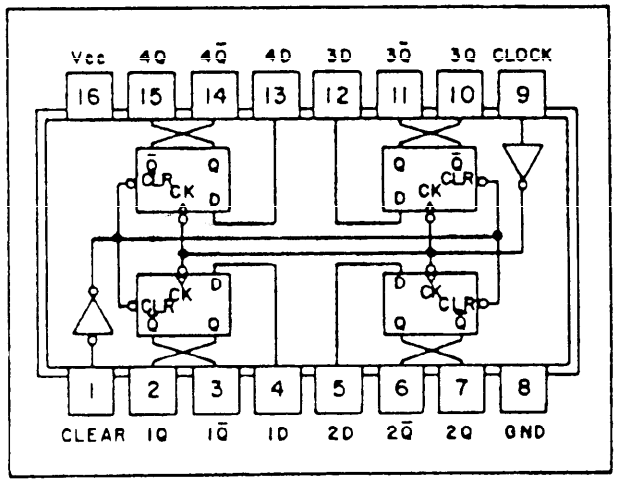
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DEVICE TYPE 07

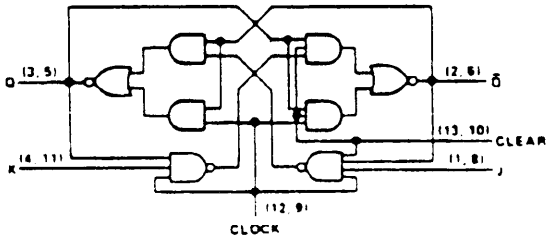


DEVICE TYPE 07

CASES E AND F



DEVICE TYPE 08



DEVICE TYPE 08

CASES A, B, C, AND D

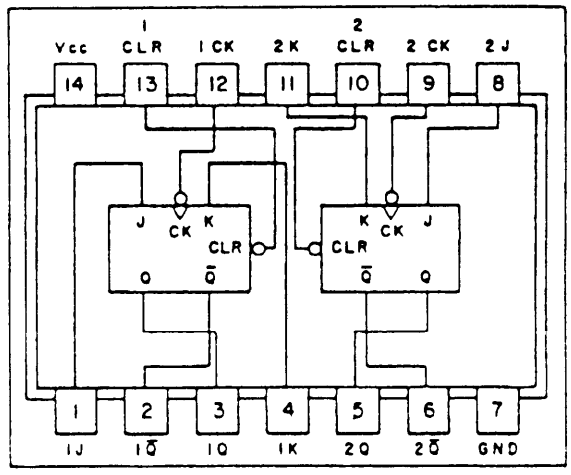
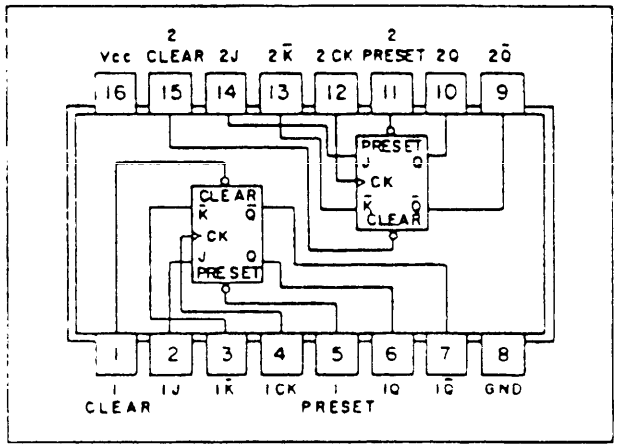


FIGURE 1. Logic diagrams and terminal connections - Continued.

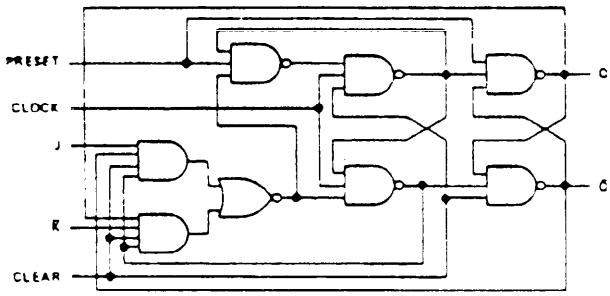
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DEVICE TYPE 09

CASES E AND F

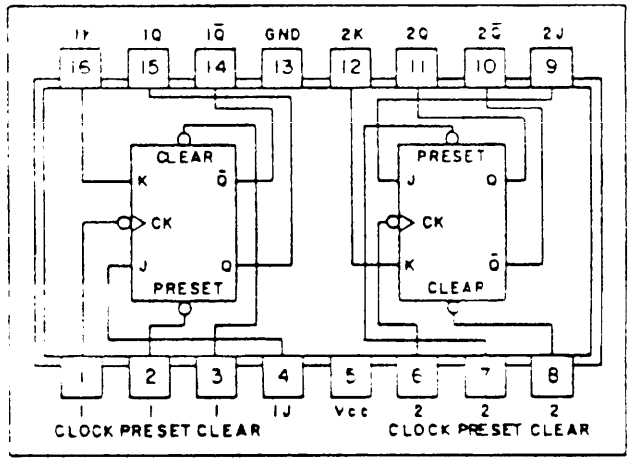


DEVICE TYPE 05



DEVICE TYPE 10

CASES E AND F



DEVICE TYPE 10

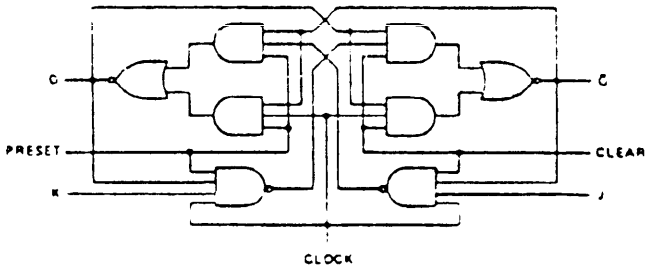
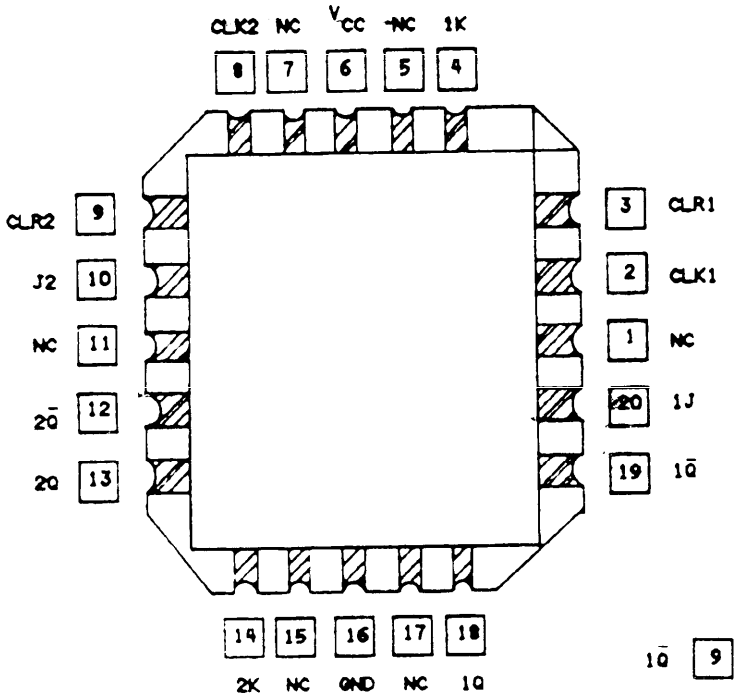


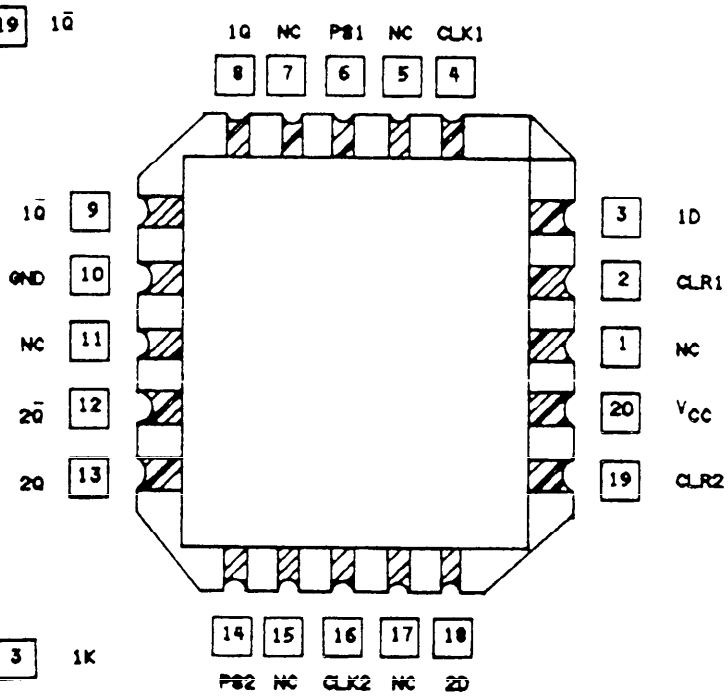
FIGURE 1. Logic diagrams and terminal connections - Continued.

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Device type 01
Cases 2 and X



Device type 02
Cases 2 and X



Device type 03
Cases 2 and X

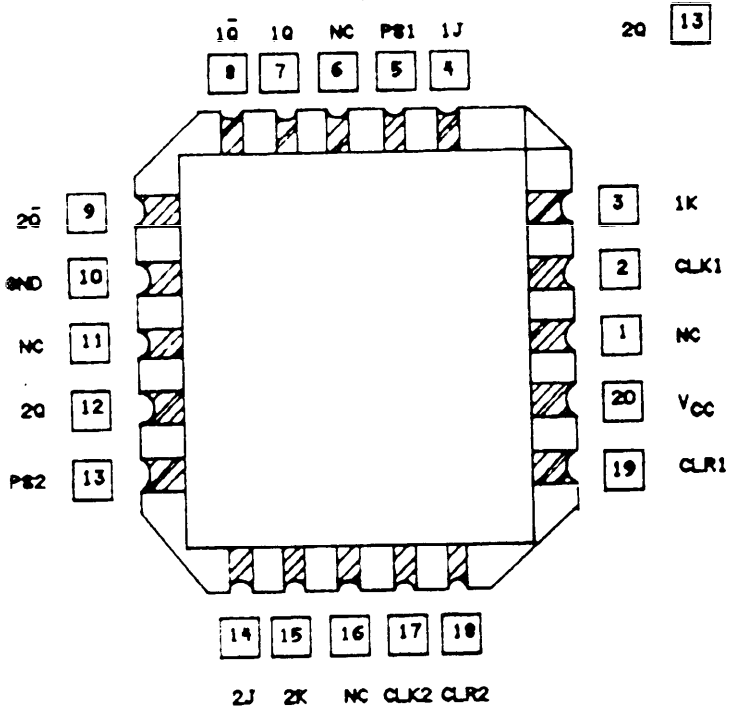
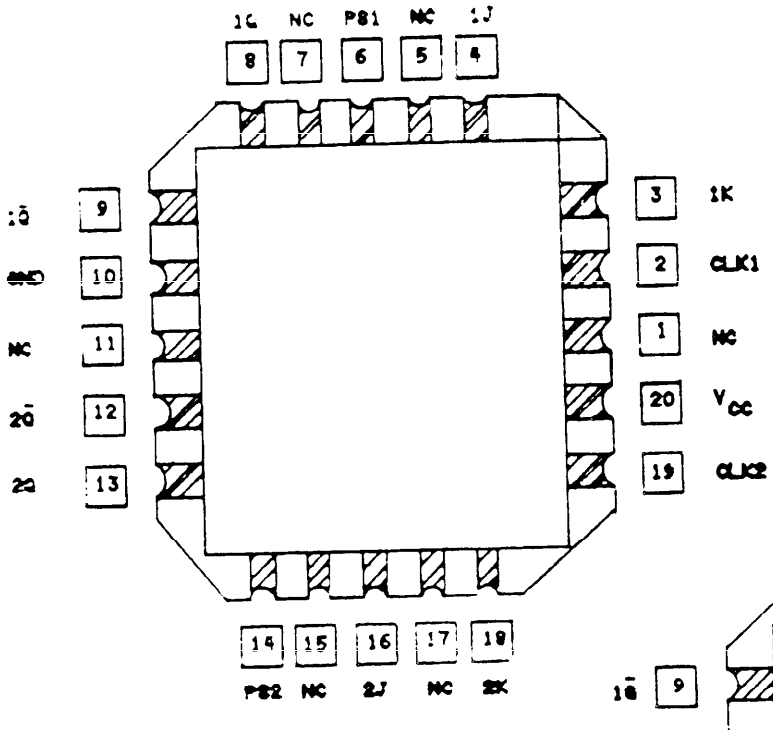


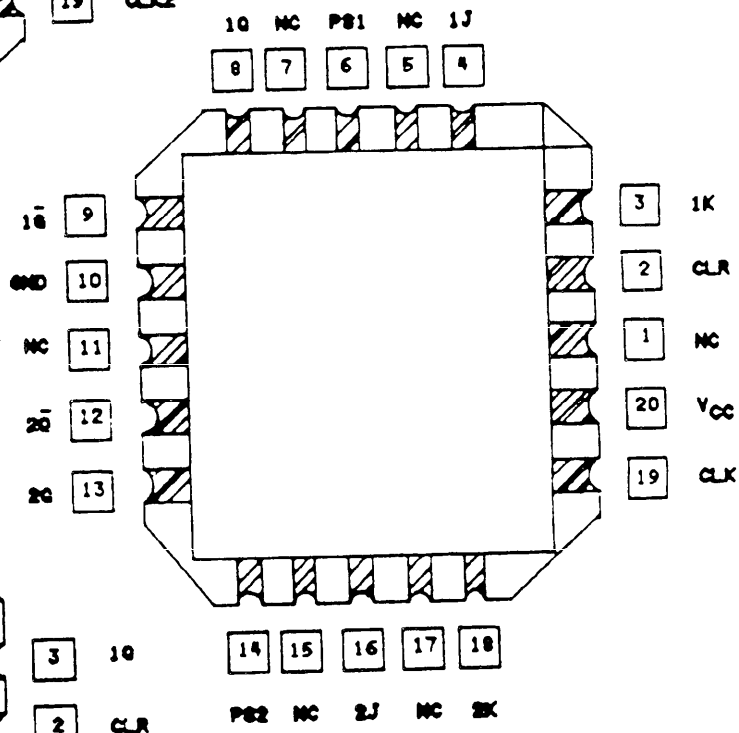
FIGURE 1. Logic diagrams and terminal connections - Continued.

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Device type 04
Cases 2 and X



Device type 05
Cases 2 and X



Device type 06
Cases 2 and X

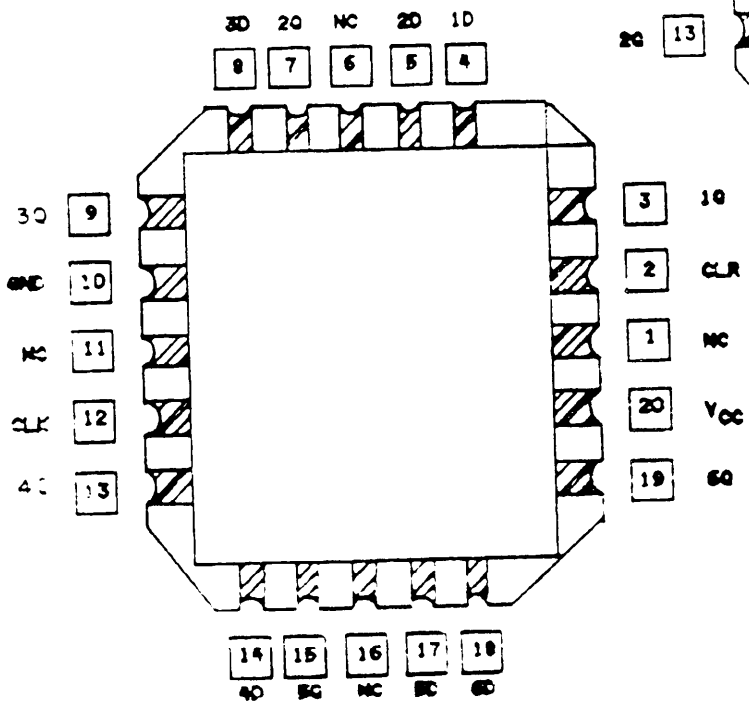
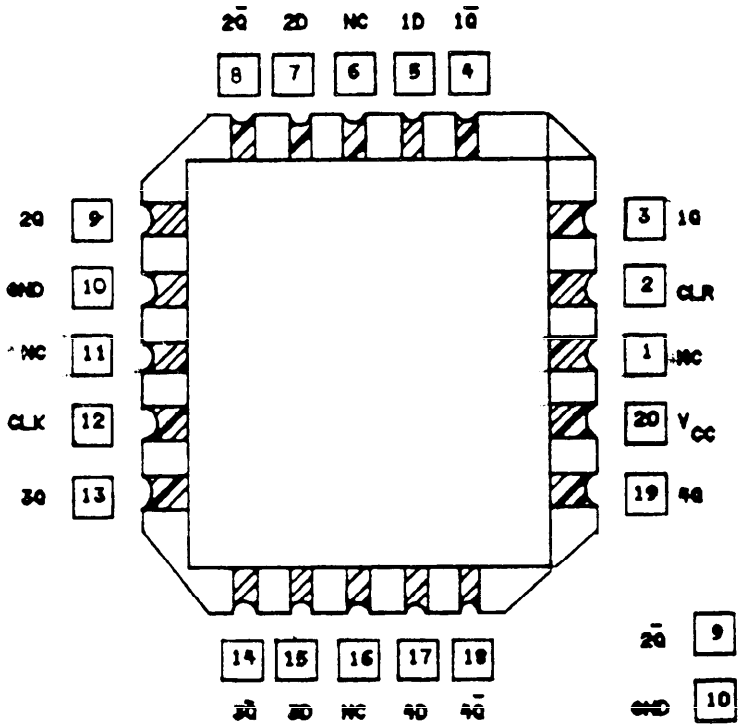


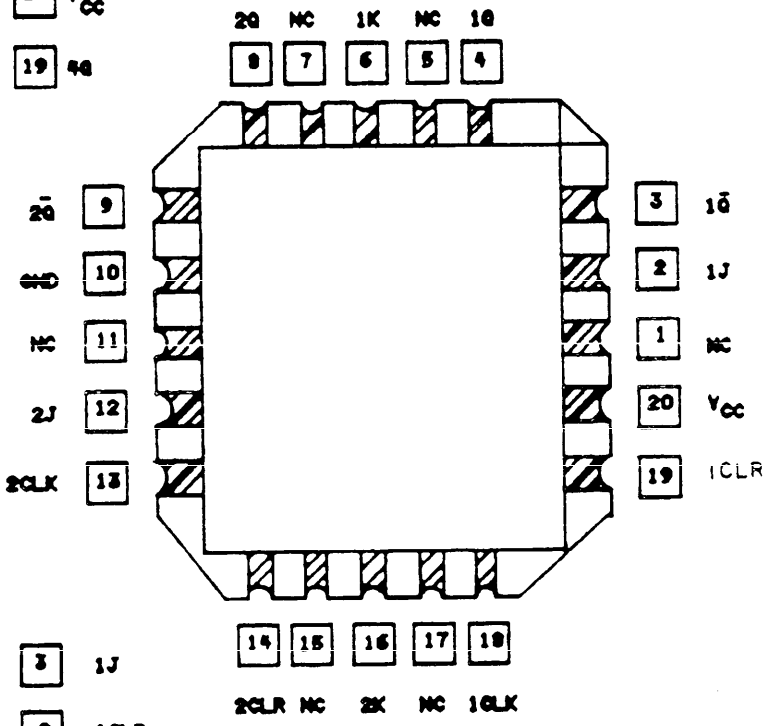
FIGURE 1. Logic diagrams and terminal connections - Continued.

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Device type 07
Cases 2 and X



Device type 08
Cases 2 and X



Device type 09
Cases 2 and X

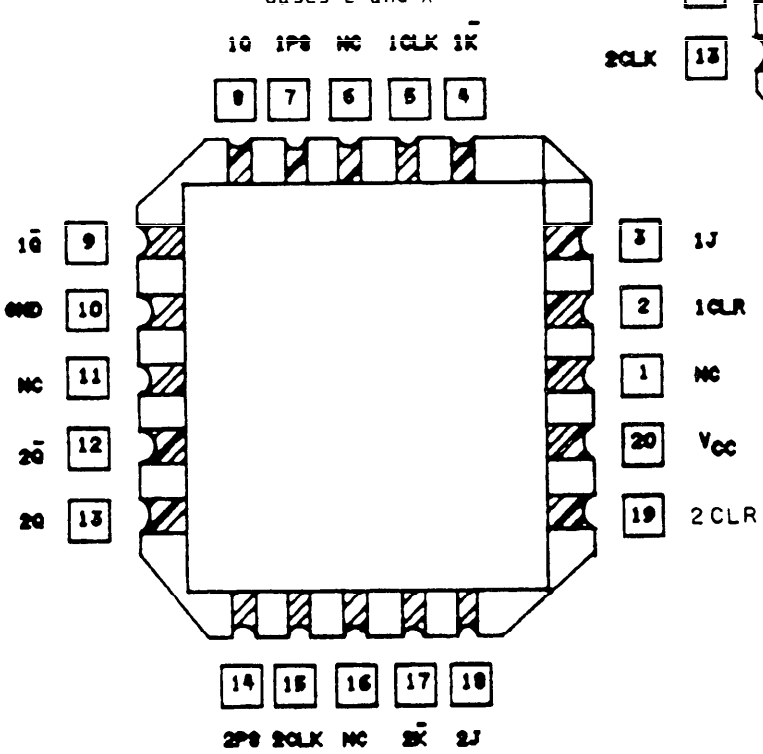


FIGURE 1. Logic diagrams and terminal connections - Continued.

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Device type 10

Cases 2 and X

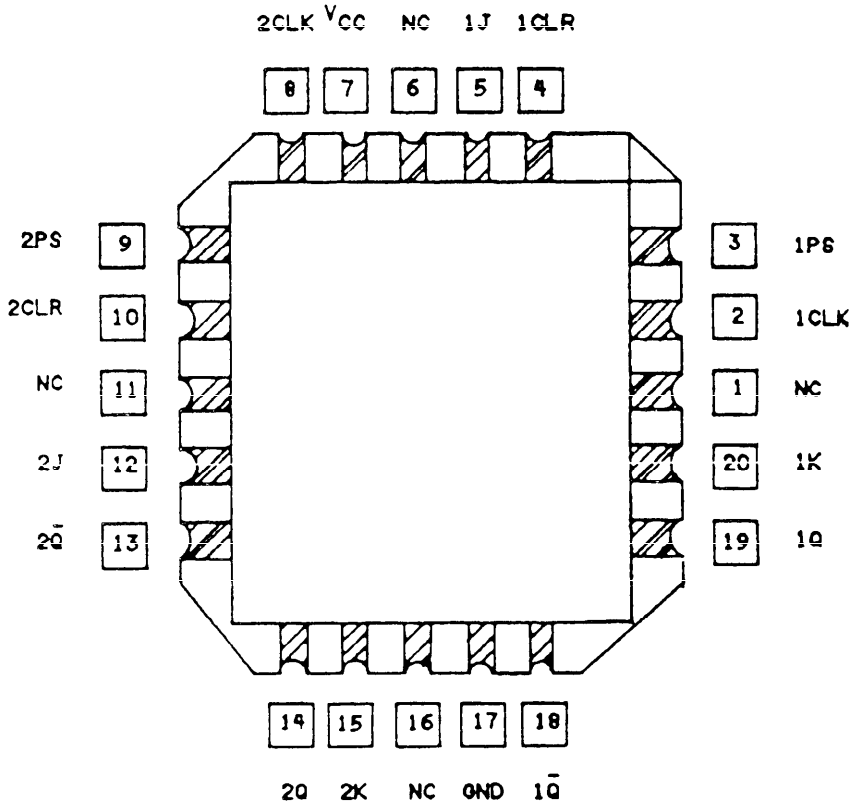


FIGURE 1. Logic diagrams and terminal connections - Continued.

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DEVICE TYPE 01

INPUTS				OUTPUTS	
CLEAR	CLOCK	J	K	Q	$\overline{Q}$
L	X	X	X	L	H
H	↓	L	L	Q_0	$\overline{Q}_0$
H	↓	H	L	H	L
H	↓	L	H	L	H
H	↓	H	H	TOGGLE	
H	H	X	X	Q_0	$\overline{Q}_0$

H = high level (steady state)
L = low level (steady state)
X = irrelevant
↓ = transition from high to low level
Q_0 = the level of Q before the indicated input conditions were established
TOGGLE: Each output changes to the complement of its previous level on each ↓ clock transition.

DEVICE TYPE 02

INPUTS				OUTPUTS	
PRESET	CLEAR	CLOCK	D	Q	$\overline{Q}$
L	H	X	X	H	L
H	L	X	X	L	H
L	L	X	X	H*	H*
H	H	↑	H	H	L
H	H	↑	L	L	H
H	H	L	X	Q_0	$\overline{Q}_0$

H = high level (steady state)
L = low level (steady state)
X = irrelevant
↑ = transition from low to high level
Q_0 = the level of Q before the indicated steady state input conditions were established.
*This configuration is nonstable; that is, it will not persist when preset and clear inputs return to their inactive (high) level.

FIGURE 2. Truth tables.

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DEVICE TYPES 03 AND 10

INPUTS					OUTPUTS	
PRESET	CLEAR	CLOCK	J	K	Q	$\overline{Q}$
L	H	X	X	X	H	L
H	L	X	X	X	L	H
L	L	X	X	X	H*	H*
H	H	↓	L	L	Q_0	$\overline{Q}_0$
H	H	↓	H	L	H	L
H	H	↓	L	H	L	H
H	H	↓	H	H	TOGGLE	
H	H	H	X	X	Q_0	$\overline{Q}_0$

H = high level (steady state)
L = low level (steady state)
X = irrelevant
↓ = transition from high to low level
Q_0 = the level of Q before the indicated steady state input conditions were established.
TOGGLE: Each output changes to the complement of its previous level on each ↓ clock transition.
*This configuration is nonstable; that is, it will not persist when preset and clear inputs return to their inactive (high) level.

DEVICE TYPE 04

INPUTS				OUTPUTS	
PRESET	CLOCK	J	K	Q	$\overline{Q}$
L	X	X	X	H	L
H	↓	L	L	Q_0	$\overline{Q}_0$
H	↓	H	L	H	L
H	↓	L	H	L	H
H	↓	H	H	TOGGLE	
H	H	X	X	Q_0	$\overline{Q}_0$

H = high level (steady state)
L = low level (steady state)
X = irrelevant
↓ = transition from high to low level
Q_0 = the level of Q before the indicated steady state input conditions were established.
TOGGLE: Each output changes to the complement of its previous level of each ↓ clock transition.

FIGURE 2. Truth tables - Continued.

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DEVICE TYPE 05

INPUTS					OUTPUTS	
PRESET	CLEAR	CLOCK	J	K	Q	$\overline{Q}$
L	H	X	X	X	H	L
H	L	X	X	X	L	H
L	L	X	X	X	H*	H*
H	H	↓	L	L	Q_0	$\overline{Q}_0$
H	H	↓	H	L	H	L
H	H	↓	L	H	L	H
H	H	↓	H	H	TOGGLE	
H	H	H	X	X	Q_0	$\overline{Q}_0$

H = high level (steady state)
L = low level (steady state)
X = irrelevant
↓ = transition from high to low level
Q_0 = the level of Q before the indicated steady state input conditions were established.
TOGGLE: Each output changes to the complement of its previous level on each ↓ clock transition.
*This configuration is nonstable; that is, it will not persist when preset and clear inputs return to their inactive (high) level.

DEVICE TYPE 06

INPUTS			OUTPUT
CLEAR	CLOCK	D	Q
L	X	X	L
H	↑	H	H
H	↑	L	L
H	L	X	Q_0

H = high level (steady state)
L = low level (steady state)
X = irrelevant
↑ = transition from low to high level
Q_0 = the level of Q before the indicated steady state input conditions were established.

FIGURE 2. Truth tables - Continued.

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DEVICE TYPE 07

INPUTS			OUTPUTS	
CLEAR	CLOCK	D	Q	$\overline{Q}$
L	X	X	L	H
H	↑	H	H	L
H	↑	L	L	H
H	L	X	Q_0	$\overline{Q}_0$

H = high level (steady state)
L = low level (steady state)
X = irrelevant
↑ = transition from low to high level
Q_0 = the level of Q before the indicated steady state input conditions were established.

Device type 08, Truth Table

INPUTS				OUTPUTS	
CLEAR	CLOCK	J	K	Q	$\overline{Q}$
L	X	X	X	L	H
H	↑	L	L	Q_0	$\overline{Q}_0$
H	↑	H	L	H	L
H	↑	L	H	L	H
H	↑	H	H	TOGGLE	

H = high level (steady state)
L = low level (steady state)
X = irrelevant
↑ = transition from low to high level
Q_0 = the level of Q before the indicated input conditions were established.
TOGGLE: Each output changes to the complement of its previous level on each clock transition.

FIGURE 2. Truth tables - Continued.

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DEVICE TYPE 09

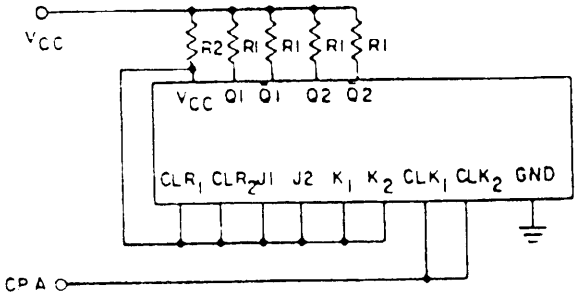
INPUTS					OUTPUTS	
PRESET	CLEAR	CLOCK	J	$\overline{K}$	Q	$\overline{Q}$
L	H	X	X	X	H	L
H	L	X	X	X	L	H
L	L	X	X	X	H*	H*
H	H	↓	L	L	L	H
H	H	↓	H	L	TOGGLE	
H	H	↓	L	H	Q_0	$\overline{Q}_0$
H	H	↓	H	H	H	L
H	H	L	X	X	Q_0	$\overline{Q}_0$

H = high level (steady state)
L = low level (steady state)
X = irrelevant
↓ = transition from low to high level
Q_0 = the level of Q before the indicated steady state input conditions were established.
TOGGLE: Each output changes to the complement of its previous level on each ↓ clock transition.
*This configuration is nonstable; that is it will not presist when preset and clear inputs return to their inactive (high) level.

FIGURE 2. Truth tables - Continued.

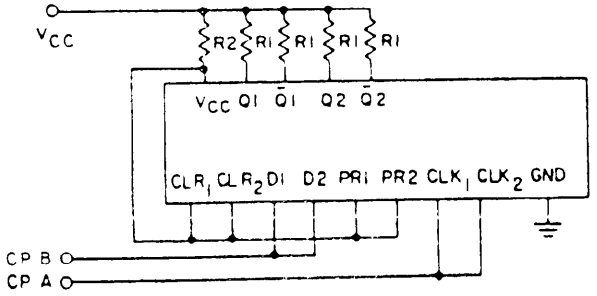
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:02Z
Check the source to verify that this is the current version before use.

Device types 01 and 08
Condition D or E



NOTES:
1. CP A = 100 kHz ±50% square wave; duty cycle = 50 ±15%; V_{IL} = -0.5 V minimum to +0.7 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.
2. R1 = 1 kΩ ±5%; R2 = 27Ω ±5%.
3. V_{CC} shall be high enough to insure that 5.0 V minimum is present at V_{CC} device terminal.

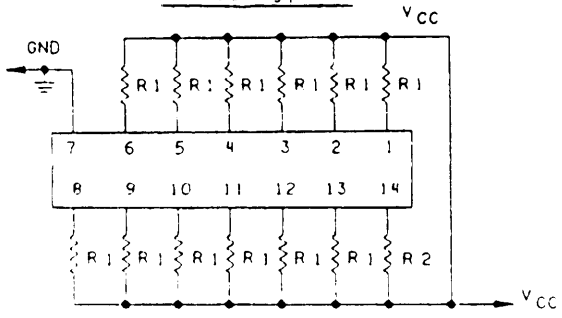
Device type 02
Condition D or E



NOTES:
1. CP A = 100 kHz ±50% square wave; duty cycle = 50 ±15%; V_{IL} = -0.5 V minimum to +0.7 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.
   CP B = Same as CP A, synchronized with CP A, except 50 kHz ±50% square wave.
2. R1 = 1 kΩ ±5%; R2 = 27Ω ±5%.
3. V_{CC} shall be high enough to insure that 5.0 V minimum is present at V_{CC} device terminal.

FIGURE 3. Burn-in and life test circuit.

Device type 02

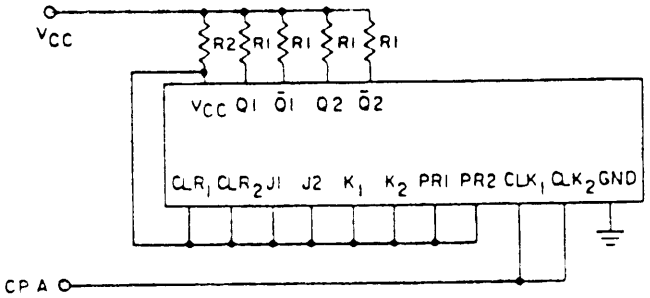


NOTES:
1. High temperature (175°C) condition F burn-in circuit only.
2. R1 = 2 kΩ, R2 = 2Ω
3. V_{CC} shall be chosen such that 5.0 V minimum is present at device terminal.

FIGURE 3. Burn-in and life test circuit.

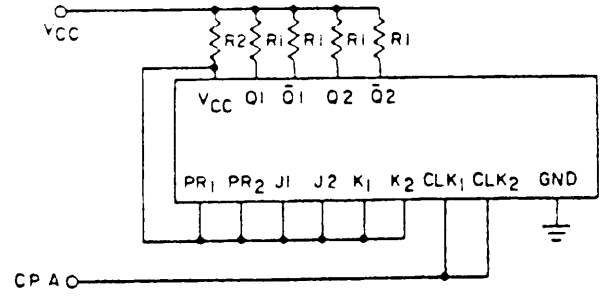
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Device types 03 and 10
Condition D or E



NOTES:
1. CP A = 100 kHz ±50% square wave; duty cycle = 50 ±15%; V_{IL} = -0.5 V minimum to +0.7 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.
2. R1 = 1 kΩ ±5%; R2 = 27Ω ±5%.
3. V_{CC} shall be high enough to insure that 5.0 V minimum is present at V_{CC} device terminal.

Device type 04
Condition D or E

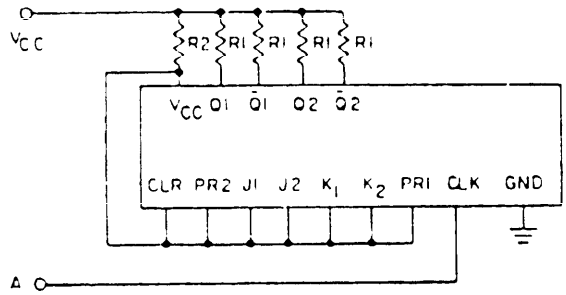


NOTES:
1. CP A = 100 kHz ±50% square wave; duty cycle = 50 ±15%; V_{IL} = -0.5 V minimum to +0.7 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.
2. R1 = 1 kΩ ±5%; R2 = 27Ω ±5%.
3. V_{CC} shall be high enough to insure that 5.0 V minimum is present at V_{CC} device terminal.

FIGURE 3. Burn-in and life test circuit - Continued.

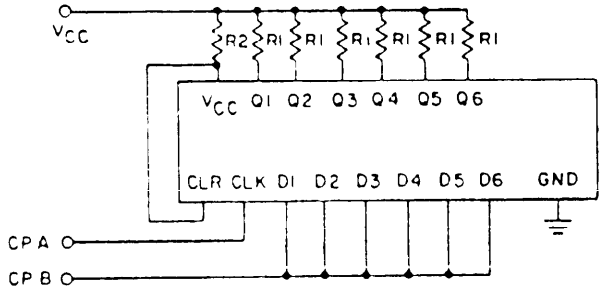
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Device type 05
Condition D or E



NOTES:
1. CP A = 100 kHz ±50% square wave; duty cycle = 50 ±15%; V_{IL} = -0.5 V minimum to +0.7 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.
2. R1 = 1 kΩ ±5%; R2 = 27Ω ±5%.
3. V_{CC} shall be high enough to insure that 5.0 V minimum is present at V_{CC} device terminal.

Device type 06
Condition D or E



NOTES:
1. CP A = 100 kHz ±50% square wave; duty cycle = 50 ±15%; V_{IL} = -0.5 V minimum to +0.7 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.
   CP B = Same as CP A, synchronized with CP A, except 50 kHz ±50% square wave.
2. R1 = 1 kΩ ±5%; R2 = 27Ω ±5%.
3. V_{CC} shall be high enough to insure that 5.0 V minimum is present at V_{CC} device terminal.

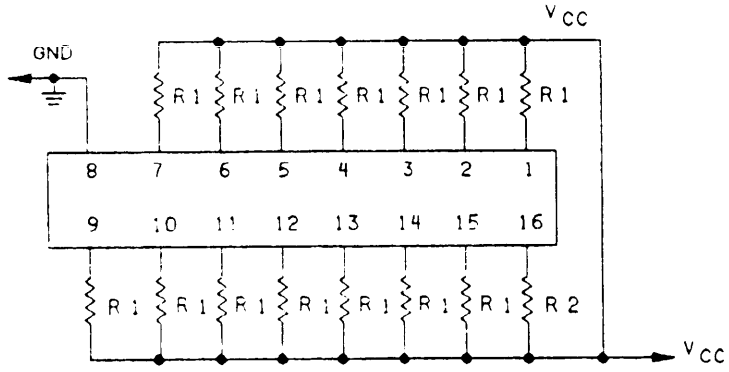
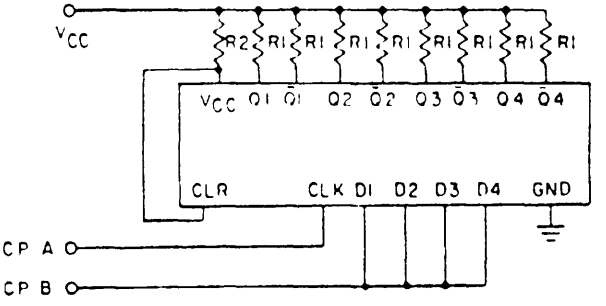


FIGURE 3. Burn-in and life test circuit - Continued.

NOTES:
1. High temperature (175°C) condition F burn-in circuit only.
2. R1 = 2 kΩ, R2 = 2Ω
3. V_{CC} shall be chosen such that 5.0 V minimum is present at device terminal.

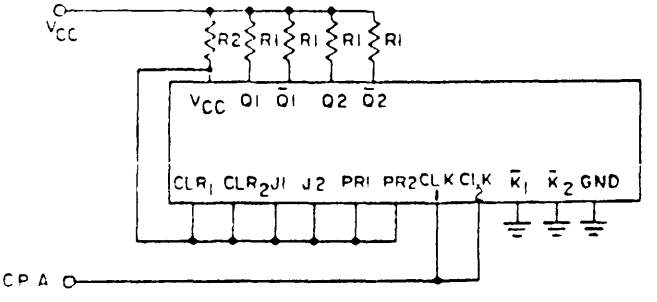
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Device type 07
Condition D or E



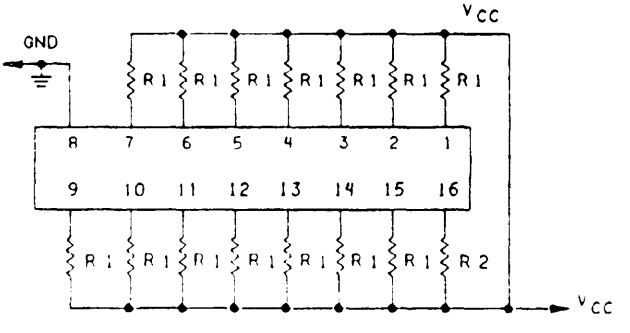
NOTES:
1. CP A = 100 kHz ±50% square wave; duty cycle = 50 ±15%; V_{IL} = -0.5 V minimum to +0.7 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.
   CP B = Same as CP A, synchronized with CP A, except 50 kHz ±50% square wave.
2. R1 = 1 kΩ ±5%; R2 = 27Ω ±5%.
3. V_{CC} shall be high enough to insure that 5.0 V minimum is present at V_{CC} device terminal.

Device type 09
Condition D or E



NOTES:
1. CP A = 100 kHz ±50% square wave; duty cycle = 50 ±15%; V_{IL} = -0.5 V minimum to +0.7 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.
2. R1 = 1 kΩ ±5%; R2 = 27Ω ±5%.
3. V_{CC} shall be high enough to insure that 5.0 V minimum is present at V_{CC} device terminal.

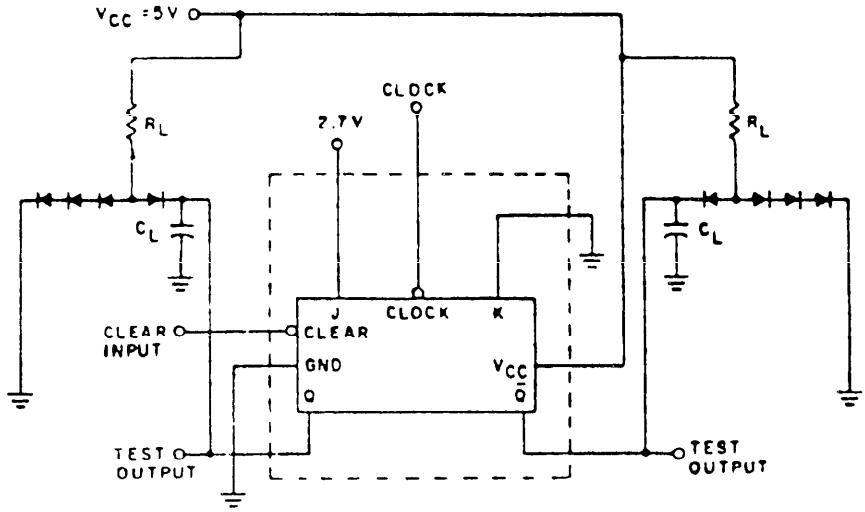
Device types 07 and 09



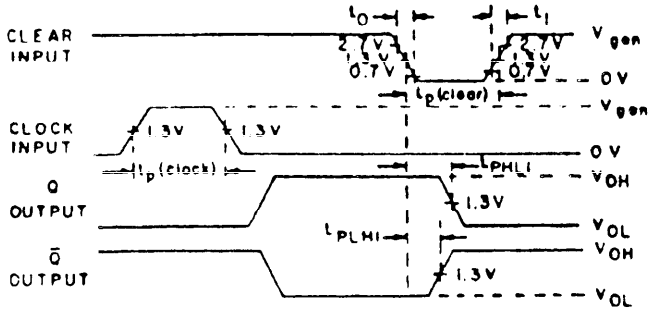
NOTES:
1. High temperature (175°C) condition F burn-in circuit only.
2. R1 = 2 kΩ, R2 = 2Ω
3. V_{CC} shall be chosen such that 5.0 V minimum is present at device terminal.

FIGURE 3. Burn-in and life test circuit - Continued.

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TEST CIRCUIT

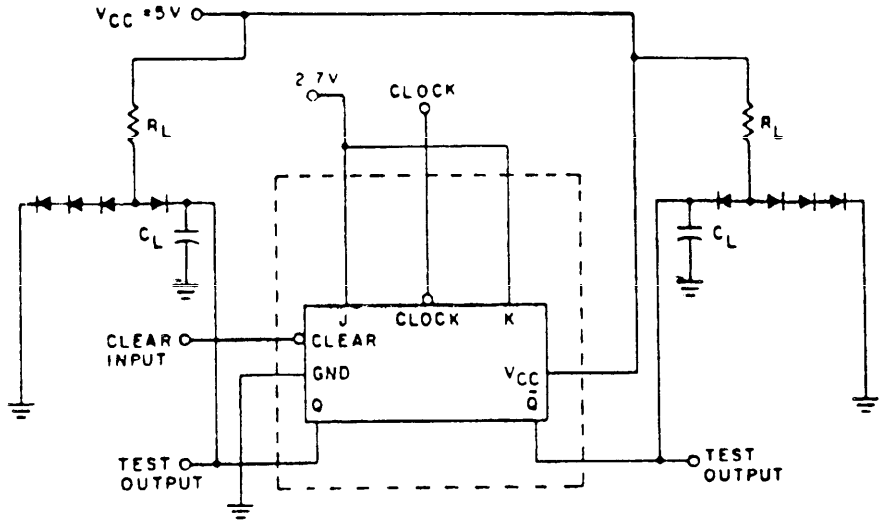


VOLTAGE WAVEFORMS

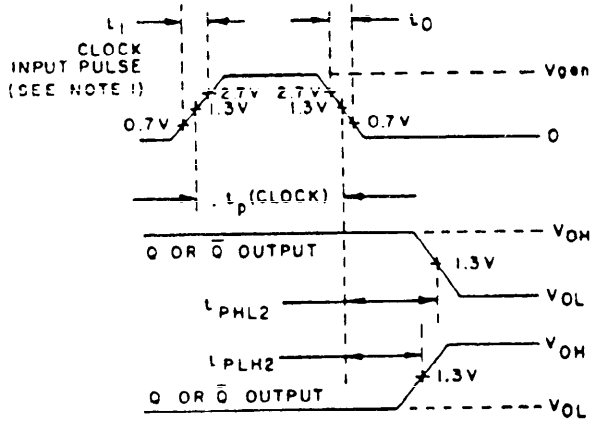
NOTES:
1. Clear inputs dominate regardless of the state of clock or J-K inputs.
2. Clear input pulse characteristics: V_{gen} = 3 V, $t_1 \leq 15$ ns, $t_0 \leq 6$ ns, t_p (clear) = 30 ns, PRR $\leq$ 1 MHz.
3. All diodes are 1N3064, or equivalent.
4. C_L = 50 pF ±10% (including jig and probe capacitance).
5. R_L = 2 kΩ ±5%.
6. Clock input pulse characteristics: V_{gen} = 3 V, t_p (clock) = 25 ns, PRR $\leq$ 1 MHz.

FIGURE 4. Clear switching test circuit and waveforms for device types 01 and 08.

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TEST CIRCUIT

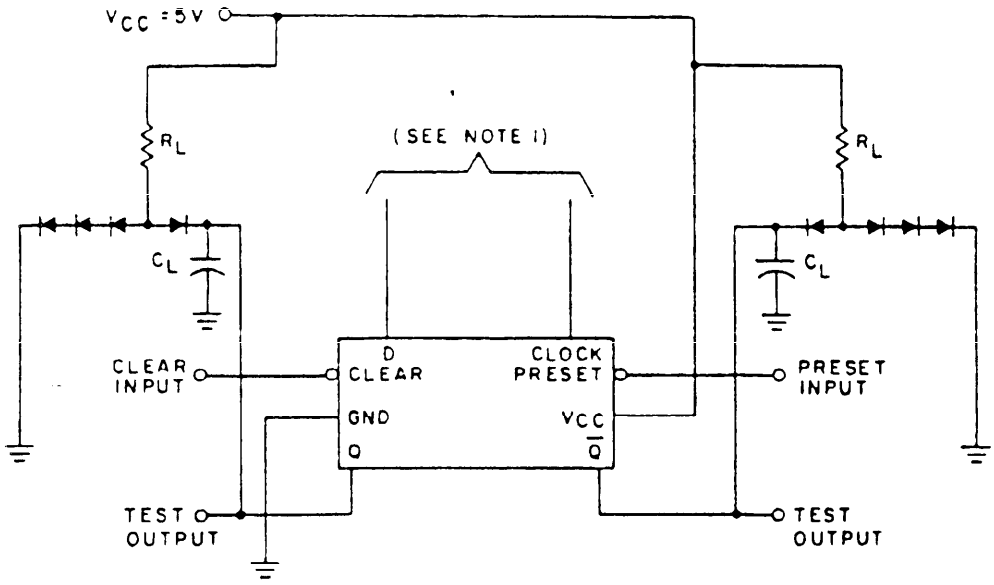


VOLTAGE WAVEFORMS

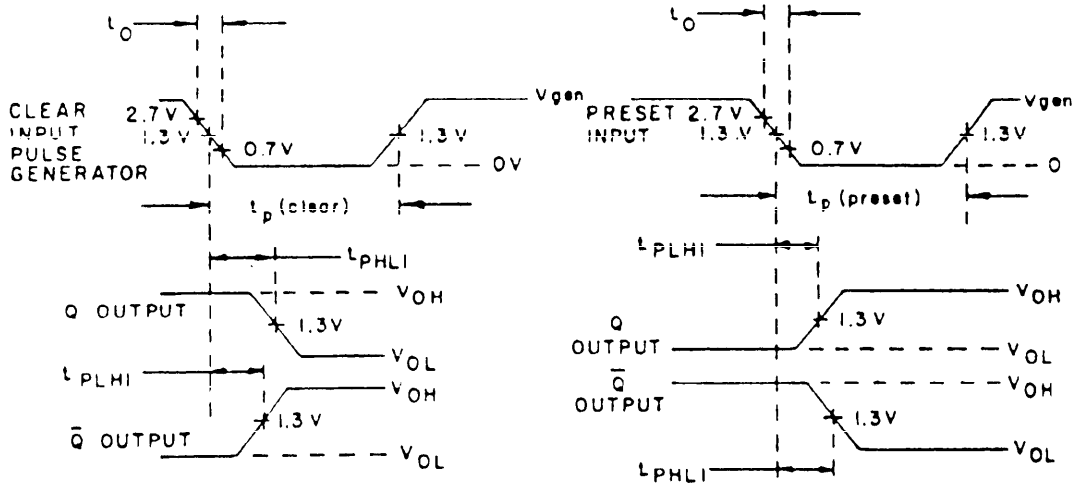
NOTES:
1. Clock input characteristics for t_{PLH}, t_{PHL} (clock to output), V_{gen} = 3 V, $t_1 \leq 15$ ns, $t_0 \leq 6$ ns, t_p (clock) = 25 ns, and PRR $\leq$ 1 MHz.
   When testing f_{MAX} the clock input characteristics are V_{gen} = 3 V, $t_1 = t_0 \leq 6$ ns, t_p (clock) $\leq$25 ns, and PRR = see table III.
2. All diodes are 1N3064, or equivalent.
3. C_L = 50 pF ±10% (including jig and probe capacitance).
4. R_L = 2 kΩ ±5%.

FIGURE 5. Synchronous switching test circuit for device types 01 and 08.

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TEST CIRCUIT

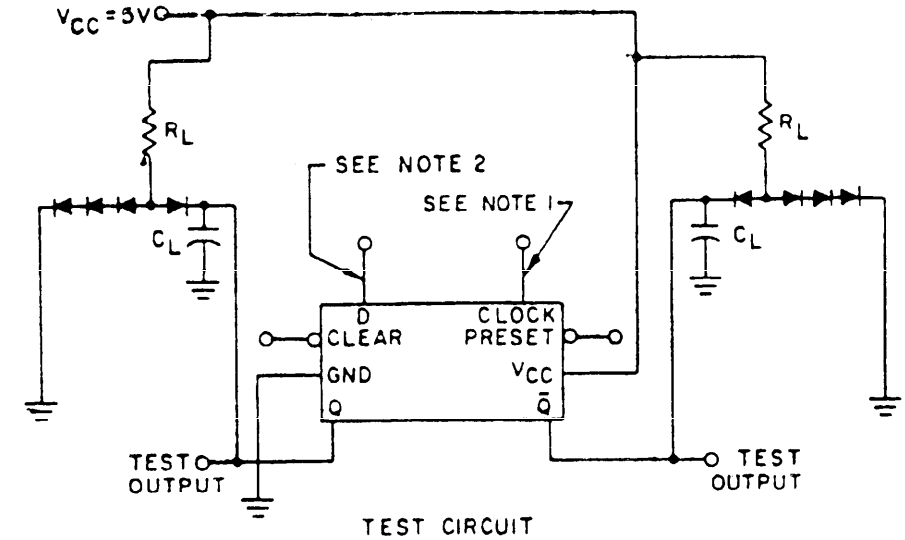


VOLTAGE WAVEFORMS

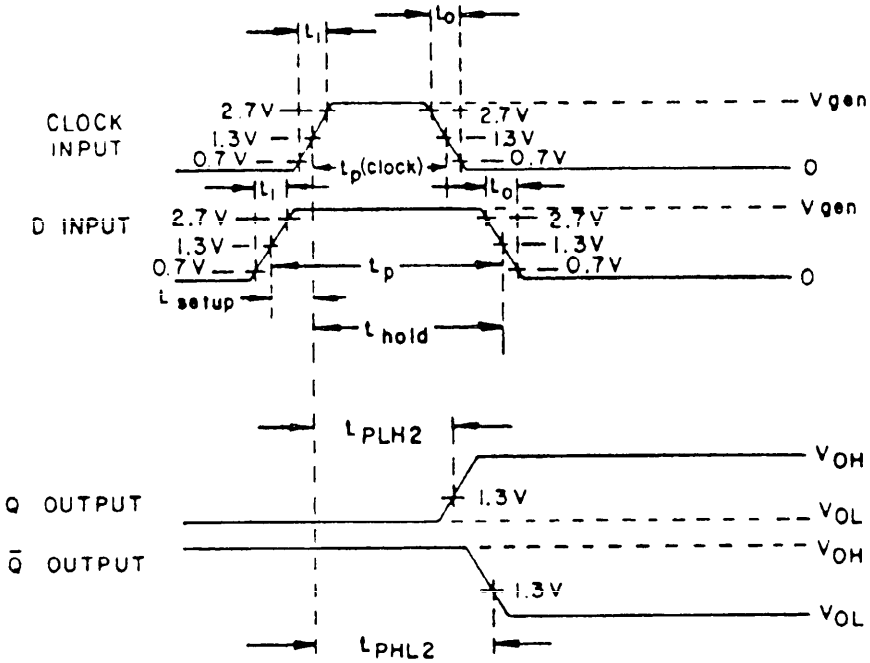
NOTES:
1. Clear and preset inputs dominate regardless of the state of clock or D inputs.
2. All diodes are 1N3064, or equivalent.
3. Clear or preset input pulse characteristics: V_{gen} = 3 V, $t_0 \leq 6$ ns, t_p (clear) = t_p (preset) = 35 ns, and PRR $\leq$ 1 MHz.
4. C_L = 50 pF ±10% (including jig and probe capacitance).
5. R_L = 2 kΩ ±5%.
6. When testing clear to output switching, preset input shall have a logical "1" voltage applied. When testing preset to output switching, clear input shall have a logical "1" voltage applied (see table III).

FIGURE 6. Clear and preset switching test circuit and waveforms for device type 02.

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TEST CIRCUIT

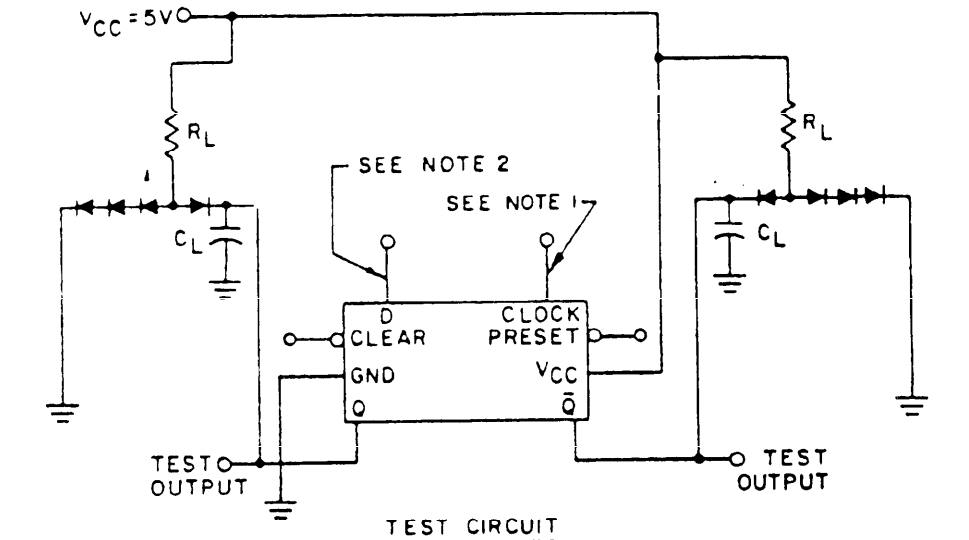


VOLTAGE WAVEFORMS

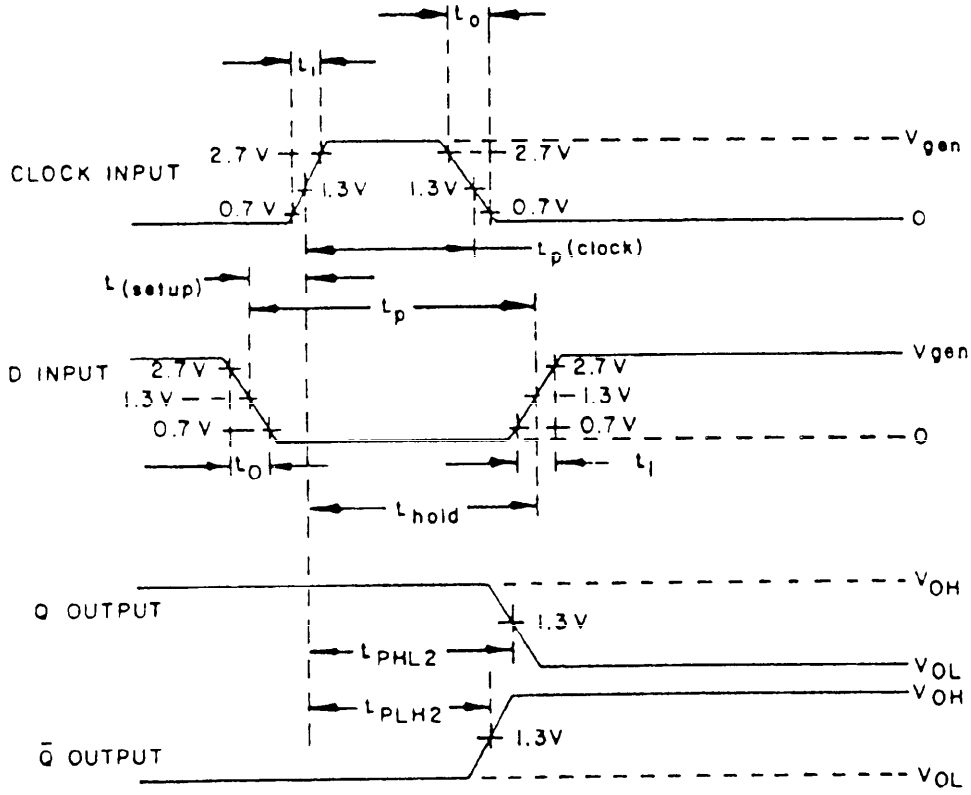
NOTES:
1. Clock input pulse has the following characteristics: V_{gen} = 3 V, $t_1 \leq 15$ ns, $t_0 \leq 6$ ns, t_p (clock) = 30 ns, and PRR $\leq$ 1 MHz. When testing f_{MAX}, PRR = see table III, t_p (clock) $\leq$30 ns, and $t_0 = t_1 \leq 6$ ns.
2. D input has the following characteristics: V_{gen} = 3 V, $t_1 \leq 15$ ns, $t_0 \leq 6$ ns, t_{setup} = 20 ns, t_{hold} = 5 ns, t_p = 25 ns, and PRR is 50% of the clock PRR. For f_{MAX}, $t_0 = t_1 \leq 6$ ns.
3. All diodes are 1N3064, or equivalent.
4. C_L = 50 pF ±10% (including jig and probe capacitance).
5. R_L = 2 kΩ ± 5%.

FIGURE 7. Synchronous switching test circuit (high-level data) for device type 02.

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TEST CIRCUIT

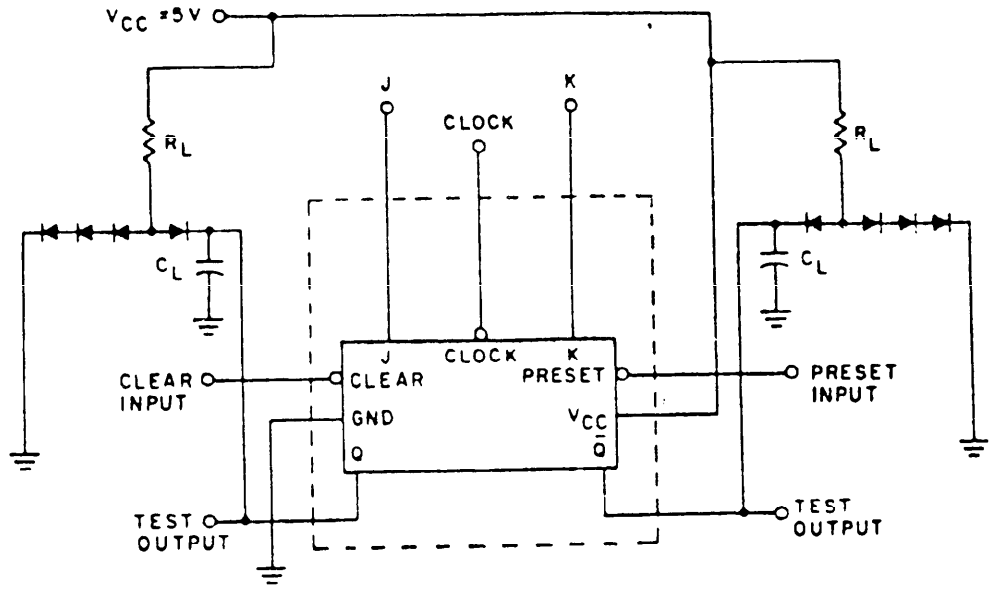


VOLTAGE WAVEFORMS

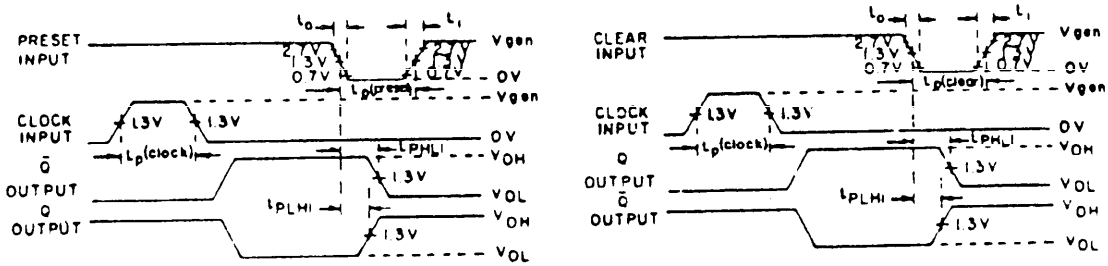
NOTES:
1. Clock input pulse has the following characteristics: V_{gen} = 3 V, $t_1 \leq$ 15 ns, $t_0 \leq$ 6 ns, t_p (clock) = 30 ns, and PRR $\leq$ 1 MHz.
2. D input has the following characteristics: V_{gen} = 3 V, $t_1 \leq$ 15 ns, $t_0 \leq$ 6 ns, t_{setup} = 20 ns, t_{hold} = 5 ns, t_p = 25 ns, and PRR is 50% of the clock PRR.
3. All diodes are 1N3064, or equivalent.
4. C_L = 50 pF ±10% (including jig and probe capacitance).
5. R_L = 2 kΩ ±5%.

FIGURE 8. Synchronous switching test circuit (low-level data) for device type 02.

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TEST CIRCUIT

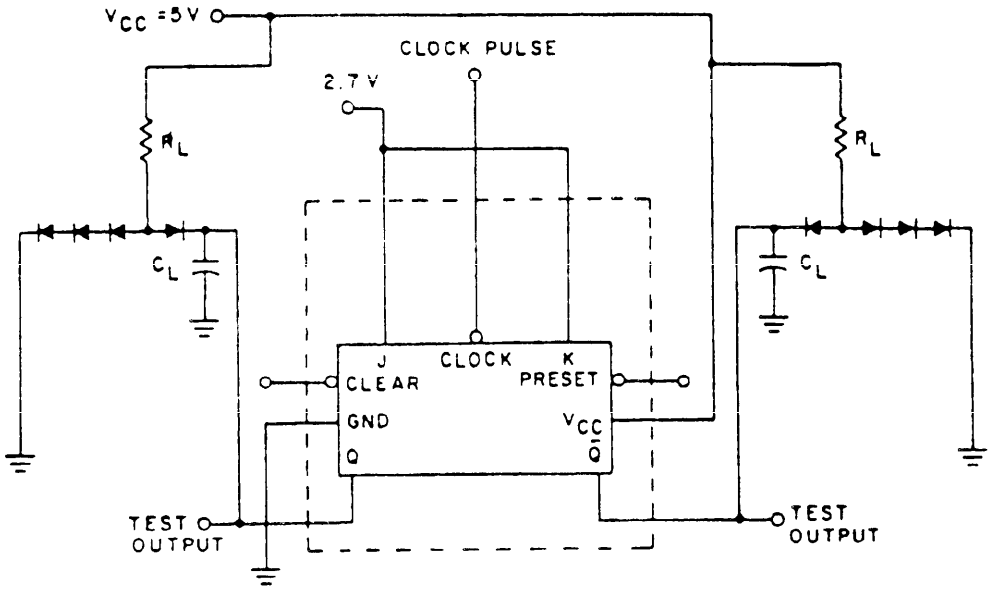


VOLTAGE WAVEFORMS

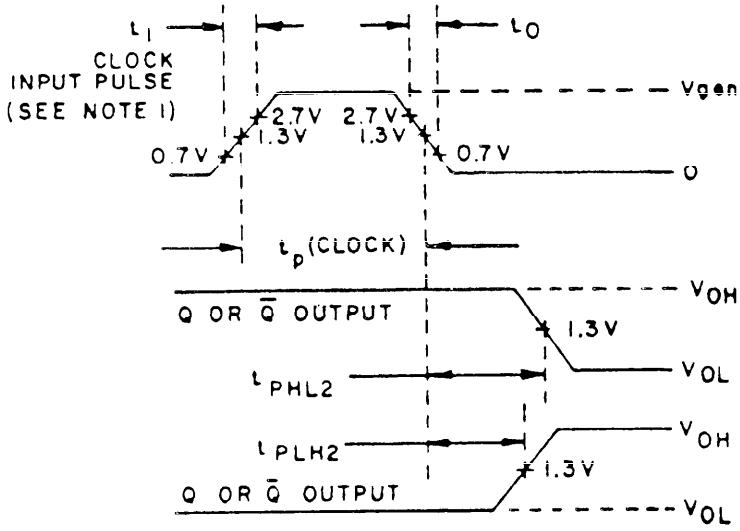
NOTES:
1. Clear or preset inputs dominate regardless of the state of clock or J-K inputs.
2. Clear or preset input pulse characteristics: V_{gen} = 3 V, $t_1 \leq 15$ ns, $t_0 \leq 6$ ns, t_p (clear) = t_p (preset) = 30 ns, PRR ≤ 1 MHz, and Z_{out} ≈ 50Ω.
3. C_L = 50 pF ±10% (including jig and probe capacitance).
4. R_L = 2 kΩ ±5%.
5. All diodes are 1N3064, or equivalent.
6. When testing clear to output switching, preset input shall have a logical "1" voltage applied. When testing preset to output switching, clear input shall have a logical "1" voltage applied. (see table III).
7. Clock input pulse characteristics: t_p (clock) ≥ 25 ns, V_{gen} = 3 V, PRR ≤ 1 MHz.

FIGURE 9. Clear and preset switching test circuit and waveforms for device types 03, 05 and 10.

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TEST CIRCUIT

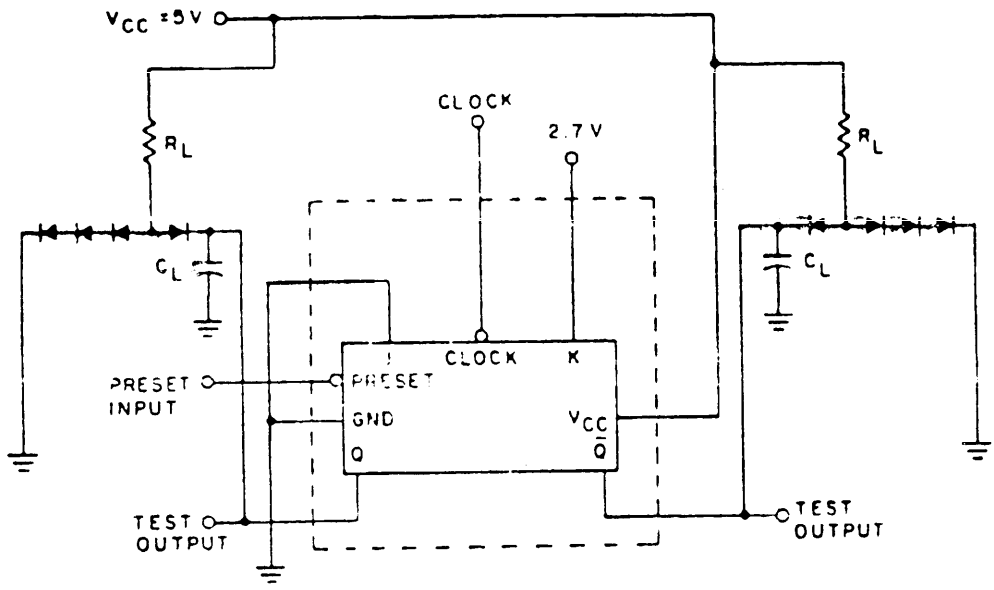


VOLTAGE WAVEFORMS

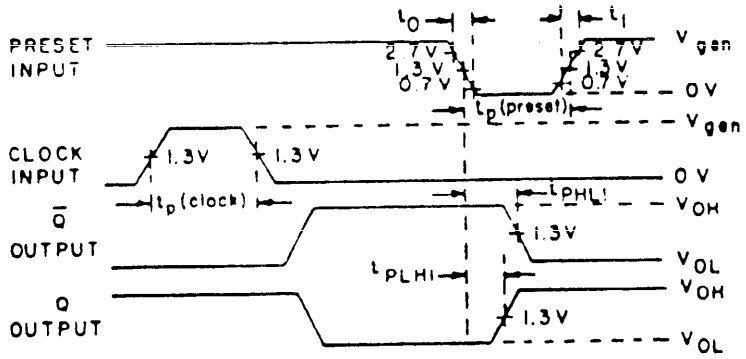
NOTES:
1. Clock input characteristics for t_{PLH}, t_{PHL} (clock to output), V_{gen} = 3 V, $t_0 \leq 6$ ns, $t_1 \leq 15$ ns, t_p (clock) = 25 ns, and PRR $\leq$ 1 MHz.
   When testing f_{MAX} the clock input characteristics are V_{gen} = 3 V, $t_1 = t_0 \leq 6$ ns, t_p (clock) $\leq$25 ns, and PRR = see table III.
2. All diodes are 1N3064, or equivalent.
3. C_L = 50 pF ±10% (including jig and probe capacitance).
4. R_L = 2 kΩ ±5%.

FIGURE 10. Synchronous switching test circuit for device types 03, 05 and 10.

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TEST CIRCUIT

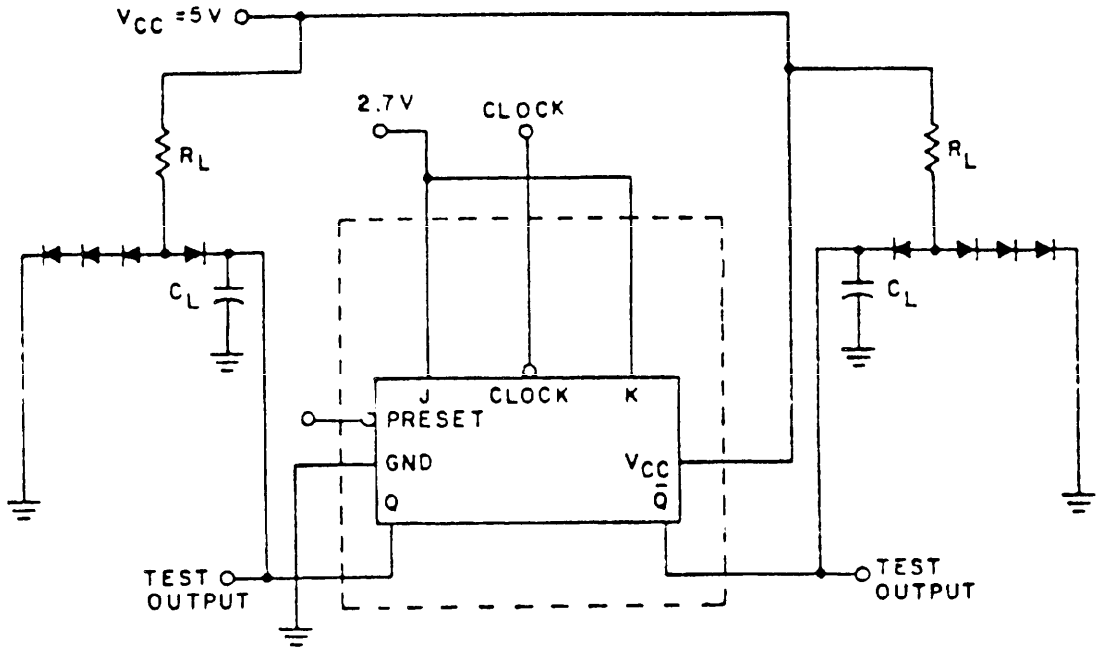


VOLTAGE WAVEFORMS

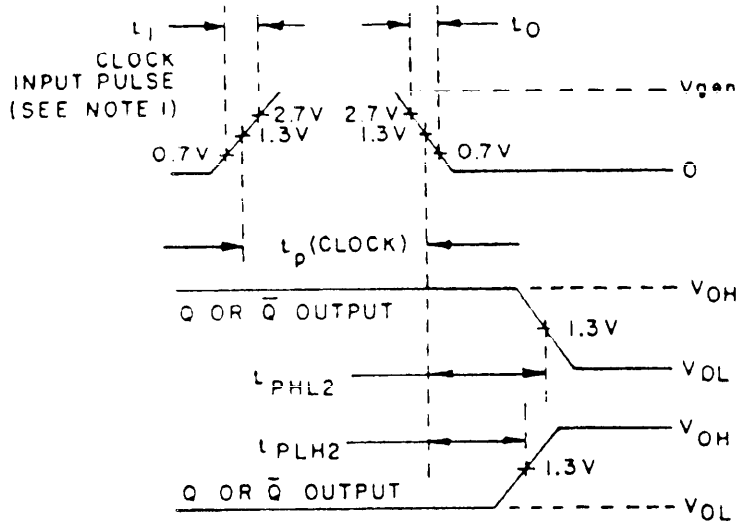
NOTES:
1. Preset inputs dominate regardless of the state of clock or J-K inputs.
2. Preset input pulse characteristics: V_{gen} = 3 V, $t_0 \leq$ 15 ns, $t_1 \leq$ 6 ns, t_p (preset) = 30 ns, PRR $\leq$ 1 MHz.
3. All diodes are 1N3064, or equivalent.
4. C_L = 50 pF ±10% (including jig and probe capacitance).
5. R_L = 2 kΩ ±5%.
5. Clock input pulse characteristics: V_{gen} = 3 V, t_p (clock) $\geq$ 25 ns, PRR $\leq$ 1 MHz.

FIGURE 11. Preset switching test circuit and waveforms for device type 04.

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TEST CIRCUIT

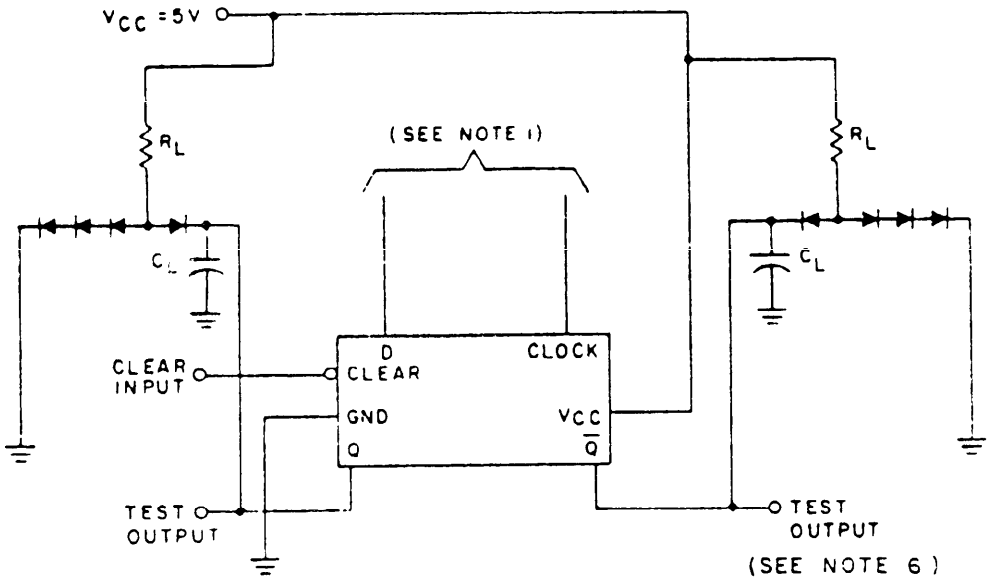


VOLTAGE WAVEFORMS

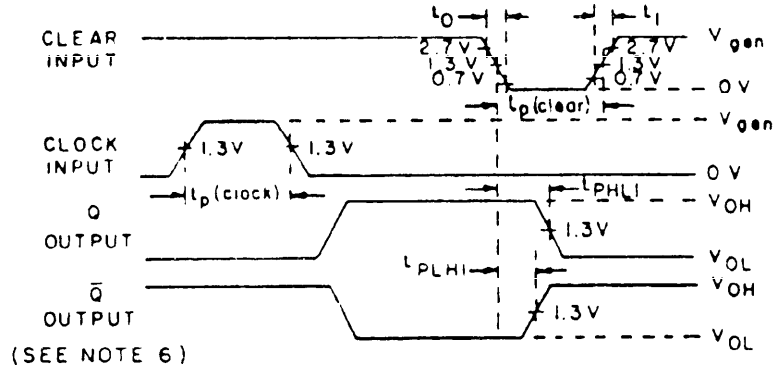
NOTES:
1. Clock input characteristics for t_{PLH}, t_{PHL} (clock to output), V_{gen} = 3 V, $t_1 \leq 15$ ns, $t_0 \leq 6$ ns, t_p (clock) = 25 ns, and PRR $\leq$ 1 MHz.
   When testing f_{MAX} the clock input characteristics are V_{gen} = 3 V, $t_1 = t_0 \leq 6$ ns, t_p (clock) $\leq$25 ns, and PRR = see table III.
2. All diodes are 1N3064, or equivalent.
3. C_L = 50 pF ±10% (including jig and probe capacitance).
4. R_L = 2 kΩ ±5%.

FIGURE 12. Synchronous switching test circuit for device type 04.

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TEST CIRCUIT

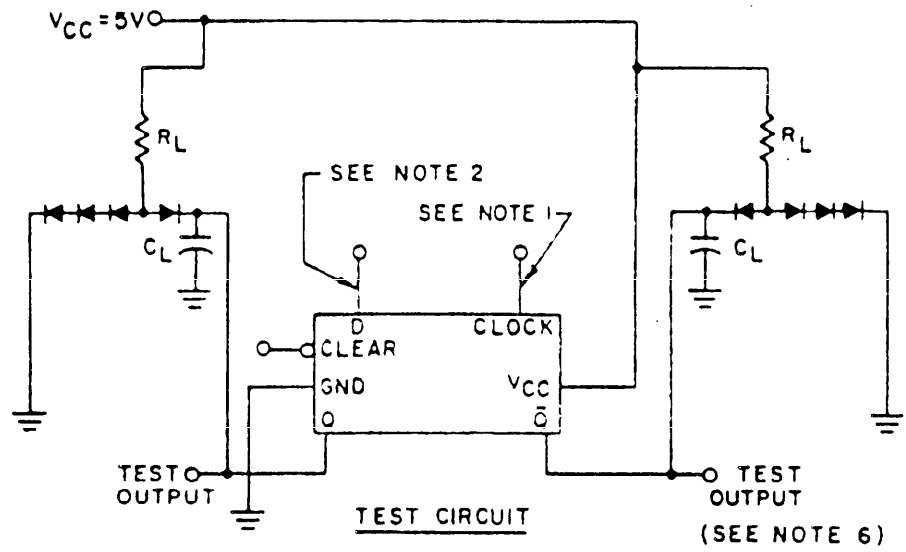


VOLTAGE WAVEFORMS

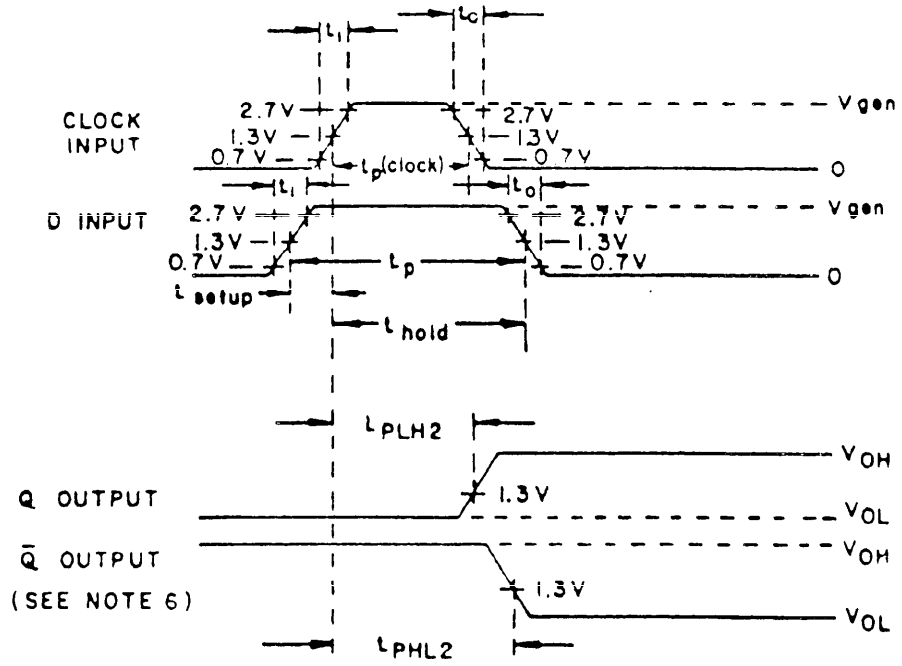
NOTES:
1. Clear input dominates regardless of the state of clock or D inputs.
2. All diodes are 1N3064, or equivalent.
3. Clear input pulse characteristics: V_{gen} = 3 V, $t_0 \leq 6$ ns, $t_1 \leq 15$ ns, t_p (clear) = 35 ns, and PRR $\leq$ 1 MHz.
4. C_L = 50 pF ±10% (including jig and probe capacitance).
5. R_L = 2 kΩ ±5%.
6. $\overline{Q}$ output applies to device type 07 only.
7. Clock input pulse characteristics: t_p (clock) $\geq$ 25 ns, V_{gen} = 3 V; PRR $\leq$ 1 MHz.

FIGURE 13. Asynchronous switching test circuit for device types 06 and 07.

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TEST CIRCUIT



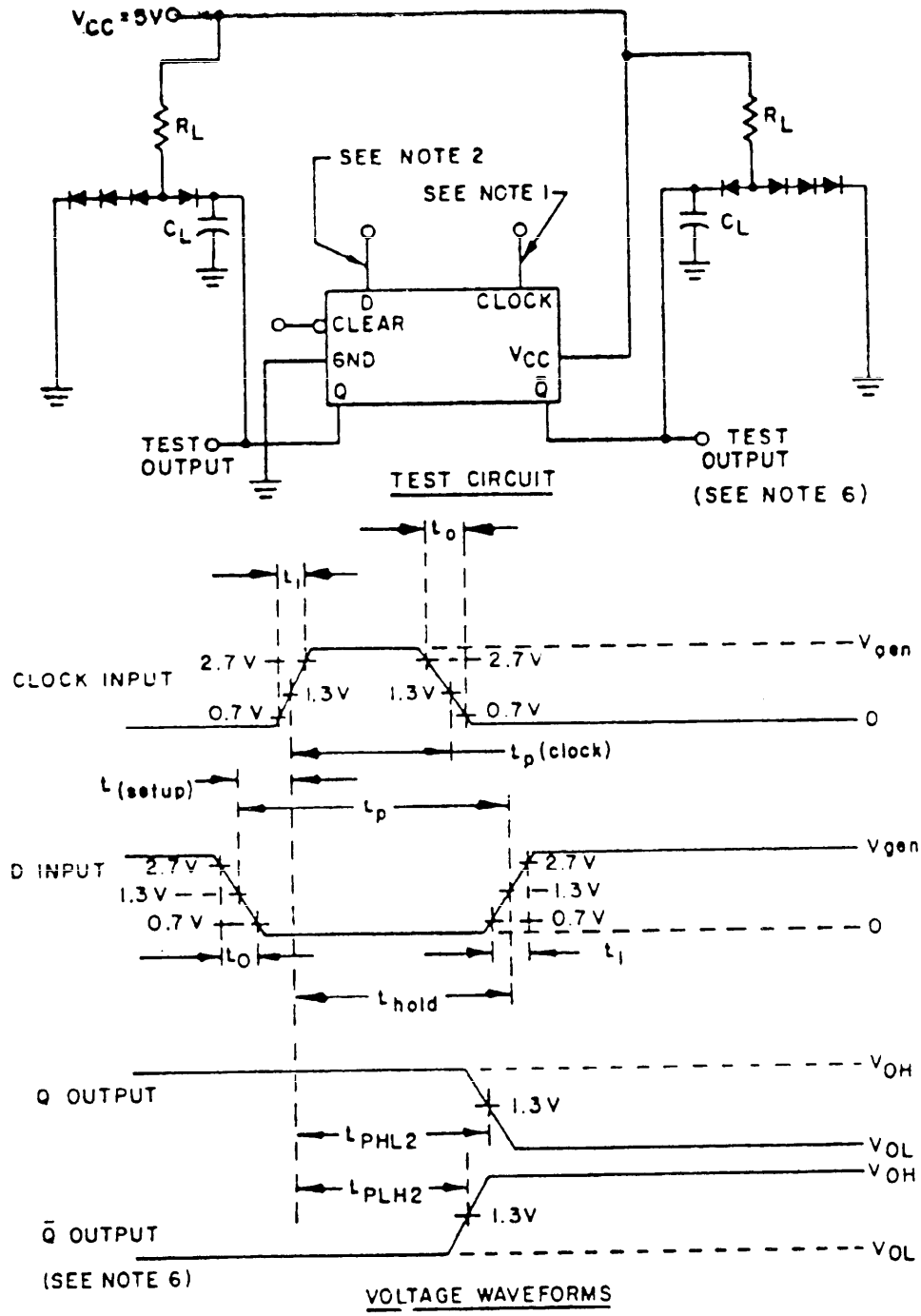
VOLTAGE WAVEFORMS

NOTES:

1. Clock input pulse has the following characteristics: V_{gen} = 3 V, $t_1 \leq 15$ ns, $t_0 \leq 6$ ns, t_p (clock) = 30 ns, and PRR $\leq$ 1 MHz. When testing f_{MAX}, PRR = see table III, t_p (clock) $\leq$30 ns, and $t_0 = t_1 \leq 6$ ns.
2. D input has the following characteristics: V_{gen} = 3 V, $t_1 \leq 15$ ns, $t_0 \leq 6$ ns, t_{setup} = 20 ns, t_{hold} = 5 ns, t_p = 25 ns, and PRR is 50% of the clock PRR. For f_{MAX}, $t_0 = t_1 \leq 6$ ns.
3. All diodes are 1N3064, or equivalent.
4. C_L = 50 pF ±10% (including jig and probe capacitance).
5. R_L = 2 kΩ ±5%.
6. $\bar{Q}$ output applies to device type 07 only.

FIGURE 14. Synchronous switching test circuit (worst-level data) for device types 06 and 07.

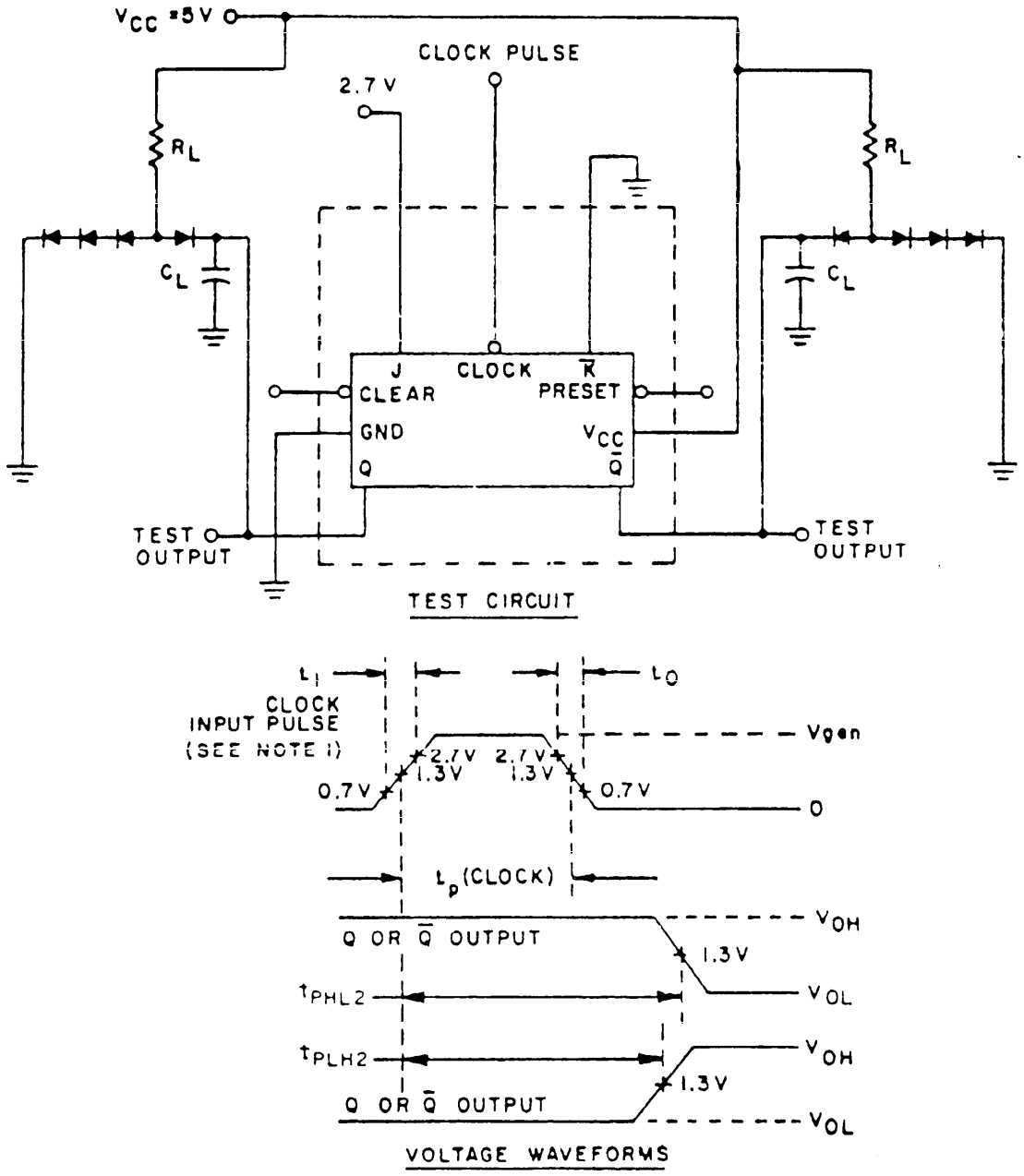
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NOTES:
1. Clock input pulse has the following characteristics: V_{gen} = 3 V, $t_1 \leq 15$ ns, $t_0 \leq 6$ ns, t_p (clock) = 30 ns, and PRR $\leq$ 1 MHz.
2. D input has the following characteristics: V_{gen} = 3 V, $t_1 \leq 15$ ns, $t_0 \leq 6$ ns, t_{setup} = 20 ns, t_{hold} = 5 ns, t_p = 25 ns, and PRR is 50% of the clock PRR.
3. All diodes are 1N3064, or equivalent.
4. C_L = 50 pF ±10% (including jig and probe capacitance).
5. R_L = 2 kΩ ±5%.
6. $\bar{Q}$ output applies to device type 07 only.

FIGURE 15. Synchronous switching test circuit (low-level data) for device types 06 and 07.

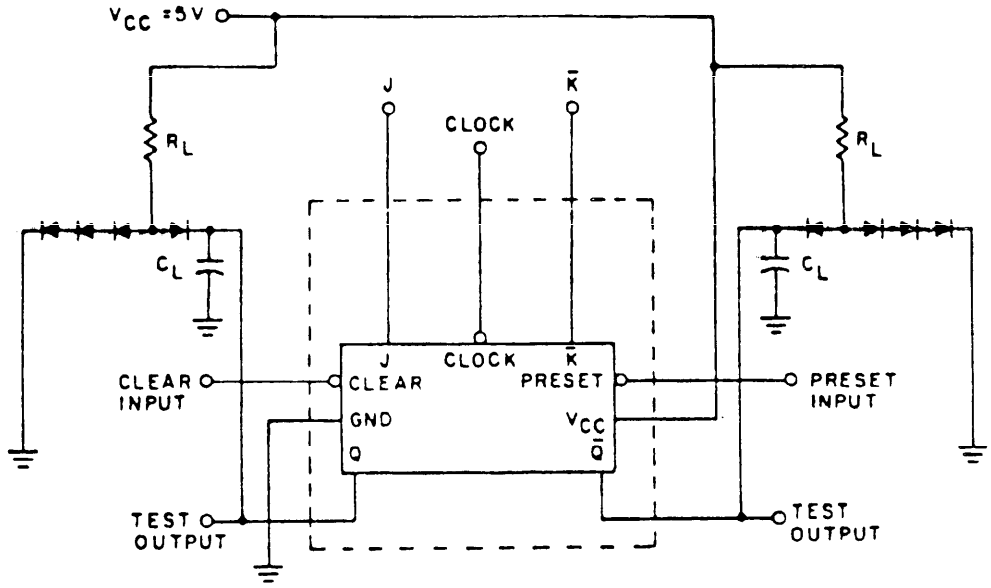
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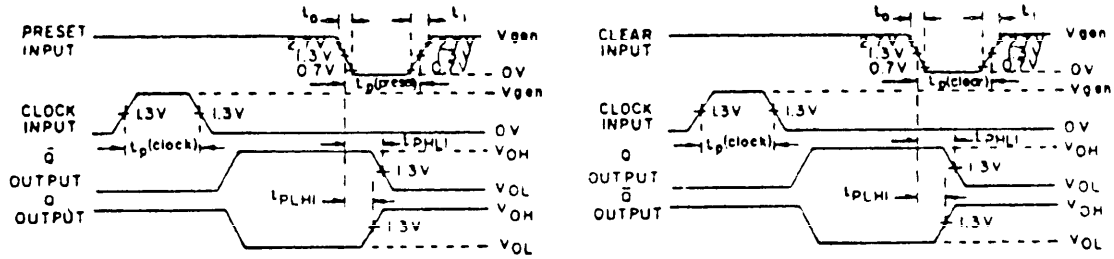
NOTES:
1. Clock input characteristics for t_{PLH}, t_{PHL} (clock to output), V_{gen} = 3 V, $t_0 \leq 6$ ns, $t_1 \leq 15$ ns, t_p (clock) = 25 ns, and PRR $\leq$ 1 MHz. When testing f_{MAX} the clock input characteristics are V_{gen} = 3 V, $t_1 = t_0 \leq 6$ ns, t_p (clock) $\leq$ 25 ns, and PRR = see table III.
2. All diodes are 1N3064, or equivalent.
3. C_L = 50 pF ±10% (including jig and probe capacitance).
4. R_L = 2 kΩ ±5%.

FIGURE 16. Synchronous switching test circuit for device type 09.

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TEST CIRCUIT



VOLTAGE WAVEFORMS

NOTES:
1. Clear or preset inputs dominate regardless of the state of clock or J-$\bar{K}$ inputs.
2. Clear or preset input pulse characteristics: V_{gen} = 3 V, $t_1 \leq 15$ ns, $t_0 \leq 6$ ns, t_p (clear) = t_p (preset) = 30 ns, PRR $\leq$ 1 MHz, and $Z_{out} \approx 50\Omega$.
3. C_L = 50 pF ±10% (including jig and probe capacitance).
4. R_L = 2 kΩ ±5%.
5. All diodes are 1N3064, or equivalent.
6. When testing clear to output switching, preset input shall have a logical "1" voltage applied. When testing preset to output switching, clear input shall have a logical "1" voltage applied (see table III).
7. Clock input pulse characteristics: t_p (clock) $\geq$ 25 ns, V_{gen} = 3 V, PRR $\leq$ 1 MHz.

FIGURE 17. Clear and preset switching test circuit and waveforms for device type 03.

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TABLE III. Group A inspection for device types 01 and 08.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 1/ 2,X / Cases A,B,C,D Test no.	* 2 / ** 18 / * 1 / ** 12 CLK1	3 / 19 / 2 / 13 CLR1	4 / 6 / 3 / 4 K1	6 / 20 / 4 / 14 V_{CC}	8 / 13 / 5 / 9 CLK2	9 / 14 / 6 / 10 CLR2	10 / 12 / 7 / 8 J2	12 / 9 / 8 / 6 $\overline{Q}2$	13 / 8 / 9 / 5 Q2	14 / 16 / 10 / 11 K2	16 / 10 / 11 / 7 GND	18 / 4 / 12 / 3 Q1	19 / 3 / 13 / 2 $\overline{Q}1$	20 / 2 / 14 / 1 J1	Measured terminal	Test limits Min	Test limits Max	Unit
1 T_C = +25°C	V_{OH}	3006	1	2.0 V	0.7 V	2.0 V	4.5 V							GND		-.4 mA	2.0 V	$\overline{Q}1$	2.5		V
	"	"	2	2/	2.0 V	2.0 V	"							"		-.4 mA	0.7 V	$\overline{Q}1$	"		"
	"	"	3	2/	2.0 V	0.7 V	"							"	-.4 mA		2.0 V	Q1	"		"
	"	"	4				"	2.0 V	0.7 V	2.0 V	-.4 mA		2.0 V	"				$\overline{Q}2$	"		"
	"	"	5				"	2/	2.0 V	0.7 V	-.4 mA		2.0 V	"				$\overline{Q}2$	"		"
	"	"	6				"	"	"	2.0 V		-.4 mA	0.7 V	"				Q2	"		"
	V_{OL}	3007	7				"	"	"	2.0 V	4 mA		0.7 V	"				$\overline{Q}2$		0.4	"
	"	"	8				"	"	"	0.7 V		4 mA	2.0 V	"				Q2		"	"
	"	"	9				"	2.0 V	0.7 V	2.0 V		4 mA	2.0 V	"				Q2		"	"
	"	"	10	2.0 V	0.7 V	2.0 V	"							"	4 mA		2.0 V	Q1		"	"
	"	"	11	2/	2.0 V	2.0 V	"							"	4 mA		0.7 V	Q1		"	"
	"	"	12	2/	2.0 V	0.7 V	"							"		4 mA	2.0 V	$\overline{Q}1$		"	"
	V_{IC}		13	-18 mA			"							"				CLK1		-1.5	"
	"		14		-18 mA		"							"				CLR1		"	"
	"		15			-18 mA	"							"				K1		"	"
	"		16				"	-18 mA						"				CLK2		"	"
	"		17				"		-18 mA					"				CLR2		"	"
	"		18				"			-18 mA				"				J2		"	"
	"		19				"						-18 mA	"				K2		"	"
	"		20				"							"			-18 mA	J1		"	"
	I_{IL1}	3009	21	3/	4.5 V	0.4 V	5.5 V							"			4.5 V	K1	4/	4/	mA
	"	"	22	4.5 V	3/	4.5 V	"							"			0.4 V	J1	"	"	"
	"	"	23				"	4.5 V	3/	0.4 V			4.5 V	"				J2	"	"	"
	"	"	24				"	3/	4.5 V	4.5 V			0.4 V	"				K2	"	"	"
	I_{IL3}	"	25	0.4 V	3/	4.5 V	"							"			4.5 V	CLK1	"	"	"
	"	"	26				"	0.4 V	3/	4.5 V			4.5 V	"				CLK2	"	"	"
	I_{IL4}	"	27	4.5 V	0.4 V	4.5 V	"							"			4.5 V	CLR1	"	"	"
	"	"	28				"	4.5 V	0.4 V	4.5 V			4.5 V	"				CLR2	"	"	"
	I_{IH1}	3010	29	GND	GND	2.7 V	"							"			4.5 V	K1		20	µA
	"	"	30	GND	GND	4.5 V	"							"			2.7 V	J1		"	"
	"	"	31				"	GND	GND	2.7 V			4.5 V	"				J2		"	"
	"	"	32				"	GND	GND	4.5 V			2.7 V	"				K2		"	"
	I_{IH2}	"	33				"	GND	GND	4.5 V			5.5 V	"				K2		100	"
	"	"	34				"	GND	GND	5.5 V			4.5 V	"				J2		"	"
	"	"	35	GND	GND	5.5 V	"							"			4.5 V	K1		"	"
	"	"	36	GND	GND	4.5 V	"							"			5.5 V	J1		"	"
	I_{IH5}	"	37	GND	2.7 V	4.5 V	"							"			GND	CLR1		60	"
	"	"	38				"	GND	2.7 V	GND			4.5 V	"				CLR2		60	"
	I_{IH6}	"	39				"	GND	5.5 V	GND			4.5 V	"				CLR2		300	"
	"	"	40	GND	5.5 V	4.5 V	"							"			GND	CLR1		300	"

See footnotes at end of device types 01 and 08

MIL-M-38510/30102

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TABLE III. Group A inspection for device types 01 and 08 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 1/ 2,X	* 2	3	4	5	8	9	10	12	13	14	16	18	19	20	Measured terminal	Test limits		Unit
				** 18	19	6	20	13	14	12	9	8	16	10	4	3	2				
			Cases A,B,C,D	* 1	2	3	4	5	6	7	8	9	10	11	12	13	14				
			Test no.	** 12	13	4	14	9	10	8	6	5	11	7	3	2	1		Min	Max	
				CLK1	CLR1	K1	V_{CC}	CLK2	CLR2	J2	Q2	$\overline{Q2}$	K2	GND	Q1	$\overline{Q1}$	J1				
1 T_C = +25°C	I_{IH7}	3010	41	2.7 V	GND	GND	5.5 V							GND			GND	CLK1		80	µA
		"	42				"	2.7 V	GND	GND			GND	"				CLK2		80	"
	I_{IH8}	"	43				"							"				CLK2		400	"
		"	44	5.5 V	GND	GND	"	5.5 V	GND	GND			GND	"			GND	CLK1		400	"
	I_{OS}	3011	45	GND	GND	4.5 V	"							"		GND	GND	$\overline{Q1}$	-15	-100	mA
		"	46	2/	4.5 V	GND	"							"	2.25 V	GND	4.5 V	Q1	5/	5/	"
		"	47				"	GND	GND	GND	GND		4.5 V	"				Q2	-15	-100	"
		"	48				"	2/	4.5 V	4.5 V	GND	2.25 V	GND	"				$\overline{Q2}$	5/	5/	"
	I_{CC}	3005	49	GND	GND	GND	"	GND	GND	GND			GND	"			GND	V_{CC}		8.0	"
	I_{CC}	3005	50	2/	5.5 V	GND	"	2/	5.5 V	5.5 V			GND	"			5.5 V	V_{CC}		8.0	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} test are omitted.																				
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																				
7 6///	Truth table tests	3014	51	B	B	B	4.5 V	B	B	B	H	L	B	GND	L	H	A	All outputs	See 8/		
		"	52	A	"	"	"	"	"	"	"	"	"	"	"	"	"				
		"	53	B	"	"	"	"	"	"	"	"	"	"	"	"	"				
		"	54	B	A	"	"	"	"	"	"	"	"	"	"	"	"				
		"	55	A	"	"	"	"	"	"	"	"	"	"	"	"	"				
		"	56	B	"	"	"	"	"	"	"	"	"	"	H	L	B				
		"	57	B	"	A	"	"	"	"	"	"	"	"	"	"	"				
		"	58	A	"	"	"	"	"	"	"	"	"	"	L	H	"				
		"	59	B	"	"	"	"	"	"	"	"	"	"	"	"	"				
		"	60	"	B	B	"	"	A	A	"	"	"	"	"	"	"				
		"	61	"	"	"	"	A	"	"	"	"	"	"	"	"	"				
		"	62	"	"	"	"	"	"	"	"	"	"	"	"	"	"				
		"	63	"	"	"	"	B	"	"	L	H	"	"	"	"	"				
		"	64	"	"	"	"	B	"	B	"	"	A	"	"	"	"				
		"	65	"	"	"	"	A	"	"	"	"	"	"	"	"	"				
		"	66	"	"	"	"	B	"	"	H	L	"	"	"	"	"				
		"	67	"	"	"	"	"	B	"	"	"	B	"	"	"	"				
		"	68	"	A	"	"	"	A	"	"	"	"	"	"	"	"				
		"	69	A	"	"	"	A	"	"	"	"	"	"	"	"	"				
		"	70	B	"	"	"	B	"	"	"	"	"	"	"	"	"				
		"	71	B	"	A	"	B	"	A	"	"	A	"	"	"	A				
		"	72 9/	A	"	"	"	A	"	"	"	"	"	"	"	"	"				
		"	73	B	"	"	"	B	"	"	L	H	"	"	H	L	B				
		"	74	B	"	B	"	B	"	B	"	"	B	"	"	"	"				
		"	75	A	"	"	"	A	"	"	"	"	"	"	"	"	"				
		"	76	B	"	"	"	B	"	"	"	"	"	"	"	"	"				
		"	77	B	"	A	"	B	"	A	"	"	A	"	"	"	A				
		"	78 9/	A	"	"	"	A	"	"	"	"	"	"	"	"	"				
		"	79	B	"	"	"	B	"	"	H	L	"	"	L	H	"				
8 6///	Repeat subgroup 7 at T_C = +125°C and T_C = -55°C.																				

See footnotes at end of device types 01 and 08.

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TABLE III. Group A inspection for device types 01 and 08 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 1/ 2,X	* 2	3	4	6	8	9	10	12	13	14	16	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
				** 18	19	6	20	13	14	12	9	8	16	10	4	3	2				
			Cases A,B,C,D	* 1	2	3	4	5	6	7	8	9	10	11	12	13	14				
				** 17	13	4	14	9	10	8	6	5	11	7	3	2	1				
			Test no.	CLK1	CLR1	K1	V_{CC}	CLK2	CLR2	J2	$\overline{Q2}$	Q2	K2	GND	Q1	$\overline{Q1}$	J1				
9 T_C = +25°C	f_{MAX}	Fig. 4	80	IN	2.7 V	2.7 V	5.0 V							GND	OUT		2.7 V	Q1	25		MHz
	10/	"	81	IN	2.7 V	2.7 V	"							"		OUT	2.7 V	$\overline{Q1}$	"		"
		"	82				"	IN	2.7 V	2.7 V	OUT		2.7 V	"				$\overline{Q2}$	"		"
		"	83				"	IN	2.7 V	2.7 V		OUT	2.7 V	"				Q2	"		"
	t_{PLH1}	3003	84				"	IN	IN	2.7 V	OUT		GND	"				CLR2 to $\overline{Q2}$	5	21	ns
		Fig. 5	85	IN	IN	GND	"							"		OUT	2.7 V	CLR1 to $\overline{Q1}$	"	21	"
	t_{PHL1}	"	86	IN	IN	GND	"							"	OUT		2.7 V	CLR1 to Q1	"	28	"
		"	87				"	IN	IN	2.7 V		OUT	GND	"				CLR2 to Q2	"	28	"
	t_{PLH2}	3003	88				"	IN	2.7 V	2.7 V		OUT	2.7 V	"				CLK2 to Q2	"	22	"
		Fig. 4	89				"	IN	2.7 V	2.7 V	OUT		2.7 V	"				CLK2 to $\overline{Q2}$	"	"	"
		"	90	IN	2.7 V	2.7 V	"							"	OUT		2.7 V	CLK1 to Q1	"	"	"
		"	91	"	"	"	"							"		OUT	2.7 V	CLK1 to $\overline{Q1}$	"	"	"
	t_{PHL2}	"	92	"	"	"	"							"		OUT	2.7 V	CLK1 to $\overline{Q1}$	"	30	"
		"	93	"	"	"	"							"	OUT		2.7 V	CLK1 to Q1	"	"	"
		"	94				"	IN	2.7 V	2.7 V	OUT		2.7 V	"				CLK2 to $\overline{Q2}$	"	"	"
		"	95				"	IN	2.7 V	2.7 V		OUT	2.7 V	"				CLK2 to Q2	"	"	"
10 T_C = +125°C	f_{MAX} 10/	Fig. 4	96 99																25		MHz
	t_{PLH1}	3003 Fig.5	100-101																5	32	ns
	t_{PHL1}	3003 Fig. 5	102 103	Same tests and terminal conditions as for subgroup 9, except T_C = +125°C.															"	40	"
	t_{PLH2}	3003 Fig. 4	104-107																"	32	"
	t_{PHL2}	3003 Fig. 4	108-111																"	42	"
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																				

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* Terminal numbers for device type 01.

** Terminal numbers for device type 08.

1/ Case 2 pins not referenced are NC.

2/ 2.5 V minimum/5.5 V maximum.
0 V

3/ 2.5 V minimum/5.5 V maximum.
0 V

4/ I_{IL} limits in mA are as follows:

	Min/Max limits for CKT					
	A	B	C	D	E	
I_{IL1}	-.075/ -.250	-.03/ -.30	-.11/ -.25	-.12/ -.36	-.12/ -.36	
I_{IL3}	-.15/ -.60	-.06/ -.60	-.15/ -.56	-.29/ -.72	-.24/ -.72	
I_{IL4}	-.15/ -.70	-.06/ -.70	-.29/ -.65	-.20/ -.80	-.12/ -.72	

5/ I_{OS} limits are as follows:

Test		
45 and 49	CKT's A, B, C	-.5/ -50
	CKT D	-.15/ -100

6/ Input voltages shown are A = 2.0 volts minimum and B = 0.7 volts maximum.

7/ Tests shall be performed in sequence, attributes data only.

8/ Output voltages shall be H ≥ 1.5 V and L ≤ 1.5 V.

9/ These tests may be performed as shown in table III or alternately as follows:

Test no.	CLK1	CLR1	K1	V_{CC}	CLK2	CLR2	J2	$\overline{Q2}$	Q2	K2	GND	Q1	$\overline{Q1}$	J1
72A	A	A	A	4.5 V	B	A	A	H	L	A	GND	L	H	A
72B	B	"	"	"	B	"	"	H	L	"	"	H	L	"
72C	B	"	"	"	A	"	"	H	L	"	"	H	L	"
78A	A	"	"	"	B	"	"	L	H	"	"	H	L	"
78B	B	"	"	"	B	"	"	L	H	"	"	L	H	"
78C	B	"	"	"	A	"	"	L	H	"	"	L	H	"

10/ f_{MAX} minimum limit specified is the frequency of the input pulse. The output frequency shall be one half of the input frequency.

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TABLE III Group A inspection for device type 02.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 1/ 2,X Cases A, B, C, D Test no.	2 1 CLR1	3 2 D1	4 3 CLK1	6 4 PR1	8 5 Q1	9 6 $\overline{Q1}$	10 7 GND	12 8 $\overline{Q2}$	13 9 Q2	14 10 PR2	16 11 CLK2	18 12 D2	19 13 CLR2	20 14 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
1 T_C = +25°C	V_{OH}	3006	1	0.7 V	2.0 V	GND	2.0 V	-.4 mA	-.4 mA	GND							4.5 V	$\overline{Q1}$	2.5		V
	"	"	2	2.0 V	"	GND	0.7 V	"	"	"							"	Q1	"		"
	"	"	3	"	"	2/	2.0 V	"	"	"							"	Q1	"		"
	"	"	4	"	0.7 V	2/	2.0 V	"	-.4 mA	"							"	$\overline{Q1}$	"		"
	"	"	5							"		-.4 mA	0.7 V	GND	2.0 V	2.0 V	"	Q2	"		"
	"	"	6							"	-.4 mA		2.0 V	GND	2.0 V	0.7 V	"	$\overline{Q2}$	"		"
	"	"	7							"	-.4 mA		"	2/	0.7 V	2.0 V	"	$\overline{Q2}$	"		"
	"	"	8							"		-.4 mA	"	2/	2.0 V	2.0 V	"	Q2	"		"
	V_{OL}	3007	9	2.0 V	0.7 V	2/	2.0 V	4 mA		"							"	$\overline{Q1}$		0.4	"
	"	"	10	"	2.0 V	2/	2.0 V		4 mA	"							"	Q1		"	"
	"	"	11	"	"	GND	0.7 V		4 mA	"							"	$\overline{Q1}$		"	"
	"	"	12	0.7 V	"	GND	2.0 V	4 mA		"							"	Q1		"	"
	"	"	13							"		4 mA	2.0 V	2/	0.7 V	2.0 V	"	$\overline{Q2}$		"	"
	"	"	14							"	4 mA		2.0 V	2/	2.0 V	"	"	Q2		"	"
	"	"	15							"	4 mA		0.7 V	GND	"	"	"	$\overline{Q2}$		"	"
	"	"	16							"		4 mA	2.0 V	GND	"	0.7 V	"	Q2		"	"
	V_{IC}		17	-18 mA						"							"	CLR1		-1.5	"
			18		-18 mA					"							"	D1		"	"
			19			-18 mA				"							"	CLK1		"	"
			20				-18 mA			"							"	PR1		"	"
			21							"			-18 mA				"	PR2		"	"
			22							"				-18 mA			"	CLK2		"	"
			23							"					-18 mA		"	D2		"	"
			24							"						-18 mA	"	CLR2		"	"
	I_{IL2}	3009	25	4.5 V	0.4 V	4.5 V	GND			"							5.5 V	D1	3/	3/	mA
	"	"	26							"			GND	4.5 V	0.4 V	4.5 V	"	D2	"	"	"
	I_{IL4}	"	27	4.5 V	GND	0.4 V	GND			"							"	CLK1	"	"	"
	"	"	28	GND	GND	GND	0.4 V			"							"	PR1	"	"	"
	"	"	29							"			0.4 V	GND	GND	GND	"	PR2	"	"	"
	"	"	30							"			GND	0.4 V	GND	4.5 V	"	CLR2	"	"	"

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 1/ 2, K Cases A, B, C, D Test no.	2 1 CLR1	3 2 D1	4 3 CLK1	6 4 PR1	8 5 Q1	9 6 $\overline{Q}$1	10 7 GND	12 8 Q2	13 9 $\overline{Q}$2	14 10 PR2	16 11 CLK2	18 12 D2	19 13 $\overline{CLR}$2	20 14 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
1 T_C = +25°C	I_{IL5}	3009 "	31 32	0.4 V	4.5 V	4.5 V	GND			GND "			GND	4.5 V	4.5 V	0.4 V	5.5 V "	CLR1 CLR2	3/ 3/	3/ 3/	mA mA
	I_{IH1}	3010 "	33 34	GND	2.7 V	4.5 V	4.5 V			" "			4.5 V	4.5 V	2.7 V	GND	" "	D1 D2		20 20	µA "
	I_{IH2}	" "	35 36	GND	5.5 V	4.5 V	4.5 V			" "			4.5 V	4.5 V	5.5 V	GND	" "	D2 D1		100 100	" "
	I_{IH3}	" " " "	37 38 39 40	GND 4.5 V	4.5 V 4.5 V	2.7 V 4/	4.5 V 2.7 V			" " " "			2.7 V 4.5 V	4/ 2.7 V	4.5 V 4.5 V	4.5 V GND	" " " "	CLK1 PR1 PR2 CLK2		40 " " "	" " " "
	I_{IH4}	" " " "	41 42 43 44	4.5 V GND	4.5 V 4.5 V	4/ 5.5 V	5.5 V 4.5 V			" " " "			4.5 V 5.5 V	5.5 V 4/	4.5 V 4.5 V	GND 4.5 V	" " " "	CLK2 PR2 PR1 CLK1		200 " " "	" " " "
	I_{IH5}	" "	45 46	2.7 V	GND	4/	4.5 V			" "			4.5 V	4/	GND	2.7 V	" "	CLR1 CLR2		60 60	" "
	I_{IH6}	" "	47 48	5.5 V	GND	4/	4.5 V			" "			4.5 V	4/	GND	5.5 V	" "	CLR2 CLR1		300 300	" "
	I_{O5}	3011 " "	49 50 51 52	GND			GND	GND	GND	" " " "	GND	GND	GND			GND	" " " "	$\overline{Q}$1 Q1 Q2 $\overline{Q}$2	-15 " "	-100 " " "	mA " " "
	I_{CC}	3005 3005	53 54	5.5 V GND	GND GND	GND GND	GND 5.5 V			" "			GND 5.5 V	GND GND	GND GND	5.5 V GND	" "	V_{CC} V_{CC}		8.0 8.0	" "
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} test s are omitted.																				
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																				
7/8/ T_C = +25°C	Truth table tests	3014 " " " " " " "	55 56 57 58 59 60 61	B B A " " B B	B " " " " " A	B " A B A " "	B A L H " " "	H L L " L H H	H " " L L H H	GND " " " " " "	H " " L L H H	H L L H " " "	B A A B " " "	B " " " A " "	B " " " " " A	B B A " " B B	4.5 V " " " " " "	All outputs " " " " " "	See 7/		

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 2/ 2, X	2	3	4	6	8	9	10	12	13	14	16	18	19	20	Measured terminal	Test limits		Unit
			Cases A, B, C, D	1	2	3	4	5	6	7	8	9	10	11	12	13	14		Min	Max	
			Test no.	CLR1	D1	CLK1	PR1	Q1	Q1	GND	Q2	Q2	PR2	CLK2	D2	CLR2	V_{CC}				
7 5/6/ T_C = +25°C	Truth table tests	3014	62	B	A	A	A	L	H	GND	H	L	A	A	A	B	4.5 V	All outputs	See 7/		
		"	63	A	"	"	A	L	H	"	H	L	A	"	"	A	"	"			
		"	64	"	"	"	B	H	L	"	L	H	B	"	"	"	"	"			
		"	65	"	"	"	A	"	"	"	"	"	A	"	"	"	"	"			
		"	66	"	"	B	"	"	"	"	"	"	"	B	"	"	"	"			
		"	67	"	B	B	"	"	"	"	"	"	"	B	B	"	"	"			
		"	68	"	"	A	"	L	H	"	H	L	A	A	"	"	"	"			
		"	69	"	"	"	B	H	L	"	L	H	B	"	"	"	"	"			
		"	70	B	A	"	"	"	H	"	H	"	"	"	A	B	"	"			
		"	71	"	B	"	"	"	"	"	"	"	"	"	B	"	"	"			
		"	72	"	"	"	A	L	"	"	"	L	A	"	"	A	"	"			
		"	73	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
		"	74	"	A	B	"	"	"	"	"	"	"	B	A	"	"	"			
		"	75	"	"	A	"	H	L	"	L	H	"	A	"	"	"	"			
		"	76	"	"	"	B	"	"	"	"	"	B	"	"	"	"	"			
		"	77	"	"	"	A	"	"	"	"	"	A	"	"	"	"	"			
		"	78	B	"	"	"	L	H	"	H	L	"	"	"	B	"	"			
		"	79	A	"	"	"	L	H	"	H	L	"	"	"	A	"	"			
		"	80	"	B	"	B	H	L	"	-	H	B	"	B	"	"	"			
		"	81	"	B	"	A	H	L	"	-	H	A	"	B	"	"	"			
8 4/5/	Repeat subgroup 7 at T_C = +125°C and T_C = -55°C.																				
9 T_C = +25°C	f_{MAX} 8/	Fig. 8	82	2.7 V	IN	IN	2.7 V	OUT	OUT	GND	OUT	OUT	2.7 V	IN	IN	2.7 V	5.0 V	Q1	20		MHz
		"	83	2.7 V	IN	IN	2.7 V			"			2.7 V	IN	IN	2.7 V	"	$\overline{Q1}$	"		"
		"	84							"							"	Q2	"		"
		"	85							"							"	$\overline{Q2}$			
	t_{PLH1}	3003 Fig. 6	86							"		OUT	IN			IN	"	PR2 to Q2	5	30	ns
		"	87							"			IN			IN	"	CLR2 to $\overline{Q2}$	"	"	"
		"	88	IN			IN		OUT	"	OUT						"	CLR1 to $\overline{Q1}$	"	"	"
		"	89	IN			IN	OUT		"							"	PR1 to Q1	"		"
	t_{PHL1}	"	90	IN			IN	OUT	OUT	"	OUT	OUT	IN			IN	"	CLR1 to Q1	"	46	"
		"	91	IN			IN			"			IN			IN	"	PR1 to $\overline{Q1}$	"	"	"
		"	92							"							"	PR2 to $\overline{Q2}$	"	"	"
		"	93							"							"	CLR2 to Q2	"		"
	t_{PLH2}	3003 Fig. 8	94							"		OUT	2.7 V	IN	IN	2.7 V	"	CLK2 to Q2	"	30	"
		3003 Fig. 8	95							"	OUT		2.7 V	IN	IN	2.7 V	"	CLK2 to $\overline{Q2}$	"	"	"
		3003 Fig. 9	96	2.7 V	IN	IN	2.7 V	OUT		"							"	CLK1 to Q1	"	"	"
		3003 Fig. 8	97	2.7 V	IN	IN	2.7 V		OUT	"							"	CLK1 to $\overline{Q1}$	"	"	"
		3003 Fig. 9								"											
	t_{PHL2}	3003 Fig. 8	98	2.7 V	IN	IN	2.7 V		OUT	"							"	CLK1 to $\overline{Q1}$	"	46	"
		3003 Fig. 8	99	2.7 V	IN	IN	2.7 V	OUT		"							"	CLK1 to Q1	"	"	"
		3003 Fig. 9	100							"	OUT		2.7 V	IN	IN	2.7 V	"	CLK2 to $\overline{Q2}$	"	"	"
		3003 Fig. 8	101							"		OUT	2.7 V	IN	IN	2.7 V	"	CLK2 to Q2	"	"	"
		3003 Fig. 9								"											

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 1/ 2,X Cases A, B, C, D Test no.	2 1 CLR1	3 2 D1	4 3 CLK1	6 4 PR1	8 5 Q1	9 6 $\overline{Q1}$	10 7 GND	12 8 $\overline{Q2}$	13 9 Q2	14 10 PR2	16 11 CLK2	18 12 D2	19 13 CLR2	20 14 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
10	f_{MAX} 8/	Fig. 8	102 105																20		MHz
	t_{PLH1}	3003 Fig. 7	106 109																5	39	ns
	t_{PHL1}	3003 Fig. 7	110-113																"	59	"
	t_{PLH2}	3003 Fig. 8	114																"	39	"
		3003 Fig. 9	115	Same tests and terminal conditions as for subgroup 9, except T_C = +125°C.															"	"	"
		3003 Fig. 8	116																"	"	"
		3003 Fig. 9	117																"	"	"
	t_{PHL2}	3003 Fig. 8	118																"	59	"
		3003 Fig. 9	119																"	"	"
		3003 Fig. 8	120																"	"	"
		3003 Fig. 9	121																"	"	"
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																				

1/ Case 2 pins not referenced are NC.

2/ ⎍⎍ 2.5 V minimum/5.5 V maximum.
0 V

3/ I_{IL} limits in mA are as follows:

I_{IL2}	Min/Max limits for CKT					
A	B	C	D	E	F
-.075/-.250	-.030/-.300	-.095/-.210	-.097/-.207	-.135/-.370	-.160/-.400

I_{IL4}	Min/Max limits for CKT					
A	B	C	D	E	F
-.150/-.500 for tests 27, 30 -.200/-.800 for tests 28, 29	-.060/-.700	-.160/-.400 for tests 27, 30 -.350/-.760 for tests 28, 29	-.160/-.400 for tests 27, 30 -.355/-.759 for tests 28, 29	-.120/-.360 for tests 27, 30 -.280/-.760 for tests 28, 29	-.320/-.800 (All)

I_{IL5}	Min/Max limits for CKT					
A	B	C	D	E	F
-.200/-.800	-.060/-.700	-.350/-.760	-.480/-1.200	-.280/-.760	-.480/-1.200

4/ ⎍⎍ 2.5 V minimum/5.5 V maximum.
0 V

5/ Input voltages shown are A = 2.0 volts minimum and B = 0.7 volt maximum.

6/ Tests shall be performed in sequence, attributes data only

7/ Output voltages shall be H >1.5 V and L <1.5 V.

8/ f_{MAX} minimum limit specified is the frequency of the input pulse. The output frequency shall be one half of the input frequency

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TABLE III. Group A inspection for device type 04.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.7 V, or open).

Subgroup	Symbol	MIL-STD 883 method	Cases I/ 2,X Cases A, B, C, D Test no.	2 1 CLK1	3 2 K1	4 3 J1	6 4 PR1	8 5 Q1	9 6 $\overline{Q1}$	10 7 GND	12 8 $\overline{Q2}$	13 9 Q2	14 10 PR2	16 11 J2	18 12 K2	19 13 CLK2	20 14 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
1 T_C = +25°C	V_{OH}	3006	1	2.0 V	2.0 V	2.0 V	0.7 V	-.4 mA		GND							4.5 V	Q1	2.5		V
		"	2	2/	2.0 V	0.7 V	2.0 V		-.4 mA	"							"	$\overline{Q1}$	"		"
		"	3	2/	0.7 V	2.0 V	2.0 V	-.4 mA		"							"	Q1	"		"
		"	4							"	-.4 mA		2.0 V	0.7 V	2.0 V	2/	"	Q2	"		"
		"	5							"		-.4 mA	2.0 V	2.0 V	0.7 V	2/	"	Q2	"		"
		"	6							"		-.4 mA	0.7 V	"	2.0 V	2.0 V	"	Q2	"		"
	V_{OL}	3007	7							"	4 mA		0.7 V	"	2.0 V	2.0 V	"	Q2		0.4	"
		"	8							"	4 mA		2.0 V	"	0.7 V	2/	"	$\overline{Q2}$		"	"
		"	9							"		4 mA	2.0 V	0.7 V	2.0 V	2/	"	Q2		"	"
		"	10	2.0 V	2.0 V	2.0 V	0.7 V		4 mA	"							"	$\overline{Q1}$		"	"
		"	11	2/	0.7 V	2.0 V	2.0 V		4 mA	"							"	Q1		"	"
		"	12	2/	2.0 V	0.7 V	2.0 V	4 mA		"							"	Q1		"	"
	V_{IC}		13	-18 mA						"							"	CLK1		-1.5	"
			14		-18 mA					"							"	K1		"	"
			15			-18 mA				"							"	J1		"	"
			16				-18 mA			"							"	PR1		"	"
			17							"			-18 mA				"	PR2		"	"
			18							"				-18 mA			"	J2		"	"
			19							"					-18 mA		"	K2		"	"
			20							"						-18 mA	"	CLK2		"	"
	I_{IL1}	3009	21	4.5 V	0.4 V	GND	3/			"							5.5 V	K1	4/	4/	mA
		"	22	3/	4.5 V	0.4 V	4.5 V			"							"	J1	"	"	"
		"	23							"			3/	GND	0.4 V	4.5 V	"	K2	"	"	"
		"	24							"			4.5 V	0.4 V	4.5 V	3/	"	J2	"	"	"
	I_{IL3}	"	25							"							"				
		"	26	0.4 V	4.5 V	4.5 V	3/			"			3/	4.5 V	4.5 V	0.4 V	"	CLK2	"	"	"
																		CLK1			
	I_{IL4}	"	27	4.5 V	4.5 V	4.5 V	0.4 V			"							"	PR1	"	"	"
		"	28							"			0.4 V	4.5 V	4.5 V	4.5 V	"	PR2	"	"	"
	I_{IH1}	3010	29	GND	2.7 V	GND	GND			"							"	K1		20	μA
		"	30	2/	GND	2.7 V	4.5 V			"							"	J1		"	"
		"	31							"			4.5 V	2.7 V	GND	2/	"	J2		"	"
		"	32							"			GND	GND	2.7 V	GND	"	K2		"	"

See footnotes at end of device type 04.

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TABLE III. Group A inspection for device type 04 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 1/ 2,X Cases A, B, C, D Test no.	2 1 CLK1	3 2 K1	4 3 J1	6 4 PR1	8 5 Q1	9 6 $\overline{Q}1$	10 7 GND	12 8 $\overline{Q}2$	13 9 Q2	14 10 PR2	16 11 J2	18 12 K2	19 13 CLK2	20 14 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
1 T_C = +25°C	I_{IH2}	3010	33							GND			GND	GND	5.5 V	GND	5.5 V	K2		100	µA
		"	34							"			4.5 V	5.5 V	GND	2/	"	J2		"	"
		"	35	2/	GND	5.5 V	4.5 V			"							"	J1		"	"
		"	36	GND	5.5 V	GND	GND			"							"	K1		"	"
	I_{IH5}	"	37	GND	GND	4.5 V	2.7 V	5/		"							"	PR1		60	"
		"	38							"		5/	2.7 V	4.5 V	GND	GND	"	PR2		60	"
	I_{IH6}	"	39							"		5/	5.5 V	4.5 V	GND	GND	"	PR2		300	"
		"	40	GND	GND	4.5 V	5.5 V	5/		"							"	PR1		300	"
	I_{IH7}	"	41	2.7 V	GND	GND	GND			"							"	CLK1		80	"
		"	42							"			GND	GND	GND	2.7 V	"	CLK2		80	"
	I_{IH8}	"	43							"			GND	GND	GND	5.5 V	"	CLK2		400	"
		"	44	5.5 V	GND	GND	GND			"							"	CLK1		400	"
	I_{OS}	3011	45				GND	GND		"							"	Q1	-15	-100	mA
		"	46	2/	4.5 V	GND	4.5 V	GND	GND 6/	"							"	$\overline{Q}1$	6/	6/	"
		"	47							"		GND	GND				"	Q2	-15	-100	"
		"	48							"	GND 6/	GND	4.5 V	GND	4.5 V	2/		$\overline{Q}2$	6/	6/	"
	I_{CC}	3005	49	2/	5.5 V	GND	5.5 V			"			5.5 V	GND	5.5 V	2/	"	V_{IC}		8.0	"
		3005	50	5.5 V	5.5 V	5.5 V	GND			"			GND	5.5 V	5.5 V	5.5 V	"	V_{CC}		8.0	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} testss are omitted.																				
7/8/ T_C = +25°C	Truth table tests	3014	51	B	A	B	B	H	L	GND	L	H	B	B	B	B	4.5 V	All outputs	See 9/		
		"	52	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	
		"	53	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	
		"	54	B	"	"	A	"	"	"	"	"	"	"	"	"	"	"	"	"	
		"	55	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	
		"	56	B	"	"	"	L	H	"	"	"	"	"	"	"	"	"	"	"	
		"	57	B	B	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	
		"	58	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	
		"	59	B	"	"	"	H	L	"	"	"	"	"	"	"	"	"	"	"	
		"	60	"	"	B	B	"	"	"	"	"	"	"	A	"	"	"	"	"	
		"	61	"	"	"	"	"	"	"	"	"	A	"	"	"	"	"	"	"	
		"	62	"	"	"	"	"	"	"	"	"	"	"	"	A	"	"	"	"	
		"	63	"	"	"	"	"	"	"	H	L	"	"	"	B	"	"	"	"	
		"	64	"	"	"	"	"	"	"	"	"	A	B	"	B	"	"	"	"	
		"	65	"	"	"	"	"	"	"	"	"	"	"	"	A	"	"	"	"	
		"	66	"	"	"	"	"	"	"	L	H	"	"	"	B	"	"	"	"	
		"	67	"	"	"	"	"	"	"	"	"	B	B	"	"	"	"	"	"	
		"	68	"	"	"	A	"	"	"	"	"	A	"	"	"	"	"	"	"	
		"	69	A	"	"	"	"	"	"	"	"	"	"	"	A	"	"	"	"	
		"	70	B	"	"	"	"	"	"	"	"	"	"	"	B	"	"	"	"	
		"	71	B	A	A	"	"	"	"	"	"	A	A	B	"	"	"	"	"	
		"	72	A	"	"	"	"	"	"	"	"	"	"	A	"	"	"	"	"	
		"	73	B	"	"	"	L	H	"	H	L	"	"	B	"	"	"	"	"	
		"	74	B	B	B	"	"	"	"	"	"	B	B	B	"	"	"	"	"	
		"	75	A	"	"	"	"	"	"	"	"	"	"	A	"	"	"	"	"	
		"	76	B	"	"	"	"	"	"	"	"	"	"	B	"	"	"	"	"	
		"	77	B	A	A	"	"	"	"	"	"	A	A	B	"	"	"	"	"	
		"	78	A	"	"	"	"	"	"	"	"	"	"	A	"	"	"	"	"	
		"	79	B	"	"	"	H	L	"	L	H	"	"	B	"	"	"	"	"	

See footnotes at end of device type 04.

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TABLE III. Group A inspection for device type 04 - Continued.
Terminal conditions (pins not designated may be high >2.0 V, low <0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 1/ 2,K Cases A, B, C, D Test no.	2 CLK1	3 K1	4 J1	6 PR1	8 Q1	9 $\overline{Q1}$	10 GND	12 $\overline{Q2}$	11 Q2	14 PR2	16 J2	18 K2	19 CLK2	20 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
8	5/7/	Repeat subgroup 7 at T_C = +125°C and T_C = -55°C.																			
9 T_C = +25°C	f_{MAX} 8/	Fig. 9	90	IN	2.7 V	2.7 V	2.7 V	OUT	OUT	GND	OUT	OUT	2.7 V	2.7 V	2.7 V	IN	5.0 V	Q1	25		MHz
		"	81	IN	2.7 V	2.7 V	2.7 V			"			2.7 V	2.7 V	2.7 V	IN	"	$\overline{Q1}$	"		"
		"	82							"							"	Q2	"		"
		"	83							"											
	t_{PLH1}	3003 Fig. 10	84							"							"	PR2 to Q2	5	21	ns
	t_{PHL1}		85	IN	2.7 V	GND	IN	OUT		"			IN	GND	2.7 V	IN	"	PR1 to $\overline{Q1}$	"	21	"
	t_{PHL1}	"	86	IN	2.7 V	GND	IN		OUT	"	OUT		IN	GND	2.7 V	IN	"	PR1 to $\overline{Q1}$	"	28	"
		"	87							"							"	PR2 to $\overline{Q2}$	"	28	"
	t_{PLH2}	3003 Fig. 9	88							"							"	CLK2 to $\overline{Q2}$	"	22	"
		"	89							"							"	CLK2 to Q2	"	"	"
		"	90	IN	2.7 V	2.7 V	2.7 V		OUT	"	OUT		2.7 V	2.7 V	2.7 V	IN	"	CLK1 to Q1	"	"	"
		"	91	"	"	"	"	OUT		"			2.7 V	2.7 V	2.7 V	IN	"	CLK1 to $\overline{Q1}$	"	"	"
	t_{PHL2}	"	92	"	"	"	"	OUT		"							"	CLK1 to $\overline{Q1}$	"	30	"
		"	93	"	"	"	"			"							"	CLK1 to Q1	"	"	"
		"	94						OUT	"	OUT		2.7 V	2.7 V	2.7 V	IN	"	CLK2 to Q2	"	"	"
		"	95							"		OUT	2.7 V	2.7 V	2.7 V	IN	"	CLK2 to $\overline{Q2}$	"	"	"
10	f_{MAX} 10/	Fig. 9	96-99																25		MHz
	t_{PLH1}	3003 Fig.10	100-101																5	17	ns
	t_{PHL1}	3003 Fig.10	102-103	Same tests and terminal conditions as for subgroup 9, except T_C = +125°C.															"	40	"
	t_{PLH2}	3003 Fig. 9	104-107																"	12	"
	t_{PHL2}	3003 Fig. 9	108-111																"	47	"
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																				

1/ Case 2 pins not referenced are NC.

2/ 2.5 V minimum/5.5 V maximum. 0 V

3/ 2.5 V minimum/5.5 V maximum. 0 V

4/ I_{IL} limits in mA are in follows:

	Min/max limits for CRT				
	A	B	C	D and F	E
I_{IL1}	-.075/-.250	-.030/-.300	-.110/-.250	-.120/-.360	-.120/-.360
I_{IL2}	-.175/-.550	-.060/-.600	-.150/-.560	-.240/-.720	-.280/-.760
I_{IL3}	-.200/-.800	-.060/-.700	-.290/-.650	-.120/-.720	-.320/-.800

5/ Momentary GND, then open.

6/ I_{OS} limits in mA are as follows:

Test no.	A	B	C	D and E	F
46,48	-7.5/-50	-7.5/-50	-30/-110	-15/-100	-7.5/-50
46,48 D_1, D_2	2.25 V	2.25 V	---	---	2.25 V

7/ Input voltages shown are A = 2.0 volts minimum and B = 0.7 volt maximum.

8/ Tests shall be performed in sequence, attributes data only.

9/ Output voltages shall be H >1.5 V and L <1.5 V.

10/ f_{MAX} minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency.

MIL-M-38510/304B

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TABLE III. Group A inspection for device type 05.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 1/ 2,X Cases A, B, C, D Test no.	1 CLR	2 K1	3 J1	4 PR1	5 Q1	6 $\overline{Q1}$	7 GND	8 $\overline{Q2}$	9 Q2	10 PR2	11 J2	12 K2	13 CLK2	14 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
1 T_C = +25°C	V_{OH}	3006	1	2.0 V	0.7 V	2.0 V	0.7 V	-.4 mA		GND						2.0 V	4.5 V	Q1	2.5		V
		"	2	0.7 V	"	"	2.0 V		-.4 mA	"						2.0 V	"	Q1	"		"
		"	3	2.0 V	"	"	"	-.4 mA		"						2/	"	Q1	"		"
		"	4	"	2.0 V	0.7 V	"		-.4 mA	"						"	"	$\overline{Q1}$	"		"
		"	5	"	"					"	-.4 mA		2.0 V	0.7 V	2.0 V	"	"	$\overline{Q2}$	"		"
		"	6	"	"					"		-.4 mA	"	2.0 V	0.7 V	"	"	$\overline{Q2}$	"		"
		"	7	0.7 V						"	-.4 mA		"	"	"	2.0 V	"	$\overline{Q2}$	"		"
		"	8	2.0 V						"		-.4 mA	0.7 V	"	"	"	"	$\overline{Q2}$	"		"
	V_{OL}	3007	9	2.0 V						"	4 mA		0.7 V	"	"	"	"	$\overline{Q2}$		0.4	"
		"	10	0.7 V						"		4 mA	2.0 V	"	"	"	"	Q2		"	"
		"	11	2.0 V						"	4 mA		"	"	"	2/	"	$\overline{Q2}$		"	"
		"	12	"						"		4 mA	"	0.7 V	2.0 V	"	"	Q2		"	"
		"	13	"	0.7 V	2.0 V	2.0 V		4 mA	"						"	"	$\overline{Q1}$		"	"
		"	14	"	2.0 V	0.7 V	2.0 V	4 mA		"						"	"	Q1		"	"
		"	15	"	"	"	0.7 V		4 mA	"						2.0 V	"	$\overline{Q1}$		"	"
		"	16	0.7 V	"	"	2.0 V	4 mA		"						2.0 V	"	Q1		"	"
	V_{IC}		17	-18 mA						"							"	CLR		-1.5	"
			18		-18 mA					"							"	K1		"	"
			19			-18 mA				"							"	J1		"	"
			20				-18 mA			"							"	PR1		"	"
			21							"			-18 mA				"	PR2		"	"
			22							"				-18 mA			"	J2		"	"
			23							"					-18 mA		"	K2		"	"
			24							"						-18 mA	"	CLK		"	"
	I_{IL1}	3009	25	4.5 V	0.4 V	GND	3/			"						4.5 V	5.5 V	K1	4/	4/	μA
		"	26	3/	GND	0.4 V	4.5 V			"						"	"	J1	"	"	"
		"	27	3/						"			4.5 V	0.4 V	4.5 V	"	"	J2	"	"	"
		"	28	4.5 V						"			3/	4.5 V	0.4 V	"	"	K2	"	"	"
	I_{IL4}	"	29							"			0.4 V	4.5 V	4.5 V	"	"	PR2	"	"	"
		"	30		4.5 V	4.5 V	0.4 V			"						"	"	PR1	"	"	"
	I_{IL6}	"	31	3/	"	"	4.5 V			"			4.5 V	4.5 V	4.5 V	0.4 V	"	CLK	"	"	"
		"	32	4.5 V	"	"	3/			"			3/		4.5 V	0.4 V	"	CLK	"	"	"

See footnotes at end of device type 05.

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Terminal conditions (pins not designated may be high >2.0 V, low <0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 1/ 2, K Cases A, B, C, D Test no.	2	3	4	6	8	9	10	12	13	14	16	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
				1	2	3	4	5	6	7	8	9	10	11	12	13	14				
				CLR	K1	J1	PR1	Q1	$\overline{Q1}$	GND	$\overline{Q2}$	Q2	PR2	J2	K2	CLK2	V_{CC}				
1 T_C = +25°C	I_{IL7}	3009	33	0.4 V	4.5 V	4.5 V	4.5 V			GND			4.5 V	4.5 V	4.5 V	4.5 V	5.5 V	CLR	4/	4/	mA
	I_{IH1}	3010	34	GND	2.7 V	GND	GND			"						GND	"	K1		20	µA
		"	35	"	GND	2.7 V	GND			"			GND	2.7 V	GND	"	"	J1		"	"
		"	36	"						"			"	GND	2.7 V	"	"	J2		"	"
		"	37	"						"								K2		"	"
	I_{IH2}	"	38	"						"			"	GND	5.5 V	"	"	K2		100	"
		"	39	"						"			"	5.5 V	GND	"	"	J2		"	"
		"	40	"	GND	5.5 V	GND			"			"			"	"	J1		"	"
		"	41	"	5.5 V	GND	GND			"								K1		"	"
	I_{IH5}	"	42	"	GND	GND	2.7 V			"			2.7 V	GND	GND	"	"	PR1		60	"
		"	43	"						"						"	"	PR2		60	"
	I_{IH6}	"	44	"	GND	GND	5.5 V			"			5.5 V	GND	GND	"	"	PR2		300	"
		"	45	"						"						"	"	PR1		300	"
	I_{IH9}	"	46	2.7 V	"	"	GND			"			GND	GND	GND	"	"	CLR		120	"
	I_{IH10}	"	47	5.5 V	"	"	"			"			"	"	"	"	"	CLR		600	"
	I_{IH11}	"	48	GND	"	"	"			"			"	"	"	2.7 V	"	CLK		160	"
	I_{IH12}	"	49	"	"	"	"			"			"	"	"	5.5 V	"	CLK		800	"
	I_{OS}	3011	50	"			4.5 V		GND	"			4.5 V				"	$\overline{Q1}$	-15	-100	mA
		"	51	"						"			GND				"	Q2	"	"	"
		"	52	4.5 V						"	GND	GND					"	Q2	"	"	"
		"	53	4.5 V			GND	GND		"							"	Q1	"	"	"
	I_{CC}	3005	54	GND	5.5 V	5.5 V	5.5 V			"			5.5 V	5.5 V	5.5 V	GND	"	V_{CC}		8.0	"
			55	5.5 V	5.5 V	5.5 V	GND			"			GND	5.5 V	5.5 V	GND	"	V_{CC}		8.0	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																				
7 5/ 6/ T_C = +25°C	Truth table tests	3014	56	B	B	A	A	L	H	GND	H	L	A	A	B	A	4.5 V	All outputs	See 7/		
		"	57	"	"	"	"	"	"	"	"	"	B	B	"	B	"	"			
		"	58	"	"	"	"	"	"	"	L	H	"	"	"	B	"	"			
		"	59	A	"	"	"	"	"	"	"	"	"	"	"	B	"	"			
		"	60	"	"	"	"	H	L	"	"	"	"	"	"	A	"	"			
		"	61	"	"	"	"	"	"	"	"	"	"	"	"	B	"	"			
		"	62	"	"	B	"	"	"	"	"	"	"	"	"	A	"	"			
		"	63	"	"	"	"	"	"	"	"	"	"	"	"	B	"	"			
		"	64	"	"	"	"	"	"	"	"	"	"	"	"	A	"	"			
		"	65	"	A	"	"	"	"	"	"	"	"	"	"	B	"	"			
		"	66	"	"	"	"	L	H	"	"	"	"	"	"	A	"	"			
		"	67	"	"	"	"	L	H	"	"	"	"	"	"	A	"	"			
		"	68	"	"	"	B	H	L	"	"	"	"	"	"	B	"	"			
		"	69	"	"	"	B	H	L	"	"	"	"	"	"	A	"	"			

See footnotes at end of device type 05.

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TABLE III. Group A inspection for device type 05 - Continued.
Terminal conditions (pins not designated may be high >2.0 V, or low <0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 1/ 2,X Cases A, B, C, D Test no.	2 I CLK1	3 J1	4 K1	5 PR1	6 Q1	7 $\overline{Q1}$	8 GND	9 $\overline{Q2}$	10 Q2	11 PR2	12 J2	13 K2	14 CLK2	15 V_{CC}	16 Measured terminal	Test limits Min	Test limits Max	Unit
7 5/ 6/ T_C = +25°C	Truth table tests	3014	70	B	B	B	A	L	H	GND	H	L	A	A	B	A	4.5 V	All outputs	See 11		
		"	71	A	"	"	B	H	L	"	H	L	"	"	"	A	"	"			
		"	72	"	"	"	"	"	"	"	L	H	"	"	"	B	"	"			
		"	73	"	"	"	"	"	"	"	"	"	"	"	"	A	"	"			
		"	74	"	"	"	"	"	"	"	"	"	"	B	"	A	"	"			
		"	75	"	"	"	"	"	"	"	"	"	"	"	"	B	"	"			
		"	76	"	"	"	"	"	"	"	"	"	"	"	"	A	"	"			
		"	77	"	"	"	"	"	"	"	"	"	"	"	A	A	"	"			
		"	78	"	"	"	"	"	"	"	H	L	"	"	"	B	"	"			
		"	79	"	"	"	"	"	"	"	H	L	"	"	"	A	"	"			
		"	80	"	"	"	"	"	"	"	L	H	B	"	"	A	"	"			
		"	81	"	"	"	"	"	"	"	L	"	"	"	"	B	"	"			
		"	82	B	A	A	"	H	"	"	H	"	"	A	"	A	"	"			
		"	83	B	B	B	A	L	"	"	"	L	A	B	B	"	"	"			
		"	84	A	"	"	"	"	"	"	"	"	"	"	"	B	"	"			
		"	85	"	"	"	"	"	"	"	"	"	"	"	"	B	"	"			
		"	86	"	A	A	"	"	"	"	"	"	"	A	A	B	"	"			
		"	87	"	"	"	"	"	"	"	"	"	"	"	"	A	"	"			
		"	88	"	"	"	H	L	"	"	L	H	"	"	"	B	"	"			
		"	89	"	B	B	"	"	"	"	"	"	"	B	B	B	"	"			
		"	90	"	"	"	"	"	"	"	"	"	"	"	"	A	"	"			
		"	91	"	"	"	"	"	"	"	"	"	"	"	"	B	"	"			
		"	92	"	A	A	"	"	"	"	"	"	"	A	A	B	"	"			
		"	93	"	"	"	"	"	"	"	"	"	"	"	"	A	"	"			
		"	94	"	"	"	"	L	H	"	H	L	"	"	"	B	"	"			
8 4/ 5/ Repeat subgroup 7 at T_C = 125°C and T_C = -55°C																					
9 T_C = +25°C	f_{MAX}	Fig. 11	95	5.0 V	2.7 V	2.7 V	2.7 V	OUT	OUT	GND	OUT	OUT	2.7 V	2.7 V	2.7 V	IN	5.0 V	Q1	25		MHz
	"	"	96	"	2.7 V	2.7 V	2.7 V	"	"	"	"	"	2.7 V	2.7 V	2.7 V	"	"	Q1	"		"
	11	"	97	"								OUT				"	"	Q2	"		"
		"	98	"									OUT			"	"	Q2	"		"
	t_{PLH1}	3003 Fig. 12	99	IN	GND	2.7 V	2.7 V		OUT	"						"	"	CLR to Q1	5	21	ns
		"	100	IN						"		OUT	2.7 V	2.7 V	GND	"	"	CLR to Q2	"	"	"
		"	101	2.7 V						"			IN	GND	2.7 V	"	"	PR2 to Q2	"	"	"
		"	102	"	2.7 V	GND	IN	OUT		"						"	"	PR1 to Q1	"	"	"
	t_{PHL1}	"	103	"	2.7 V	GND	IN		OUT	"						"	"	PR1 to $\overline{Q1}$	"	28	"
		"	104	"						"		OUT	IN	GND	2.7 V	"	"	PR2 to $\overline{Q2}$	"	"	"
		"	105	IN						"			2.7 V	2.7 V	GND	"	"	CLR to Q2	"	"	"
		"	106	IN	GND	2.7 V	2.7 V	OUT		"						"	"	CLR to Q1	"	"	"
	t_{PLH2}	3003 Fig. 11	107	2.7 V	2.7 V	2.7 V	2.7 V	OUT		"						"	"	CLK to Q1	"	22	"
		"	108	"	2.7 V	2.7 V	2.7 V		OUT	"						"	"	CLK to $\overline{Q1}$	"	"	"
		"	109	"						"		OUT	2.7 V	2.7 V	2.7 V	"	"	CLK to $\overline{Q2}$	"	"	"
		"	110	"						"			2.7 V	2.7 V	2.7 V	"	"	CLK to Q2	"	"	"
	t_{PHL2}	"	111	"						"		OUT	2.7 V	2.7 V	2.7 V	"	"	CLK to Q2	"	30	"
		"	112	"						"		OUT	2.7 V	2.7 V	2.7 V	"	"	CLK to $\overline{Q2}$	"	"	"
		"	113	"	2.7 V	2.7 V	2.7 V		OUT	"						"	"	CLK to $\overline{Q1}$	"	"	"
		"	114	"	2.7 V	2.7 V	2.7 V	OUT		"						"	"	CLK to Q1	"	"	"

See footnotes at end of device type 05.

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TABLE III. Group A inspection for device type 05 - Continued.
Terminal conditions (pins not designated may be high >2.0 V, or low <0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 1/ 2,X Cases A, B, C, D Test no.	2	3	4	6	8	9	10	12	13	14	16	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
				1	2	3	4	5	6	7	8	9	10	11	12	13	14				
				CLR	K1	J1	PR1	Q1	$\overline{Q}_1$	GND	$\overline{Q}_2$	Q2	PR2	J2	K2	CLK2	V_{CC}				
10	f_{MAX} 8/	Fig. 11	115-118																25		MHz
	t_{PLH1}	3003 Fig. 12	119 122	Same tests and terminal conditions as for subgroup 9, except T_C = +125°C.															5	32	ns
	t_{PHL1}	3003 Fig. 12	123 126																"	40	"
	t_{PLH2}	3003 Fig. 11	127-130																"	32	"
	t_{PHL2}	3003 Fig. 11	131-134																"	42	"
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																				

1/ Case 2 pins not referenced are NC.

2/ ---2.5 V minimum/5.5 V maximum.
---0 V

3/ 2.5 V minimum/5.5 V maximum.
---0 V

4/ I_{IL} limits in mA are as follows:

	Min/Max limits for CKT					
	A	B	C	D	E	
I_{IL1}	-.075/-.750	-.030/-.300	-.110/-.2701	-.120/-.3601	-.120/-.1601	
I_{IL4}	-.200/-.800	-.060/-.7001	-.290/-.6501	-.120/-.7201	-.320/-.8001	
I_{IL6}	-.300/-1.0001	-.120/-1.0001	-.300/-1.1201	-.240/-1.4401	-.560/-1.5201	
I_{IL7}	-.450/-1.3001	-.120/-1.0001	-.580/-1.30	-.120/-1.5001	-.640/-1.6001	

5/ Input voltages shown are A = 2.0 volts minimum and B = 0.7 volt maximum.

6/ Tests shall be performed in sequence, attributes data only.

7/ Output voltages shall be H >1.5 V and L <1.5 V.

8/ f_{MAX} minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency.

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TABLE III. Group A inspection for device types 03 and 10.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.7 V, or open). 1/

Subgroup	Symbol	MIL STD-883 method	Cases 2/ 2,X	* 2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20	Measured terminal	Test Limits Min	Test Limits Max	Unit
			Cases E,F	** 2	20	5	3	19	18	13	17	14	9	12	15	8	10	4	7				
				* 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16				
				** 1	16	4	2	15	14	10	13	11	7	9	12	6	8	3	5				
			Test no.	CLK1	K1	J1	PR1	Q1	$\overline{Q1}$	$\overline{Q2}$	GND	Q2	PR2	J2	K2	CLK2	CLR2	CLR1	V_{CC}				
1 T_C = +25°C	V_{OH}	3006	1	2.0 V	2.0 V	2.0 V	2.0 V		-.4 mA		GND							0.7 V	4.5 V	Q1	2.5		V
		"	2	2.0 V	"	2.0 V	0.7 V	-.4 mA			"							2.0 V	"	"			
		"	3	3/	"	0.7 V	2.0 V		.4 mA		"							"	"	"			
		"	4	3/	0.7 V	2.0 V	2.0 V	-.4 mA			"							"	"	Q1			
		"	5								"	-.4 mA	0.7 V	2.0 V	2.0 V	2.0 V	2.0 V		"	Q2			
		"	6							-.4 mA	"		2.0 V	"	2.0 V	2.0 V	0.7 V		"	Q2			
		"	7								"	-.4 mA	"	"	0.7 V	3/	2.0 V		"	Q2			
		"	8							.4 mA	"		"	0.7 V	2.0 V	"	"		"	Q2			
	V_{OL}	3007	9								"	.4 mA	"	0.7 V	2.0 V	"	"		"	Q2		0.4	
		"	10							.4 mA	"		"	2.0 V	0.7 V	"	"		"	Q2		"	
		"	11								"	.4 mA	"	2.0 V	2.0 V	2.0 V	0.7 V		"	Q2		"	
		"	12							.4 mA	"		0.7 V	2.0 V	2.0 V	2.0 V	2.0 V		"	Q2		"	
		"	13	2.0 V	2.0 V	2.0 V	2.0 V	.4 mA			"							0.7 V	"	Q1		"	
		"	14	2.0 V	"	2.0 V	0.7 V		.4 mA		"							2.0 V	"	Q1		"	
		"	15	3/	"	0.7 V	2.0 V	.4 mA			"							"	"	Q1		"	
		"	16	3/	0.7 V	2.0 V	2.0 V		.4 mA		"								"	Q1		"	
	V_{IC}		17	-.18 mA							"								"	CLK1		-1.5	
			18		-.18 mA						"								"	K1		"	
			19			-.18 mA					"								"	J1		"	
			20				-.18 mA				"								"	PR1		"	
			21								"		-.18 mA						"	PR2		"	
			22								"			-.18 mA					"	J2		"	
			23								"				-.18 mA				"	K2		"	
			24								"					-.18 mA			"	CLK2		"	
			25								"						-.18 mA		"	CLR2		"	
			26								"							-.18 mA	"	CLR1		"	
	I_{IL1}	3009	27	4.5 V	0.4 V	4.5 V	4/				"							4.5 V	5.5 V	K1	5/	5/	uA
		"	28	4.5 V	4.5 V	0.4 V	4.5 V				"							4/	"	J1	"	"	
		"	29								"		4.5 V	0.4 V	4.5 V	4/			"	J2	"	"	
		"	30								"		4/	4.5 V	0.4 V	4.5 V			"	K2	"	"	
	I_{IL3}	"	31								"		4/	4.5 V	4.5 V	0.4 V	4.5 V		"	CLK2		"	
		"	32								"		4.5 V	4.5 V	4.5 V	0.4 V	4/		"	CLK2		"	
		"	33	0.4 V	4.5 V	4.5 V	4.5 V				"							4/	"	CLK1		"	
		"	34	0.4 V	4.5 V	4.5 V	4/				"							4.5 V	"	CLK1		"	

See footnotes at end of device types 03 and 10.

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TABLE III. Group A inspection for device types 03 and 10 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.7 V, or open). 1/

Subgroup	Symbol	MIL-STD-883 method	Cases 2/ 2,X Cases F,F Test no.	2 / 2 / 1 / 1 CLK1	3 / 20 / 2 / 16 K1	4 / 5 / 3 / 4 J1	5 / 3 / 4 / 2 PR1	7 / 19 / 5 / 15 Q1	8 / 18 / 6 / 14 $\overline{Q1}$	9 / 13 / 7 / 16 $\overline{Q2}$	10 / 17 / 8 / 17 GND	12 / 14 / 9 / 11 Q2	13 / 9 / 10 / 7 PR2	14 / 12 / 11 / 9 J2	15 / 15 / 12 / 12 K2	17 / 8 / 13 / 6 CLK2	18 / 10 / 14 / 8 CLR2	19 / 4 / 15 / 3 CLR1	20 / 7 / 16 / 5 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
1 T_C +25°C	I_{IL4}	3009	35	4.5 V	4.5 V	4.5 V	0.4 V				GND							4.5 V	5.5 V	PR1	5/	5/	mA
		"	36	4.5 V	4.5 V	4.5 V	4.5 V				"							0.4 V	"	CLR1	"	"	"
		"	37								"		4.5 V	4.5 V	4.5 V	4.5 V	0.4 V		"	CLR2	"	"	"
		"	38								"		0.4 V	4.5 V	4.5 V	4.5 V	4.5 V		"	PR2	"	"	"
	I_{IH1}	3010	39								"		GND	4.5 V	2.7 V	GND	4.5 V		"	K2		20	µA
		"	40								"		4.5 V	2.7 V	4.5 V	GND	GND		"	J2		"	"
		"	41	GND	4.5 V	2.7 V	4.5 V				"							GND	"	J1		"	"
		"	42	GND	2.7 V	4.5 V	GND				"							4.5 V	"	K1		"	"
	I_{IH2}	"	43	GND	5.5 V	4.5 V	GND				"							4.5 V	"	K1		100	"
		"	44	GND	4.5 V	5.5 V	4.5 V				"							GND	"	J1		"	"
		"	45								"		4.5 V	5.5 V	4.5 V	GND	GND		"	J2		"	"
		"	46								"		GND	4.5 V	5.5 V	"	4.5 V		"	K2		"	"
	I_{IH5}	"	47								"		4/	GND	4.5 V	"	2.7 V		"	CLR2		60	"
		"	48								"		2.7 V	4.5 V	GND	"	4/		"	PR2		"	"
		"	49	GND	4.5 V	GND	4/				"							2.7 V	"	CLR1		"	"
		"	50	GND	GND	4.5 V	2.7 V				"							4/	"	PR1		"	"
	I_{IH6}	"	51	GND	GND	4.5 V	5.5 V				"							4/	"	PR1		300	"
		"	52	GND	4.5 V	GND	4/				"							5.5 V	"	CLR1		"	"
		"	53								"		5.5 V	4.5 V	GND	GND	4/		"	PR2		"	"
		"	54								"		4/	GND	4.5 V	GND	5.5 V		"	CLR2		"	"
	I_{IH7}	"	55								"		GND	GND	GND	2.7 V	GND		"	CLK2		80	"
		"	56	2.7 V	GND	GND	GND				"							GND	"	CLK1		80	"
	I_{IH8}	"	57	5.5 V	GND	GND	GND				"							GND	"	CLK1		400	"
			58								"		GND	GND	GND	5.5 V	GND		"	CLK2		400	"
	I_{OS}	3011	59	GND	GND	GND	GND	GND										4.5 V	"	Q1	-15	-100	mA
		"	60	GND	GND	GND	4.5 V		GND									GND	"	Q1	"	"	"
		"	61									GND	GND	GND	GND	GND	4.5 V		"	Q2	"	"	"
		"	62							GND			4.5 V	"	"	"	GND		"	Q2	"	"	"
	I_{CC}	3005	63	GND	GND	GND	GND				"		GND	"	"	"	15.5 V	5.5 V	"	V_{CC}		8.0	"
			64	GND	GND	GND	5.5 V				"		5.5 V	"	"	"	GND	GND	"	V_{CC}		8.0	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						

See footnotes at end of device types 03 and 10.

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TABLE III. Group A inspection for device types 03 and 10 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.7 V, or open). 1/

Subgroup	Symbol	MIL-STD-883 method	Cases 2/ 2,X / Cases E,F / Test no.	* 2 / ** 2 / * 1 / ** 1	3 / 20 / 2 / 16	4 / 5 / 3 / 4	5 / 3 / 4 / 2	7 / 19 / 5 / 15	8 / 18 / 6 / 14	9 / 13 / 7 / 10	10 / 17 / 8 / 13	12 / 14 / 9 / 11	13 / 9 / 10 / 7	14 / 12 / 11 / 9	15 / 15 / 12 / 12	17 / 8 / 13 / 6	18 / 10 / 14 / 8	19 / 4 / 15 / 3	20 / 7 / 16 / 5	Measured terminal	Test limits Min	Test limits Max	Unit
				CLK1	K1	J1	PR1	Q1	$\bar{Q}1$	$\bar{Q}2$	GND	Q2	PR2	J2	K2	CLK2	CLR2	CLR1	V_{CC}				
7 6/7/ T_C = +25°C	Truth table tests	3014	65	B	B	A	A	L	H	H	GND	L	A	A	B	B	B	B	4.5 V	All outputs	See 8/		
		"	66	A	"	"	"	"	"	"	"	"	"	"	"	A	"	"	"	"	"		
		"	67	B	"	"	"	"	"	"	"	"	"	"	"	B	"	"	"	"	"		
		"	68	B	A	B	B	H	L	L	"	H	B	B	A	B	A	A	"	"	"		
		"	69	A	"	"	"	"	"	"	"	"	"	"	"	A	"	"	"	"	"		
		"	70	B	"	"	"	"	"	"	"	"	"	"	"	B	"	"	"	"	"		
		"	71	"	B	"	A	L	H	H	"	L	A	"	B	"	B	B	"	"	"		
		"	72	"	"	"	"	"	"	"	"	"	"	"	"	"	A	A	"	"	"		
		"	73	A	"	"	"	"	"	"	"	"	"	"	"	A	"	"	"	"	"		
		"	74	B	"	"	"	"	"	"	"	"	"	"	"	B	"	"	"	"	"		
		"	75	"	"	"	B	H	L	L	"	H	B	"	"	"	"	"	"	"	"		
		"	76	"	"	"	A	"	"	"	"	"	A	"	"	"	"	"	"	"	"		
		"	77	A	"	"	"	"	"	"	"	"	"	"	"	A	"	"	"	"	"		
		"	78	B	"	"	"	"	"	"	"	"	"	"	"	B	"	"	"	"	"		
		"	79	"	"	A	"	L	H	H	"	L	"	A	"	"	B	B	"	"	"		
		"	80	"	"	"	"	"	"	"	"	"	"	"	"	"	A	A	"	"	"		
		"	81	A	"	"	"	"	"	"	"	"	"	"	"	A	"	"	"	"	"		
		"	82	B	"	"	"	H	L	L	"	H	"	"	"	B	"	"	"	"	"		
		"	83	"	A	B	B	"	"	"	"	"	B	B	A	"	"	"	"	"	"		
		"	84	"	"	"	A	"	"	"	"	"	A	"	"	"	"	"	"	"	"		
		"	85	A	"	"	"	"	"	"	"	"	"	"	"	A	"	"	"	"	"		
		"	86	B	"	"	"	L	H	H	"	L	"	"	"	B	"	"	"	"	"		
		"	87	A	"	A	B	H	"	"	"	H	B	A	"	A	B	B	"	"	"		
		"	88	B	"	"	A	L	"	"	"	L	A	"	"	B	B	B	"	"	"		
		"	89	B	"	"	"	"	"	"	"	"	"	"	"	B	A	A	"	"	"		
		"	90	A	"	"	"	"	"	"	"	"	"	"	"	A	"	"	"	"	"		
		"	91	B	"	"	"	H	L	L	"	H	"	"	"	B	"	"	"	"	"		
		"	92	A	"	"	"	H	L	L	"	H	"	"	"	A	"	"	"	"	"		
		"	93	B	"	"	"	L	H	H	"	L	"	"	"	B	"	"	"	"	"		
8 6/7/	Repeat subgroup 7 at T_C = +125°C and T_C = -55°C.																						
9 T_C = +25°C	f_{MAX} 4/	Fig. 11	94	IN	2.7 V	2.7 V	2.7 V	OUT			GND							2.7 V	5.0 V	Q1	25		MHz
		"	95	IN	2.7 V	2.7 V	2.7 V		OUT		"							2.7 V	"	$\bar{Q}1$	"		"
		"	96							OUT	"		2.7 V	2.7 V	2.7 V	IN	2.7 V		"	Q2	"		"
		"	97								"	OUT	2.7 V	2.7 V	2.7 V	IN	2.7 V		"	$\bar{Q}2$	"		"
	t_{PLH1}	3003	98 10/	IN	GND	2.7 V	2.7 V		OUT		"							IN	"	CLR1 to $\bar{Q}1$	5	21	ns
		Fig. 12	99	IN	2.7 V	GND	IN	OUT			"							2.7 V	"	PR1 to Q1	"	"	"
		"	100							OUT	"		2.7 V	2.7 V	GND	IN	IN		"	CLR2 to Q2	"	"	"
		"	101								"	OUT	IN	GND	2.7 V	IN	2.7 V		"	PR2 to $\bar{Q}2$	"	"	"
	t_{PHL1}	"	102 10/								"	OUT	2.7 V	2.7 V	GND	IN	IN		"	CLR2 to Q2	"	28	"
		"	103							OUT	"		IN	GND	2.7 V	IN	2.7 V		"	PR2 to $\bar{Q}2$	"	"	"
		"	104	IN	GND	2.7 V	2.7 V	OUT			"							IN	"	CLR1 to $\bar{Q}1$	"	"	"
		"	105	IN	2.7 V	GND	IN		OUT		"							2.7 V	"	PR1 to Q1	"	"	"
	t_{PLH2}	3003	106	IN	2.7 V	2.7 V	2.7 V		OUT		"							2.7 V	"	CLK1 to $\bar{Q}1$	"	22	"
		Fig. 11	107	IN	2.7 V	2.7 V	2.7 V	OUT			"							2.7 V	"	CLK1 to Q1	"	"	"
		"	108							OUT	"		2.7 V	2.7 V	2.7 V	IN	2.7 V		"	CLK2 to $\bar{Q}2$	"	"	"
		"	109								"	OUT	2.7 V	2.7 V	2.7 V	IN	2.7 V		"	CLK2 to Q2	"	"	"
	t_{PHL2}	"	110								"	OUT	2.7 V	2.7 V	2.7 V	IN	2.7 V		"	CLK2 to Q2	"	30	"
		"	111							OUT	"		2.7 V	2.7 V	2.7 V	IN	2.7 V		"	CLK2 to $\bar{Q}2$	"	"	"
		"	112	IN	2.7 V	2.7 V	2.7 V	OUT			"							2.7 V	"	CLK1 to Q1	"	"	"
		"	113	IN	2.7 V	2.7 V	2.7 V		OUT		"							2.7 V	"	CLK1 to $\bar{Q}1$	"	"	"

See footnotes at end of device types 03 and 10.

MIL-M-38510/503C

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TABLE III. Group A inspection for device types 03 and 10 - Continued.
Terminal conditions (pins not designated may be high >2.0 V, low <0.7 V, or open). 1/

Subgroup	Symbol	MIL-STD-883 method	Cases 2/ 2,X / Cases E,F / Test no.	* 2 / ** 1 / CLK1	3 / 20 / 5 / K1	4 / 5 / 3 / J1	5 / 3 / 4 / PR1	7 / 19 / 5 / Q1	8 / 18 / 6 / $\overline{Q1}$	9 / 11 / 7 / $\overline{Q2}$	10 / 17 / 8 / GND	12 / 14 / 9 / Q2	13 / 9 / 10 / PR2	14 / 12 / 11 / J2	15 / 15 / 12 / K2	17 / 8 / 13 / CLK2	18 / 10 / 14 / CLR2	19 / 4 / 15 / CLR1	20 / 7 / 16 / V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
10	f_{MAX} 9/	Fig. 11	114-117																		25		MHz
	t_{PLH1}	3003 Fig. 12	118-121	Same tests and terminal conditions as for subgroup 9, except T_C = +125°C.																	5	32	ns
	t_{PHL1}	3003 Fig. 12	122-125																		"	40	"
	t_{PLH2}	3003 Fig. 11	126-129																		"	32	"
	t_{PHL2}	3003 Fig. 11	130-133																		"	42	"
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																						

* Terminal numbers for device type 03.
** Terminal numbers for device type 10.
1/ See 6.4 for special application note.
2/ Case 2 pins not referenced are NC.
3/ ⎍⎍ 2.5 V minimum/5.5 V maximum. ⎍⎍ 0 V
4/ ⎍⎍ 2.5 V minimum/5.5 V maximum. ⎍⎍ 0 V
5/ I_{IL} limits in mA are as follows:

	Min/Max limits for CKT A	B	C	D	E
I_{IL1}	-.075/-.250	-.030/-.300	-.150/-.560	-.120/-.360	-.120/-.360
I_{IL3}	-.150/-.500	-.060/-.600	-.250/-.560	-.240/-.720	-.280/-.760
I_{IL4}	-.200/-.800	-.060/-.700	-.290/-.650	-.120/-.720	-.320/-.800

6/ Input voltages shown are A = 2.0 volts minimum and B = 0.7 volt maximum.
7/ Tests shall be performed in sequence, attributes data only.
8/ Output voltages shall be H >1.5 V and L <1.5 V.
9/ f_{MAX} minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency.
10/ These tests may be performed as shown in table III or alternately as follows:

Test no.	CLK1	K1	J1	PR1	Q1	$\overline{Q1}$	$\overline{Q2}$	GND	Q2	PR2	J2	K2	CLK2	CLR2	CLR1	V_{CC}
98A	2.7 V	2.7 V	2.7 V	2/		OUT		GND							IN	5.0 V
99A	2.7 V	2.7 V	2.7 V	IN	OUT			"							2/	"
100A							OUT	"		2/	2.7 V	2.7 V	2.7 V	IN		"
101A								"	OUT	IN	2.7 V	2.7 V	2.7 V	2/		"
102A								"	OUT	2/	IN	2.7 V	2.7 V	IN		"
103A							OUT	"		IN	2.7 V	2.7 V	2.7 V	2/		"
104A	2.7 V	2.7 V	2.7 V	2/	OUT			"							IN	"
105A	2.7 V	2.7 V	2.7 V	IN		OUT		"							2/	"

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TABLE III. Group A inspection for device type 06.
Terminal conditions (pins not designated may be high >2.0 V, low <0.7 V, or open).

Subgroup	Symbol	MIL STD-883 method	Cases 1/ 2,X Cases E,F	1	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
			Test no.	CLR	Q1	D1	Q2	D2	Q3	D3	Q4	GND	CLK	Q5	D4	Q6	D5	D6	Q7	V_{CC}				
1	V_{OH}	3006	1	2.0 V	.4 mA	2.0 V						GND	2/							4.5 V	Q1	2.5		V
T_C = +25°C		"	2	"			2.0 V	-.4 mA				"	"							"	Q2	"		"
		"	3	"						2.0 V	-.4 mA	"	"							"	Q3	"		"
		"	4	"								"	"	-.4 mA	2.0 V					"	Q4	"		"
		"	5	"								"	"			.4 mA	2.0 V			"	Q5	"		"
		"	6	"								"	"					2.0 V	-.4 mA	"	Q6	"		"
	V_{OL}	3007	7	"								"	"					0.7 V	4 mA	"	Q6		0.4	"
	"	"	8	"								"	"			4 mA	0.7 V			"	Q5		"	"
	"	"	9	"								"	"	4 mA	0.7 V					"	Q4		"	"
	"	"	10	"						0.7 V	4 mA	"	"							"	Q3		"	"
	"	"	11	"				0.7 V	4 mA			"	"							"	Q2		"	"
	"	"	12	"	4 mA	0.7 V						"	"							"	Q1		"	"
	"	"	13	0.7 V	4 mA							"								"	Q1		"	"
	"	"	14	"					4 mA			"								"	Q2		"	"
	"	"	15	"							4 mA	"								"	Q3		"	"
	"	"	16	"								"		4 mA						"	Q4		"	"
	"	"	17	"								"				4 mA				"	Q5		"	"
	"	"	18	"								"							4 mA	"	Q6		"	"
	V_{IC}		19	-18 mA								"								"	CLR		1.5	"
	"		20			-18 mA						"								"	D1		"	"
	"		21					-18 mA				"								"	D2		"	"
	"		22							18 mA		"								"	D3		"	"
	"		23									"	18 mA							"	CLK		"	"
	"		24									"			-18 mA					"	D4		"	"
	"		25									"					-18 mA			"	D5		"	"
	"		26									"						-18 mA		"	D6		"	"
	I_{ILL}	3009	27									"						0.4 V		5.5 V	D6	3/	3/	uA
	"	"	28									"					0.4 V			"	D5	"	"	"
	"	"	29									"			0.4 V					"	D4	"	"	"
	"	"	30							0.4 V		"								"	D3	"	"	"
	"	"	31					0.4 V				"								"	D2	"	"	"
	"	"	32			0.4 V						"								"	D1	"	"	"

See footnotes at end of device type 06.

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TABLE III. Group A inspection for device type 06 - Continued.
Terminal conditions (pins not designated may be high >2.0 V, low <0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 1/ 2,X Cases E,F Test no.	2 1 CLR	3 2 Q1	4 3 D1	5 4 D2	7 5 Q2	8 6 D3	9 7 Q3	10 8 GND	12 9 CLK	13 10 Q4	14 11 D4	15 12 Q5	17 13 D5	18 14 D6	19 15 Q6	20 16 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
1 T_C = +25°C	I_{IL2}	3009	33	0.4 V							GND	0.4 V						5.5 V	CLR	3/	3/	mA
			34								"							"	CLK	3/	3/	mA
	I_{IH1}	3010	35	2.7 V														"	CLR		20	µA
		"	36			2.7 V												"	D1		"	"
		"	37				2.7 V											"	D2		"	"
		"	38						2.7 V									"	D3		"	"
		"	39									2.7 V						"	CLK		"	"
		"	40											2.7 V				"	D4		"	"
		"	41												2.7 V			"	D5		"	"
		"	42													2.7 V		"	D6		"	"
	I_{IH2}	"	43														5.5 V	"	D6		100	"
		"	44													5.5 V		"	D5		"	"
		"	45									5.5 V						"	D4		"	"
		"	46						5.5 V									"	CLK		"	"
		"	47				5.5 V											"	D3		"	"
		"	48			5.5 V												"	D2		"	"
		"	49															"	D1		"	"
		"	50	5.5 V														"	CLR		"	"
	I_{OS}	3011	51	4.5 V	GND	4.5 V						4/						"	Q1	-15	-100	mA
		"	52	"			4.5 V	GND				"						"	Q2	"	"	"
		"	53	"					4.5 V	GND		"	GND	4.5 V				"	Q3	"	"	"
		"	54	"								"			GND	4.5 V		"	Q4	"	"	"
		"	55	"								"					4.5 V	"	Q5	"	"	"
		"	56	"								"						"	Q6	"	"	"
	I_{CC}	3005	57	5.5 V		5.5 V	5.5 V		5.5 V		"	"		5.5 V		5.5 V	5.5 V	"	V_{CC}		26	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																					
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																					
7 5/6/ T_C = 25°C	Truth table tests	3014	58	B	L	A	A	L	A	L	GND	A	L	A	L	A	A	L	4.5 V	All outputs	See 7/		
		"	59	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"	"	"	"
		"	60	"	"	"	"	"	"	"	"	A	"	"	"	"	"	"	"	"	"	"
		"	61	A	"	"	"	"	"	"	"	A	H	"	H	"	"	H	"	"	"	"
		"	62	"	H	"	"	H	"	H	"	B	"	B	"	B	B	"	"	"	"	"
		"	63	"	"	B	B	"	B	"	"	A	L	"	L	A	"	L	"	"	"	"
		"	64	"	L	"	"	L	"	L	"	B	L	A	L	"	A	L	"	"	"	"
		"	65	"	"	A	A	H	A	H	"	A	H	B	H	B	"	H	"	"	"	"
		"	66	"	H	"	"	H	"	H	"	B	H	"	L	A	A	L	"	"	"	"
		"	67	"	L	A	A	L	A	L	"	A	L	A	H	"	"	H	"	"	"	"
		"	68	"	H	"	"	L	"	H	"	B	H	B	H	B	B	H	"	"	"	"
		"	69	"	H	"	"	H	"	L	"	A	L	"	L	A	B	L	"	"	"	"
		"	70	B	L	"	"	H	"	L	"	B	L	A	L	"	A	L	"	"	"	"

See footnotes at end of device type 06.

MIL-M-38510/3010

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TABLE III. Group A inspection for device type 06 - Continued.
Terminal conditions (pins not designated may be high >2.0 V, low <0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 1/ 2,X Cases E,F Test no.	2	1	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
				$\overline{CLR}$	Q1	D1	D2	Q2	D3	$\overline{Q3}$	GND	CLK	Q4	D4	Q5	$\overline{D5}$	Q6	$\overline{Q6}$	V_{CC}				
8	Repeat subgroup 7 at T_C = +125°C and T_C = -55°C.																							
9	f_{MAX} 8/	Fig. 13	71	2.7 V	OUT	IN					GND	IN								5.0 V	Q1	25		MHz
T_C = +25°C		"	72	"			IN	OUT			"	"								"	Q2	"		"
		"	73	"					IN	OUT	"	"								"	Q3	"		"
		"	74	"							"	"	OUT	IN						"	Q4	"		"
		"	75	"							"	"			OUT	IN				"	Q5	"		"
		"	76	"							"	"					IN	OUT		"	Q6	"		"
	t_{PHL1}	3003	77	IN							"	"					2.7 V	OUT		"	CLR to Q6	5	42	ns
		Fig. 14	78	"							"	"			OUT	2.7 V				"	CLR to Q5	"	"	"
		"	79	"							"	"	OUT	2.7 V						"	CLR to Q4	"	"	"
		"	80	"					2.7 V	OUT	"	"								"	CLR to Q3	"	"	"
		"	81	"			2.7 V	OUT			"	"								"	CLR to Q2	"	"	"
		"	82	"	OUT	2.7 V					"	"								"	CLR to Q1	"	"	"
	t_{PLH2}	3003	83	2.7 V	OUT	IN					"	"								"	CLK to Q1	"	37	"
		Fig. 13	84	"			IN	OUT			"	"								"	CLK to Q2	"	"	"
		"	85	"					IN	OUT	"	"								"	CLK to Q3	"	"	"
		"	86	"							"	"	OUT	IN						"	CLK to Q4	"	"	"
		"	87	"							"	"			OUT	IN				"	CLK to Q5	"	"	"
		"	88	"							"	"					IN	OUT		"	CLK to Q6	"	"	"
	t_{PHL2}	3003	89	"							"	"					IN	OUT		"	CLK to Q6	"	40	"
		Fig. 15	90	"							"	"			OUT	IN				"	CLK to Q5	"	"	"
		"	91	"							"	"	OUT	IN						"	CLK to Q4	"	"	"
		"	92	"					IN	OUT	"	"								"	CLK to Q3	"	"	"
		"	93	"			IN	OUT			"	"								"	CLK to Q2	"	"	"
		"	94	"	OUT	IN					"	"								"	CLK to Q1	"	"	"
10	f_{MAX} 8/	Fig. 13	95-100																			25		MHz
	t_{PHL1}	3003 Fig. 14	101-106	Same tests and terminal conditions as for subgroup 9, except T_C T_C = +125°C.																		5	52	ns
	t_{PLH2}	3003 Fig. 13	107-112																			"	42	"
	t_{PHL2}	3003 Fig. 15	113-118																			"	52	"
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																							

1/ Cases 2 pins not referenced are NC.

2/ ⌐¬ ⌐¬ 2.5 V minimum/5.5 V maximum.
0 V

3/ I_{IL} limits in mA are as follows:

I_{IL1}	Min/Max limits for CKT A	B	C	D	E	F	G
	-.085/ -.270	-.100/ -.340	-.075/-.250	-.075/-.250	-.120/-.360	-.160/-.400	-.075/-.250

I_{IL2}	Min/Max limits for CKT A	B	C	D	E	F	G
	5/ -.350	-.153/ -.420	-.125/-.275 for test 33	-.120/-.360	-.120/-.360	-.150/-.380 for test 33	-.075/ -.250 for test 33
			-.160/-.400 for test 34			-.160/-.400 for test 34	-.120/ -.360 for test 34

4/ ⌐¬ ⌐¬ 2.5 V minimum/5.5 V maximum.
0 V

5/ Input voltages shown are A = 2.0 volts minimum and B = 0.7 volt maximum.

6/ Tests shall be performed in sequence, attributes data only.

7/ Output voltages shall be H >1.5 V and L <1.5 V.

8/ f_{MAX} minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency.

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TABLE III. Group A inspection for device type 07.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 1/ 2,X Cases E,F	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
			Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16				
				CLR	Q1	Q1	D1	D2	Q2	Q2	GND	CLK	Q3	Q3	D3	D4	Q4	Q4	V_{CC}				
1 T_C = +25°C	V_{OH}	3006	1	0.7 V		-.4 mA			.4 mA		GND	2/		-.4 mA		2.0 V	-.4 mA	.4 mA	4.5 V	Q1	2.5		V
			2	"		"			"		"	"		"		"	"	"	"	Q2	"		"
			3	"		"			"		"	"		"		"	"	"	"	Q3	"		"
			4	"		"			"		"	"		"		"	"	"	"	Q4	"		"
			5	2.0 V		"			"		"	"	-.4 mA	"	2.0 V	"	"	"	"	Q4	"		"
			6	"		"			"		"	"	"	"	"	"	"	"	"	Q3	"		"
			7	"		"		2.0 V	"	-.4 mA	"	"	"	"	"	"	"	"	"	Q2	"		"
			8	"	-.4 mA	"	2.0 V	"	"	"	"	"	"	"	"	"	"	"	"	Q1	"		"
			9	"	"	-.4 mA	0.7 V	0.7 V	-.4 mA	"	"	"	"	-.4 mA	0.7 V	0.7 V	-.4 mA	"	"	Q1	"		"
			10	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	Q2	"		"
			11	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	Q3	"		"
			12	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	Q4	"		"
	V_{OL}	3007	13	0.7 V	4 mA					4 mA			4 mA					4 mA		Q1		0.4	"
			14	"	"					"			"					"		Q2		"	"
			15	"	"					"			"					"		Q3		"	"
			16	"	"					"			"					"		Q4		"	"
			17	2.0 V	"			2.0 V	4 mA	"		2/	"	4 mA	2.0 V	2.0 V	4 mA	"		Q4		"	"
			18	"	"			"	"	"		"	"	"	"	"	"	"		Q3		"	"
			19	"	"			"	"	"		"	"	"	"	"	"	"		Q2		"	"
			20	"	"	4 mA	2.0 V	0.7 V	"	"		"	"	"	"	"	"	"		Q1		"	"
			21	"	4 mA	"	"	0.7 V	"	4 mA		"	4 mA	"	0.7 V	0.7 V	"	"		Q1		"	"
			22	"	"	"	"	"	"	"		"	"	"	"	"	"	"		Q2		"	"
			23	"	"	"	"	"	"	"		"	"	"	"	"	"	"		Q3		"	"
			24	"	"	"	"	"	"	"		"	"	"	"	"	"	"		Q4		"	"
	V_{IC}		25	-18 mA			-18 mA													CLR		-1.5	"
			26	"			"													D1		"	"
			27	"			"													D2		"	"
			28	"			"					-18 mA			-18 mA					CLK		"	"
			29	"			"					"			"					D3		"	"
			30	"			"					"			"					D4		"	"
	I_{IL1}	3009	31																5.5 V	D4	3/	3/	mA
			32																"	D3	"	"	"
			33					0.4 V	0.4 V						0.4 V	0.4 V			"	D2	"	"	"
			34					"	"						"	"			"	D1	"	"	"
	I_{IL2}	"	35	0.4 V								0.4 V							"	CLR	"	"	"
			36	"								"								CLK	"	"	"

See footnotes at end of device type 07.

MIL-M-38510/301D

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TABLE III. Group A inspection for device type 07 - Continued.
Terminal conditions (pins not designated may be high >2.0 V, low <0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 1/ 2,X Cases F,F Test no.	2	3	4	5	7	8	9	10	17	11	14	15	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
				1	$\overline{1}$	$\overline{1}$	$\overline{1}$	$\overline{1}$	$\overline{1}$	$\overline{1}$	$\overline{1}$	$\overline{1}$	$\overline{1}$	$\overline{1}$	$\overline{1}$	$\overline{1}$	$\overline{1}$	$\overline{1}$	$\overline{1}$				
				CLR	Q1	$\overline{Q}$1	D1	D2	$\overline{Q}$2	Q2	GND	CLK	Q3	$\overline{Q}$3	D3	D4	$\overline{Q}$4	Q4	V_{CC}				
1 T_C = +25°C	I_{IH1}	3010	37	2.7 V							GND								5.5 V	CLR		20	µA
		"	38				2.7 V												"	D1		"	"
		"	39					2.7 V											"	D2		"	"
		"	40									2.7 V							"	CLK		"	"
		"	41												2.7 V				"	D3		"	"
		"	42													2.7 V			"	D4		"	"
	I_{IH2}	"	43													5.5 V			"	D4		100	"
		"	44												5.5 V				"	D3		"	"
		"	45									5.5 V							"	CLK		"	"
		"	46					5.5 V											"	D2		"	"
		"	47				5.5 V												"	D1		"	"
		"	48	5.5 V															"	CLR		"	"
	I_{OS}	3011	49	GND		GND													"	Q1	15	100	mA
		"	50	"					GND										"	Q2	"	"	"
		"	51	"										GND					"	Q3	"	"	"
		"	52	"													GND		"	Q4	"	"	"
		"	53	4.5 V								4/				4.5 V		GND	"	Q4	"	"	"
		"	54	"								"	GND		4.5 V				"	Q3	"	"	"
		"	55	"				4.5 V		GND		"							"	Q2	"	"	"
		"	56	"	GND		4.5 V					"							"	Q1	"	"	"
	I_{CC}	3005	57	5.5 V			5.5 V	5.5 V							5.5 V	5.5 V			"	V_{CC}		18	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} test are omitted																						
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						
7 5/6/ T_C = +25°C	Truth table tests	3014	58	B	L	H	A	A	H	L	GND	B	L	H	A	A	H	L	4.5 V	All outputs	See 7/		
	"	"	59	"	"	"	"	"	"	"	"	A	"	"	"	"	"	"	"	"	"			
	"	"	60	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"	"	"			
	"	"	61	A	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"	"	"			
	"	"	62	"	H	L	"	"	L	H	"	A	H	L	"	"	"	L	H	"	"			
	"	"	63	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"	"	"			
	"	"	64	"	"	"	B	B	"	"	"	B	"	"	"	B	B	"	"	"	"			
	"	"	65	"	L	H	"	"	H	L	"	A	L	H	"	"	"	H	L	"	"			
	"	"	66	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"	"	"			
	"	"	67	"	"	"	A	A	"	"	"	B	"	"	"	A	A	"	"	"	"			
	"	"	68	"	H	L	"	"	L	H	"	A	H	L	"	"	"	L	H	"	"			
	"	"	69	"	H	L	"	"	L	H	"	B	H	L	"	"	"	L	H	"	"			
	"	"	70	B	L	H	"	"	H	L	"	B	L	H	"	"	"	H	L	"	"			
8 4/5/	Repeat subgroup 7 at T_C = 125°C and T_C = -55°C.																						
9 T_C = +25°C	f_{MAX} 8/	Fig. 13	71	2.7 V	OUT		IN				GND	IN							5.0 V	Q1	25		MHz
	"	"	72	"		OUT	IN				"	"							"	$\overline{Q}$1	"		"
	"	"	73	"				IN	OUT		"	"							"	Q2	"		"
	"	"	74	"				IN		OUT	"	"							"	$\overline{Q}$2	"		"
	"	"	75	"							"	"	OUT		IN				"	Q3	"		"
	"	"	76	"					OUT		"	"		OUT	IN				"	$\overline{Q}$3	"		"
	"	"	77	"							"	"				IN	OUT		"	Q4	"		"
	"	"	78	"							"	"				IN		OUT	"	$\overline{Q}$4	"		"
	t_{PLH1}	3003 Fig. 14	79	IN							"	"				2.7 V	OUT		"	CLR to $\overline{Q}$4	5	32	ns
	"	"	80	"							"	"		OUT	2.7 V				"	CLR to $\overline{Q}$3	"	"	"
	"	"	81	"				2.7 V	OUT		"	"							"	CLR to $\overline{Q}$2	"	"	"
	"	"	82	"		OUT	2.7 V				"	"							"	CLR to $\overline{Q}$1	"	"	"

See footnote at end of device type 07.

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TABLE III. Group A inspec(r device type 07 - Continued.
Terminal conditions (pins not designa. ay be high >2.0 V, low <0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 1/ 2,X Cases E,F	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
			Test no.	1 CLR	2 Q1	3 Q1	4 D1	5 Q2	6 Q2	7 Q2	8 GND	9 CLK	10 Q3	11 Q3	12 D3	13 D4	14 Q4	15 Q4	16 V_{CC}				
9 T_C = +25°C	t_{PHL1}	3003 Fig. 14	83	IN	OUT		2.7 V				GND	IN							5.0 V	CLR to Q1	5	45	ns
			84	"				2.7 V		OUT	"	"			2.7 V				"	CLR to Q2	"	"	"
			85	"							"	"	OUT				OUT		"	CLR to Q3	"	"	"
			86	"							"	"		2.7 V					"	CLR to Q4	"	"	"
	t_{PLH2}	3003 Fig. 13	87	2.7 V							"	"			IN		OUT		"	CLK to Q4	"	35	"
			88	"							"	"	OUT	IN					"	CLK to Q3	"	"	"
			89	"				IN		OUT	"	"							"	CLK to Q2	"	"	"
			90	"	OUT		IN				"	"							"	CLK to Q1	"	"	"
		3003 Fig. 15	91	"		OUT	IN				"	"							"	CLK to Q1	"	"	"
			92	"				IN	OUT		"	"							"	CLK to Q2	"	"	"
			93	"							"	"		OUT	IN				"	CLK to Q3	"	"	"
			94	"							"	"				IN	OUT		"	CLK to Q4	"	"	"
	t_{PHL2}	3003 Fig. 13	95	"							"	"			IN		OUT		"	CLK to Q4	"	40	"
			96	"							"	"		OUT	IN				"	CLK to Q3	"	"	"
			97	"				IN	OUT		"	"							"	CLK to Q2	"	"	"
			98	"		OUT	IN				"	"							"	CLK to Q1	"	"	"
		3003 Fig. 15	99	"	OUT		IN				"	"							"	CLK to Q1	"	"	"
			100	"				IN		OUT	"	"							"	CLK to Q2	"	"	"
			101	"							"	"	OUT	IN					"	CLK to Q3	"	"	"
			102												IN		OUT		"	CLK to Q4	"	"	"
10	f_{MAX} 8/	Fig. 13	103-110																		25		MHz
	t_{PLH1}	3003 Fig. 14	111-114																		5	51	ns
	t_{PHL1}	3003 Fig. 14	115-118																		"	55	"
	t_{PLH2}	3003 Fig. 13	119-122	Same tests and terminal conditions as for subgroup 9, except T_C = +125°C																	"	46	"
	t_{PLH2}	3003 Fig. 15	123-126																		"	46	"
	t_{PHL2}	3003 Fig. 13	127-130																		"	55	"
	t_{PHL2}	3003 Fig. 15	131-134																		"	55	"
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																						

1/ Cases 2 pins not referenced are NC.

2/ [ ] 2.5 V minimum/5.5 V maximum.
[ ] 0 V

3/ I_{IL} limits in mA are as follows:

	Min/Max limits for CKT A	B	C	D	E	F	G
I_{IL1}	-.075/-.250	-.100/-.340	-.075/-.250	-.075/-.250	-.120/-.360	-.160/-.400	-.075/-.250

	Min/Max limits for CKT A	B	C	D	E	F	G
I_{IL2}	-.085/-.270 for test 35	-.150/-.420	-.125/-.275 for test 35	-.120/-.400 for test 35	-.120/-.400	-.105/-.380 for test 35	-.075/-.250 for test 35
	-.135/-.400 for test 36		-.160/-.400 for test 36	-.120/-.360 for test 36		-.160/-.400 for test 36	-.120/-.350 for test 36

4/ [ ] 2.5 V minimum/5.5 V maximum.
[ ] 0 V

5/ Input voltages shown are A = 2.0 volts minimum and B = 0.7 volt maximum.

6/ Tests shall be performed in sequence, attributes data only.

7/ Output voltages shall be H >1.5 V and L <1.5 V.

8/ f_{MAX} minimum limit specified is the frequency of the input pulse. The output frequency shall be one half of the input frequency.

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TABLE III. Group A inspection for device type 09.
Terminal conditions (pins not designated may be high >2.0 V, low <0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 1/ 2,x Cases E,F	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
			Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16				
				CLR1	J1	K1	CLK1	PR1	Q1	$\overline{Q1}$	GND	$\overline{Q2}$	Q2	PR2	CLK2	K2	J2	CLR2	V_{CC}				
1 T_C = +25°C	V_{OH}	3006	1	0.7 V	0.7 V	0.7 V	GND	2.0 V		-.4 mA	GND								4.5 V	Q1	2.5		V
		"	2	2.0 V	"	"	GND	0.7 V	-.4 mA		"								"	Q1	"		"
		"	3	"	"	"	2/	2.0 V		-.4 mA	"								"	$\overline{Q1}$	"		"
		"	4	"	2.0 V	2.0 V	2/	2.0 V	-.4 mA		"								"	Q1	"		"
		"	5								"	-.4 mA		2.0 V	GND	0.7 V	0.7 V	0.7 V	"	$\overline{Q2}$	"		"
		"	6								"		-.4 mA	0.7 V	GND	"	"	2.0 V	"	Q2	"		"
		"	7								"	-.4 mA		2.0 V	2/	"	"	"	"	$\overline{Q2}$	"		"
		"	8								"		-.4 mA	"	2/	2.0 V	2.0 V	"	"	Q2	"		"
	V_{OL}	3007	9								"		4 mA	"	GND	0.7 V	0.7 V	0.7 V	"	Q2		0.4	"
		"	10								"	4 mA		0.7 V	GND	"	"	2.0 V	"	$\overline{Q2}$		"	"
		"	11								"		4 mA	2.0 V	2/	"	"	"	"	Q2		"	"
		"	12								"	4 mA		2.0 V	2/	2.0 V	2.0 V	"	"	$\overline{Q2}$		"	"
		"	13	0.7 V	0.7 V	0.7 V	GND	2.0 V	4 mA		"								"	Q1		"	"
		"	14	2.0 V	"	"	GND	0.7 V		4 mA	"								"	$\overline{Q1}$		"	"
		"	15	"	"	"	2/	2.0 V	4 mA		"								"	Q1		"	"
		"	16	"	2.0 V	2.0 V	2/	2.0 V		4 mA	"								"	$\overline{Q1}$		"	"
	V_{IC}		17	18 mA							"								"	CLR1		-1.5	"
			18		-18 mA						"								"	J1		"	"
			19			-18 mA					"								"	K1		"	"
			20				-18 mA				"								"	CLK1		"	"
			21					-18 mA			"								"	PR1		"	"
			22								"			-18 mA					"	PR2		"	"
			23								"				-18 mA				"	CLK2		"	"
			24								"					-18 mA			"	K2		"	"
			25								"						-18 mA		"	J2		"	"
			26								"							-18 mA	"	CLR2		"	"
	I_{IL2}	3009	27	3/	0.4 V	4.5 V	GND	4.5 V			"								5.5 V	J1	4/	4/	mA
		"	28	4.5 V	4.5 V	0.4 V	GND	3/			"								"	K1	"	"	"
		"	29								"			3/	GND	0.4 V	4.5 V	4.5 V	"	K2	"	"	"
		"	30								"			4.5 V	GND	4.5 V	0.4 V	3/	"	J2	"	"	"
	I_{IL4}	"	31								"			4.5 V	0.4 V	"	4.5 V	3/	"	CLK2	"	"	"
		"	32								"			3/	0.4 V	"	"	4.5 V	"	CLK2	"	"	"
		"	33								"			0.4 V	4.5 V	"	"	GND	"	PR2	"	"	"

See footnotes at end of device type 09.

MIL-M-38510/30D

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TABLE III. Group A inspection for device type 09 Continued.
Terminal conditions (pins not designated may be high >2.0 V, low <0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 1/ 2,X	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
			Cases E,F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16				
			Test no.	CLR1	J1	K1	CLK1	PR1	Q1	$\overline{Q1}$	GND	$\overline{Q2}$	Q2	PR2	CLK2	$\overline{K2}$	J2	CLR2	V_{CC}				
1 T_C = +25°C	I_{IL4}	3009	34	GND	4.5 V	4.5 V	4.5 V	0.4 V			GND								5.5 V	PR1	4/	4/	nA
		"	35	4.5 V	"	"	0.4 V	3/			"								"	CLK1	"	"	"
		"	36	3/	"	"	0.4 V	4.5 V			"								"	CLK1	"	"	"
	I_{IL7}	"	37	0.4 V	"	"	4.5 V	GND			"			GND	4.5 V	4.5 V	4.5 V	0.4 V	"	CLR1	"	"	"
		"	38								"								"	CLR2	"	"	"
	I_{IH1}	3010	39	GND	2.7 V	GND	4.5 V	4.5 V			"								"	J1		20	µA
		"	40	GND	GND	2.7 V	4.5 V	GND			"			GND	4.5 V	2.7 V	GND	GND	"	K1		"	"
		"	41								"			4.5 V	"	GND	2.7 V	"	"	K2		"	"
		"	42								"								"	J2		"	"
	I_{IH2}	"	43								"			4.5 V	"	GND	5.5 V	"	"	J2		100	"
		"	44								"			GND	"	5.5 V	GND	"	"	K2		"	"
		"	45	GND	GND	5.5 V	4.5 V	GND			"								"	K1		"	"
		"	46	GND	5.5 V	GND	4.5 V	4.5 V			"								"	J1		"	"
	I_{IH3}	"	47	GND	4.5 V	4.5 V	2.7 V	GND			"								"	CLK1		40	"
		"	48	4.5 V	4.5 V	4.5 V	GND	2.7 V			"			2.7 V	GND	4.5 V	4.5 V	4.5 V	"	PR1		"	"
		"	49								"			GND	2.7 V	"	"	GND	"	PR2		"	"
		"	50								"								"	CLK2		"	"
	I_{IH4}	"	51								"			GND	5.5 V	"	"	GND	"	CLK2		200	"
		"	52								"			5.5 V	GND	"	"	4.5 V	"	PR2		"	"
		"	53	4.5 V	4.5 V	4.5 V	GND	5.5 V			"								"	PR1		"	"
		"	54	GND	4.5 V	4.5 V	5.5 V	GND			"								"	CLK1		"	"
	I_{IH7}	"	55	2.7 V	4.5 V	4.5 V	GND	4.5 V			"								"	CLR1		80	"
		"	56								"			4.5 V	GND	4.5 V	4.5 V	2.7 V	"	CLR2		80	"
	I_{IH8}	"	57								"								"	CLR2		400	"
		"	58	5.5 V	4.5 V	4.5 V	GND	4.5 V			"			4.5 V	GND	4.5 V	4.5 V	5.5 V	"	CLR1		400	"
	I_{OS}	3011	59	GND				4.5 V		GND	"								"	Q1	-15	-100	mA
		"	60	4.5 V				GND	GND		"								"	Q1	"	"	"
		"	61								"		GND	GND				4.5 V	"	Q2	"	"	"
		"	62								"	GND		4.5 V				GND	"	Q2	"	"	"
	I_{CC}	3005	63	GND			GND	5.5 V			"			5.5 V	GND			GND	"	V_{CC}		8.0	"
		3005	64	5.5 V			GND	GND			"			GND	GND			5.5 V	"	V_{CC}		8.0	"

See footnotes at end of device type 09.

MIL-M-38510/301A

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TABLE III. Group A inspection for device type 09 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 1/ 2,X Cases E,F Test no.	2 CLR1	3 J1	4 $\overline{K1}$	5 CLK1	7 PR1	8 Q1	9 $\overline{Q1}$	10 GND	12 $\overline{Q2}$	13 Q2	14 PR2	15 CLK2	17 K2	18 J2	19 $\overline{CLR2}$	20 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						
7 5/6/ T_C = +25°C	Truth table tests	3014	65	A	A	A	B	B	H	L	GND	L	H	B	B	A	A	A	4.5 V	All outputs	See 7/		
	"	"	66	"	"	"	B	A	"	"	"	"	"	A	B	"	"	"	"	"	"	"		
	"	"	67	"	"	"	A	"	"	"	"	"	"	"	A	"	"	"	"	"	"	"		
	"	"	68	"	"	B	A	"	"	"	"	"	"	"	A	B	"	"	"	"	"	"		
	"	"	69	"	"	"	B	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"		
	"	"	70	"	"	"	A	"	L	H	"	H	L	"	A	"	"	"	"	"	"	"		
	"	"	71	"	"	"	B	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"		
	"	"	72	B	"	"	"	"	"	"	"	"	"	"	"	"	"	B	"	"	"	"		
	"	"	73	A	"	"	"	"	"	"	"	"	"	"	"	"	"	A	"	"	"	"		
	"	"	74	"	"	"	A	"	H	L	"	L	H	"	A	"	"	"	"	"	"	"		
	"	"	75	"	"	"	B	"	H	L	"	L	H	"	B	"	"	"	"	"	"	"		
	"	"	76	"	"	"	A	"	L	H	"	H	L	"	A	"	"	"	"	"	"	"		
	"	"	77	"	"	"	B	"	L	H	"	H	L	"	B	"	"	"	"	"	"	"		
	"	"	78	"	"	"	B	B	H	L	"	L	H	B	B	"	"	"	"	"	"	"		
	"	"	79	"	"	"	A	B	"	"	"	"	"	B	A	"	"	"	"	"	"	"		
	"	"	80	"	"	"	A	A	"	"	"	"	"	A	A	"	"	"	"	"	"	"		
	"	"	81	"	"	"	B	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"		
	"	"	82	"	"	"	A	"	L	H	"	H	L	"	A	"	"	"	"	"	"	"		
	"	"	83	"	"	"	B	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"		
	"	"	84	B	B	"	"	"	"	"	"	"	"	"	"	"	B	B	"	"	"	"		
	"	"	85	A	"	"	"	"	"	"	"	"	"	"	"	"	"	A	"	"	"	"		
	"	"	86	"	"	"	A	"	"	"	"	"	"	"	A	"	"	"	"	"	"	"		
	"	"	87	"	A	"	A	"	"	"	"	"	"	"	A	"	A	"	"	"	"	"		
	"	"	88	"	"	"	B	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"		
	"	"	89	"	"	"	"	B	H	L	"	L	H	B	"	"	"	"	"	"	"	"		
	"	"	90	"	"	"	"	A	H	L	"	L	H	A	"	"	"	"	"	"	"	"		
	"	"	91	"	"	"	A	"	L	H	"	H	L	"	A	"	"	"	"	"	"	"		
	"	"	92	"	"	"	B	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"		
	"	"	93	B	"	"	B	"	"	"	"	"	"	"	B	"	"	B	"	"	"	"		
	"	"	94	B	"	"	A	"	"	"	"	"	"	"	A	"	"	B	"	"	"	"		
	"	"	95	A	"	"	A	"	"	"	"	"	"	"	A	"	"	A	"	"	"	"		
	"	"	96	"	"	"	B	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"		
	"	"	97	"	B	A	B	"	"	"	"	"	"	"	B	A	B	"	"	"	"	"		
	"	"	98	"	B	"	A	"	"	"	"	"	"	"	A	"	B	"	"	"	"	"		
	"	"	99	"	A	"	"	B	H	L	"	L	H	B	"	"	A	"	"	"	"	"		
	"	"	100	"	"	"	"	A	H	L	"	L	H	A	"	"	"	"	"	"	"	"		
	"	"	101	B	"	"	"	A	L	H	"	H	L	A	"	"	"	B	"	"	"	"		
	"	"	102	A	B	B	B	B	H	L	"	L	H	B	B	B	B	A	"	"	"	"		
	"	"	103	A	"	"	"	A	H	L	"	L	H	A	"	"	"	A	"	"	"	"		
	"	"	104	B	"	"	"	A	L	H	"	H	L	A	"	"	"	B	"	"	"	"		
8	Repeat subgroup 7 at T_C = +125°C and T_C = -55°C.																						
9 T_C = +25°C	f_{MAX}	Fig. 16	105	2.7 V	2.7 V	GND	IN	2.7 V	OUT		GND								5.0 V	Q1	20		MHz
	"	"	106	2.7 V	2.7 V	GND	IN	2.7 V		OUT	"								"	$\overline{Q1}$	"		"
	"	"	107								"	OUT		2.7 V	IN	GND	2.7 V	2.7 V	"	Q2	"		"
	"	"	108								"		OUT	2.7 V	IN	GND	2.7 V	2.7 V	"	Q2	"		"
	t_{PLH1}	1003 Fig. 17	109	IN	2.7 V	2.7 V	IN	2.7 V		OUT	"								"	CLR1 to Q1	5	20	ns
			110	2.7 V	GND	GND	IN	IN	OUT		"								"	PR1 to Q1	"	"	"
	"	"	111								"		OUT	IN	IN	GND	GND	2.7 V	"	PR2 to Q2	"	"	"
	"	"	112								"	OUT		2.7 V	IN	2.7 V	2.7 V	IN	"	CLR2 to Q2	"	"	"
	t_{PHL1}	"	113								"		OUT	2.7 V	IN	2.7 V	2.7 V	IN	"	CLR2 to Q2	"	32	"
		"	114								"	OUT		IN	IN	GND	GND	2.7 V	"	PR2 to Q2	"	"	"
	"	"	115	2.7 V	GND	GND	IN	IN		OUT	"								"	PR1 to Q1	"	"	"
	"	"	116	IN	2.7 V	2.7 V	IN	2.7 V	OUT		"								"	CLR1 to Q1	"	"	"

See footnotes at end of device type 09.

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TABLE III. Group A inspection for device type 09 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 1/ 2,X Cases E,F Test no.	1 CLR1	2 J1	3 K1	4 CLK1	5 PR1	6 Q1	7 $\overline{Q1}$	8 GND	9 $\overline{Q2}$	10 Q2	11 PR2	12 CLK2	13 K2	14 J2	15 CLR2	16 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
9 T_C = +25°C	t_{PLH2}	3003 Fig. 16	117	2.7 V	2.7 V	GND	IN	2.7 V	OUT		GND								5.0 V	CLK1 to Q1	5	24	ns
		"	118	2.7 V	2.7 V	GND	IN	2.7 V		OUT	"								"	CLK1 to $\overline{Q1}$	"	"	"
		"	119								"	OUT		2.7 V	IN	GND	2.7 V	2.7 V	"	CLK2 to $\overline{Q2}$	"	"	"
		"	120								"		OUT	2.7 V	IN	GND	2.7 V	2.7 V	"	CLK2 to Q2	"	"	"
	t_{PHL2}	"	121								"		OUT	2.7 V	IN	GND	2.7 V	2.7 V	"	CLK2 to Q2	"	35	"
		"	122								"	OUT		2.7 V	IN	GND	2.7 V	2.7 V	"	CLK2 to $\overline{Q2}$	"	"	"
		"	123	2.7 V	2.7 V	GND	IN	2.7 V		OUT	"								"	CLK1 to $\overline{Q1}$	"	"	"
		"	124	2.7 V	2.7 V	GND	IN	2.7 V	OUT		"								"	CLK1 to Q1	"	"	"
10	f_{MAX} 8/	Fig. 16	125-128																		20		MHz
	t_{PLH1}	3003 Fig. 17	129-132	Same tests and terminal conditions as for subgroup 9, except T_C = +125°C.																	5	39	ns
	t_{PHL1}	3003 Fig. 17	133-136																		"	59	"
	t_{PLH2}	3003 Fig. 16	137-140																		"	39	"
	t_{PHL2}	3003 Fig. 16	141-144																		"	59	"
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																						

1/ Cases 2 pins not referenced are NC.

2/ 2.5 V minimum/5.5 V maximum.
0 V

3/ I_{IL} limits in mA are as follows:

I_{IL2}	Min/Max limits for CKT A	B	C	D	E	F
	-.075/-.250	-.030/-.300	-.095/-.210	-.160/-.400	-.135/-.370	-.160/-.400

I_{IL4}	Min/Max limits for CKT A	B	C	D	E	F
	-.150/-.500 for tests 31,32,35,36 -.200/-.800 for tests 33,34	-.060/-.700	-.160/-.400 for tests 31,32,35,36 -.350/-.760 for tests 33,34	-.320/-.800	-.120/-.360 for tests 31,32,35,36 -.280/-.760 for tests 33,34	-.320/-.800

I_{IL7}	Min/Max limits for CKT A	B	C	D	E	F
	-.200/-.800	-.060/-.700	-.350/-.760	-.560/-1.600	-.280/-.760	-.560/-1.600

4/ 2.5 V minimum/5.5 V maximum.
0 V

5/ Input voltages shown are A = 2.0 volts minimum and B = 0.7 volt maximum.

6/ Tests shall be performed in sequence, attributes data only.

7/ Output voltages shall be H >1.5 V and L <1.5 V.

8/ f_{MAX}, minimum limit specified is the frequency of the input pulse. The output frequency shall be one half of the input frequency.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in test (method 1015 of MIL-STD-883).

(1) Test condition D or E, using the circuit shown on figure 3, or equivalent.

(2) T_A = +125°C minimum.

b. Interim and final electrical test shall be as specified in table II, except interim electrical test prior to burn-in are optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroups 4, 5, and 6 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical tests shall be as specified in table II herein.

b. Steady-state life test (method 1005 of MIL-STD-883).

(1) Test condition D or E, using the circuit shown on figure 3.

(2) T_A = +125°C minimum.

(3) Test duration: 1,000 hours, except as permitted by method 1005 of MIL-STD-883.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be specified as follows.

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

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4.6 Inclusion with other detail specifications. For qualification and quality conformance inspection purposes, devices covered by this specification may be treated as though they were included on the same detail specification as devices covered by MIL-M-38510/331. In addition, if a manufacturer is already part I qualified for types 33106 and 33107, and if the respective devices on this specifiction (MIL-M-38510/301D) are designed and manufactured identically (same die, same process, same screening) in all respects except electrical testing, then device types 30106 and 30107 may be part I qualified by conducting only group A electrical tests, including subgroups A-10 and A-11, and submitting data in accordance with MIL-M-38510, appendix D (i.e., groups B, C, and D tests are not required).

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable, these requirements shall not affect the part number. Unless otherwise specified, these requirements shall not apply to direct purchase by or direct shipment to the Government.

h. Requirements for "JAN" marking.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND - - - - - - - - - - - - - Ground zero voltage potential.
I_{IN} - - - - - - - - - - - - Current flowing into an input.
V_{IN} - - - - - - - - - - - - Voltage level at an input terminal.

6.4 Special application note for device type 03. For system applications all outputs shall be terminated either through a 2 kΩ resistor to V_{CC} or through a 15 pF capacitor to GND through an unused input of another gate. Erroneous operation at low temperature may result if this precaution is not observed.

6.5 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2) and lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

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6.6 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	54LS73
02	54LS74A
03	54LS112
04	54LS113
05	54LS114
06	54LS174
07	54LS175
08	54LS107
09	54LS109
10	54LS76A

6.7 Manufacturer's designation. Manufacturers' circuits which form a part of this specification are designated as shown in table IV herein.

TABLE IV. Manufacturers' designation.

Device type	Manufacturers						
	Texas Instruments	Signetics Corp.	National Semiconductor Corp.	Raytehon Co.	Motorola Inc.	Fairchild Semi-conductor	Advanced Micro Devices
01	A	B	C	D	E	---	---
02	A	B	C	D	E	F	---
03	A	B	C	C	D	E	---
04	A	B	C	C	F	E	D
05	A	---	C	C	D	E	---
06	A	B	C	E	F	G	D
07	A	B	C	E	F	G	D
08	A	B	C	D	E	---	---
09	A	B	C	---	E	F	---
10	A	B	C	C	D	E	---

6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SH, TD
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

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